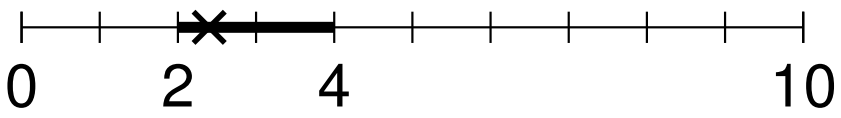
Bertrand's paradox movies

Márton Balázs

University of Bristol

Uniform distribution

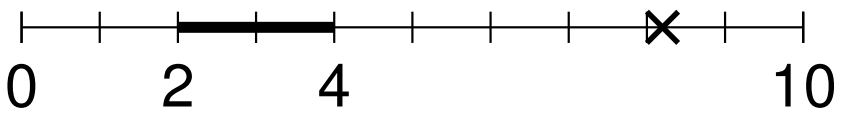
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$\mathbb{P}\{\text{—}\} = ?$

Uniform distribution

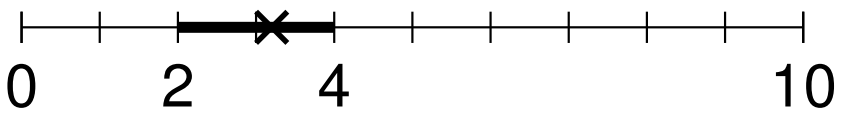
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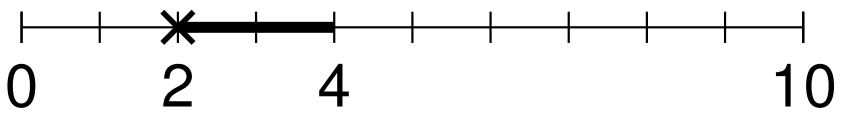
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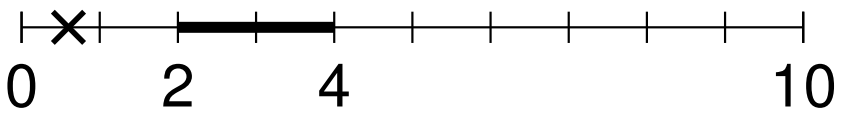
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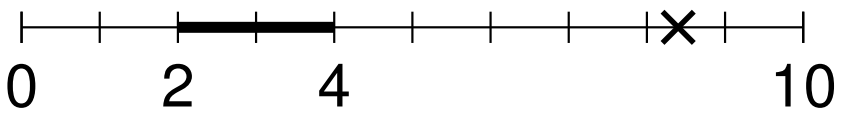
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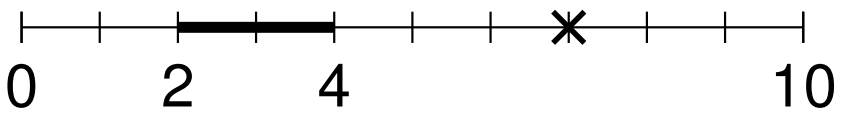
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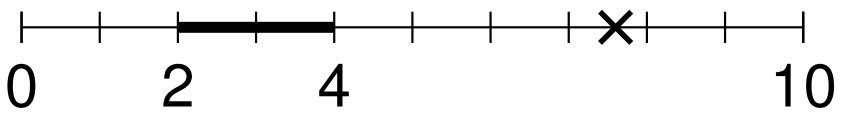
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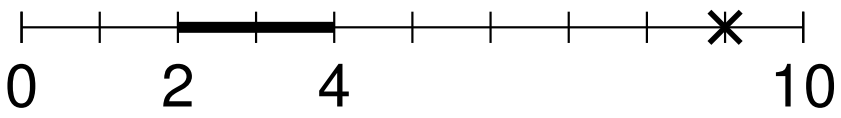
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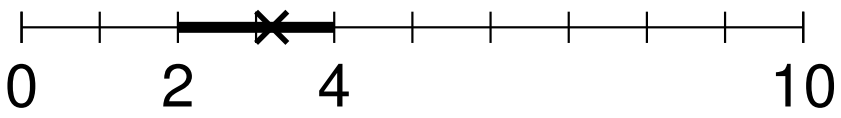
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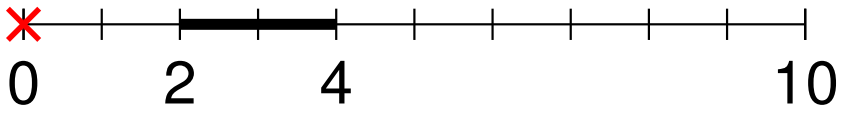
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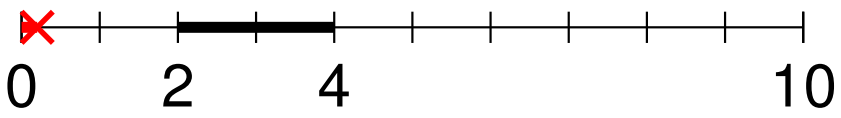
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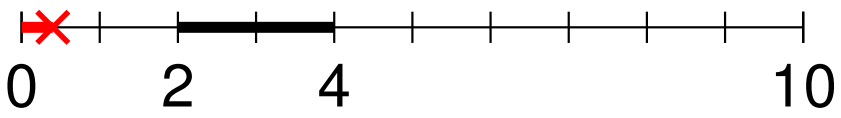
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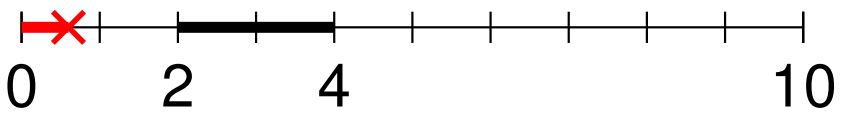
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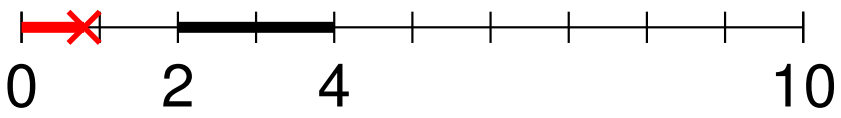
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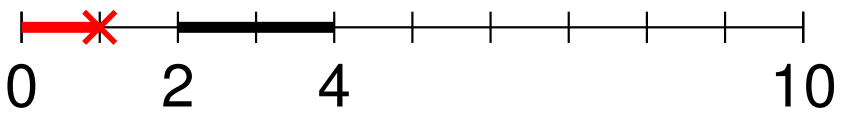
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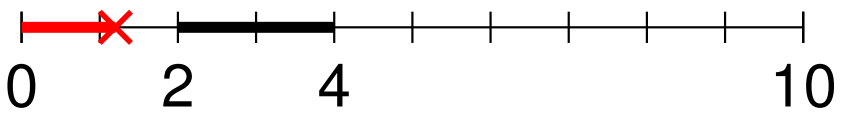
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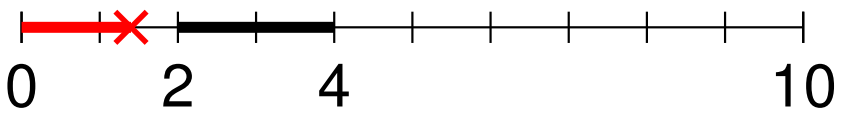
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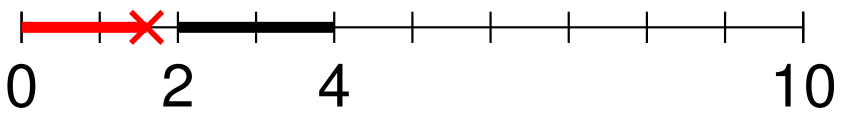
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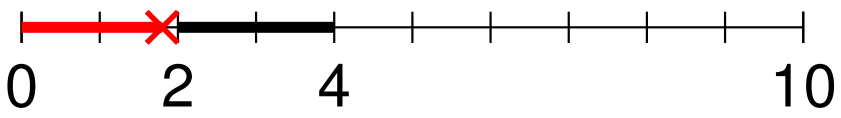
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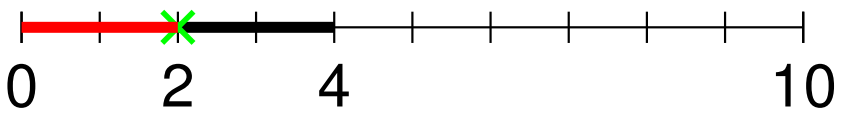
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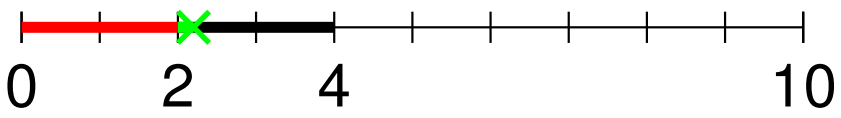
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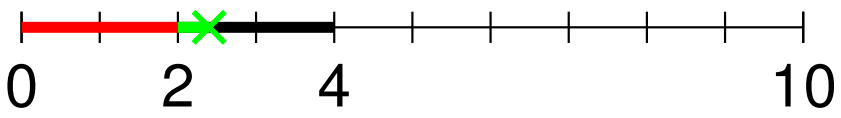
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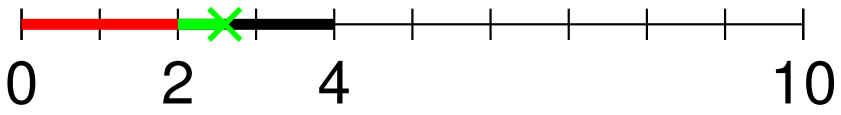
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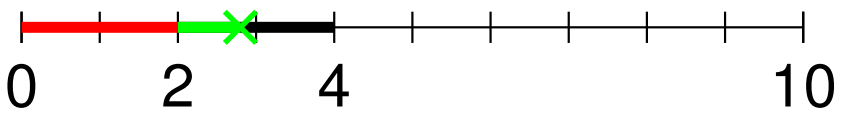
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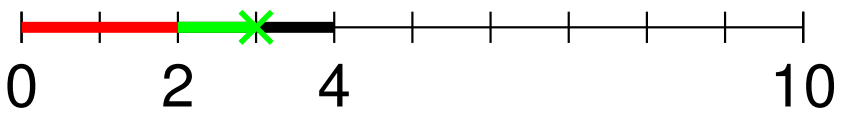
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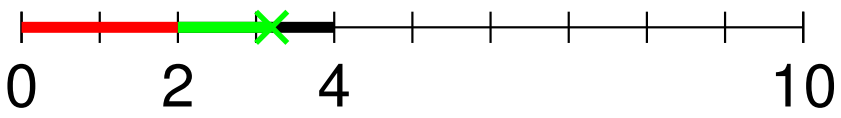
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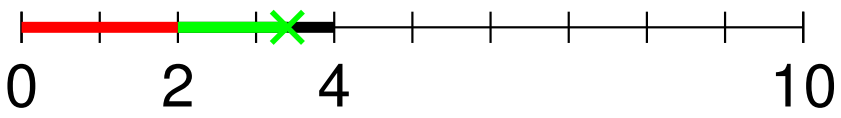
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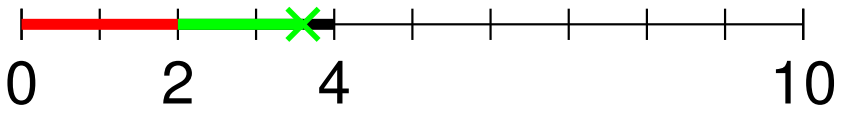
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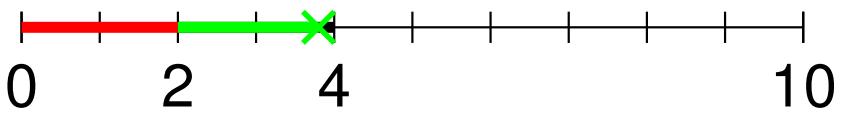
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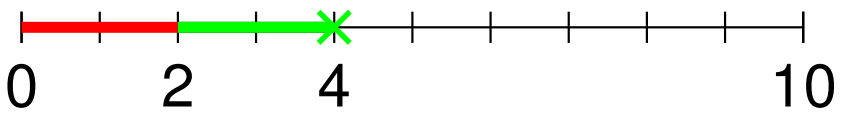
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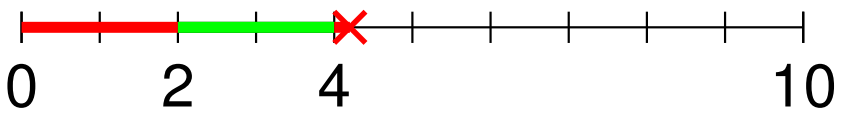
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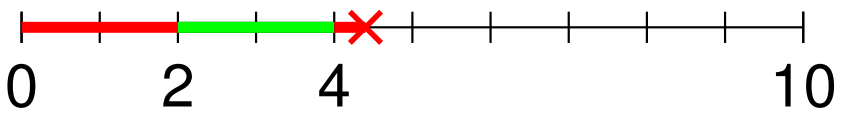
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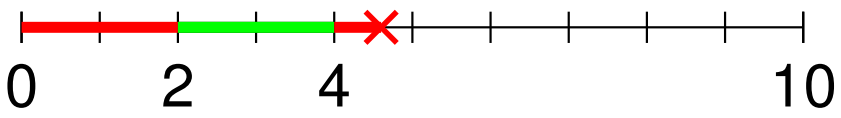
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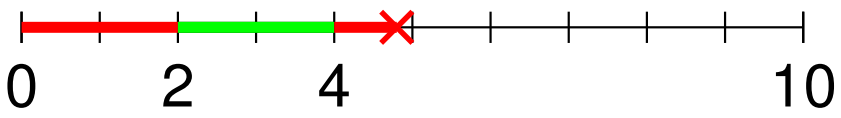
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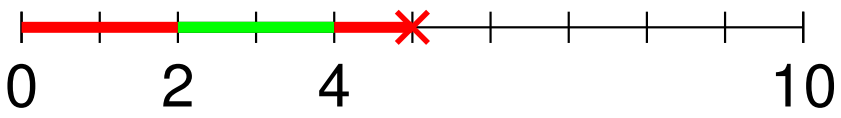
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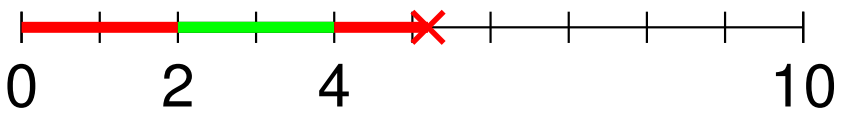
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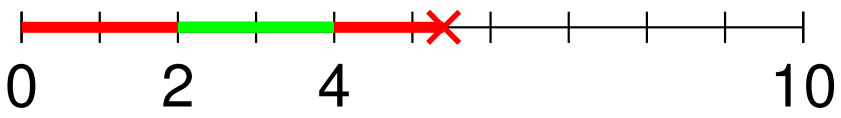
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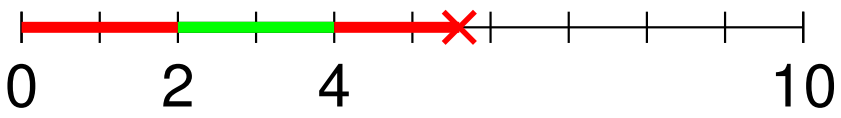
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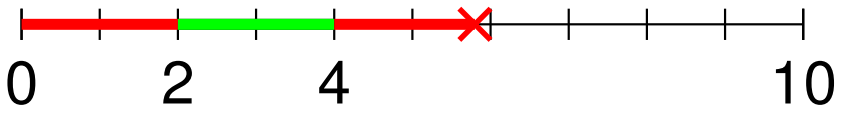
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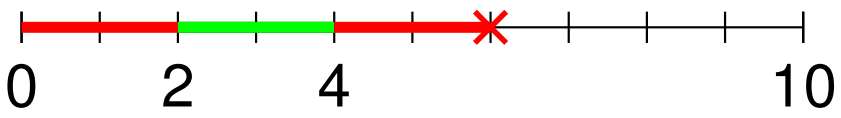
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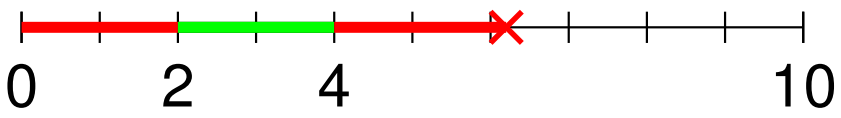
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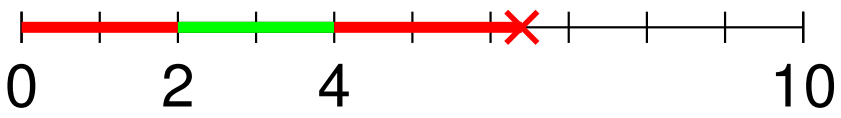
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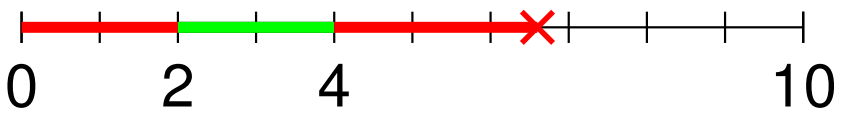
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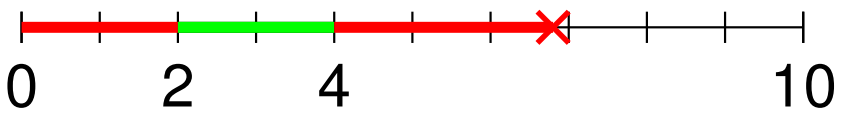
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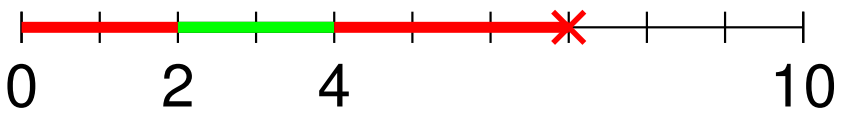
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$\mathbb{P}\{\longmapsto\longmapsto\} = ?$

Uniform distribution

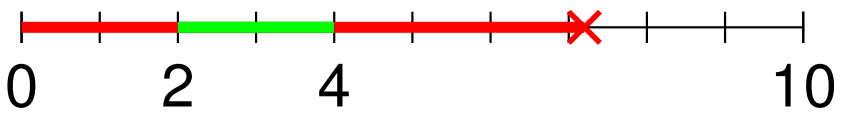
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\} = ?$

Uniform distribution

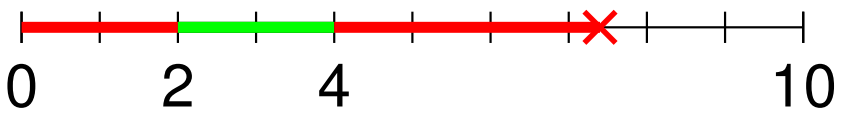
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\longmapsto\} = ?$

Uniform distribution

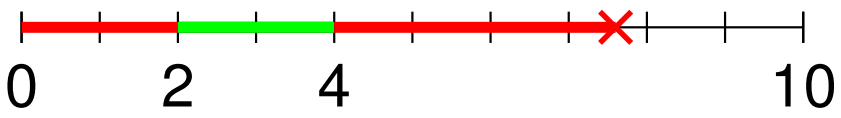
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\longmapsto\} = ?$

Uniform distribution

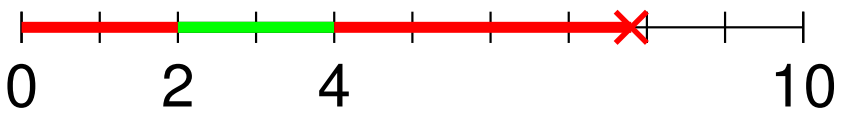
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\} = ?$

Uniform distribution

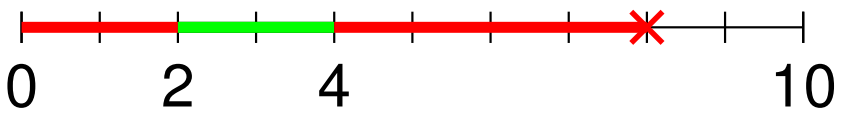
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\longmapsto\} = ?$

Uniform distribution

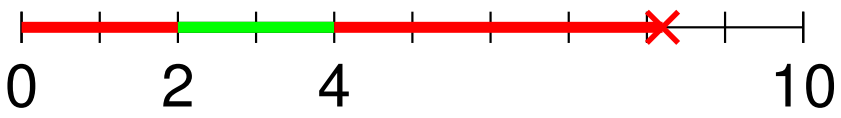
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\longmapsto\} = ?$

Uniform distribution

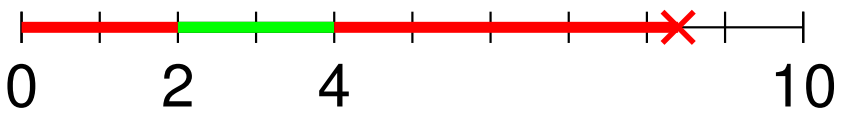
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



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Uniform distribution

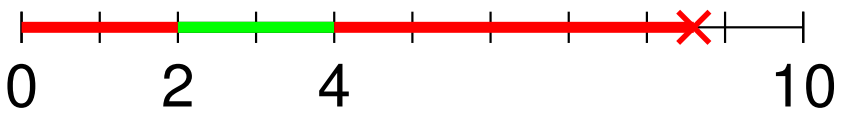
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\!\!\!\!\!\!\longmapsto\} = ?$

Uniform distribution

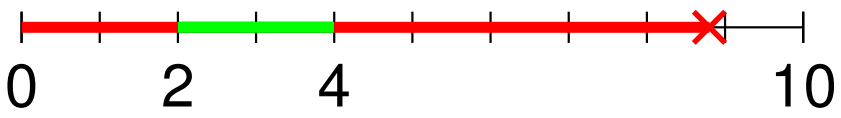
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\!\!\!\!\!\!\longmapsto\} = ?$

Uniform distribution

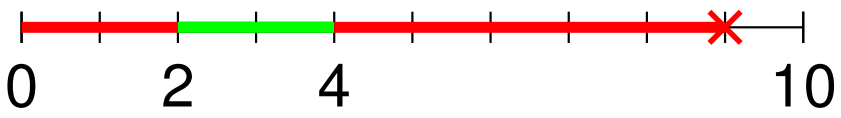
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\longmapsto\} = ?$

Uniform distribution

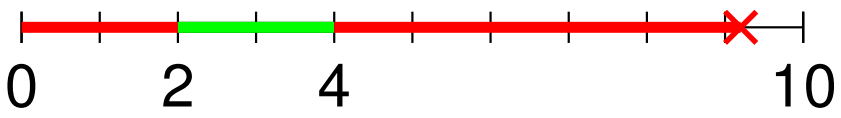
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\longmapsto\} = ?$

Uniform distribution

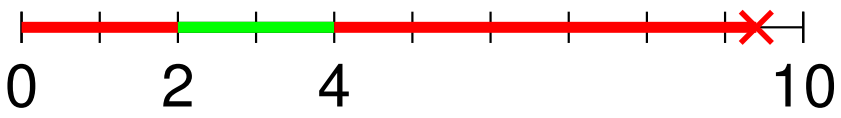
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\!\!\!\!\!\!\longmapsto\} = ?$

Uniform distribution

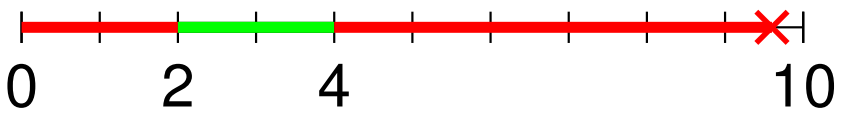
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longleftrightarrow\} = ?$

Uniform distribution

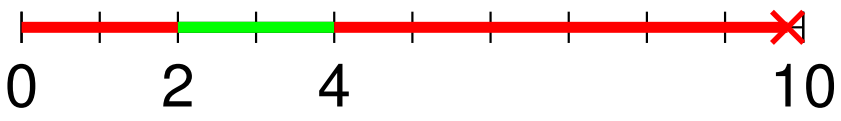
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\!\!\!

Uniform distribution

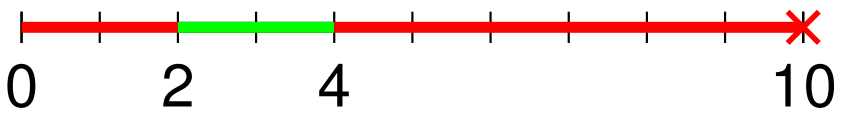
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longmapsto\!\!\!\!\!\!\longmapsto\} = ?$

Uniform distribution

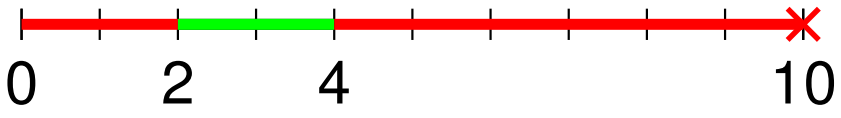
Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.



$\mathbb{P}\{\longleftrightarrow\} = ?$

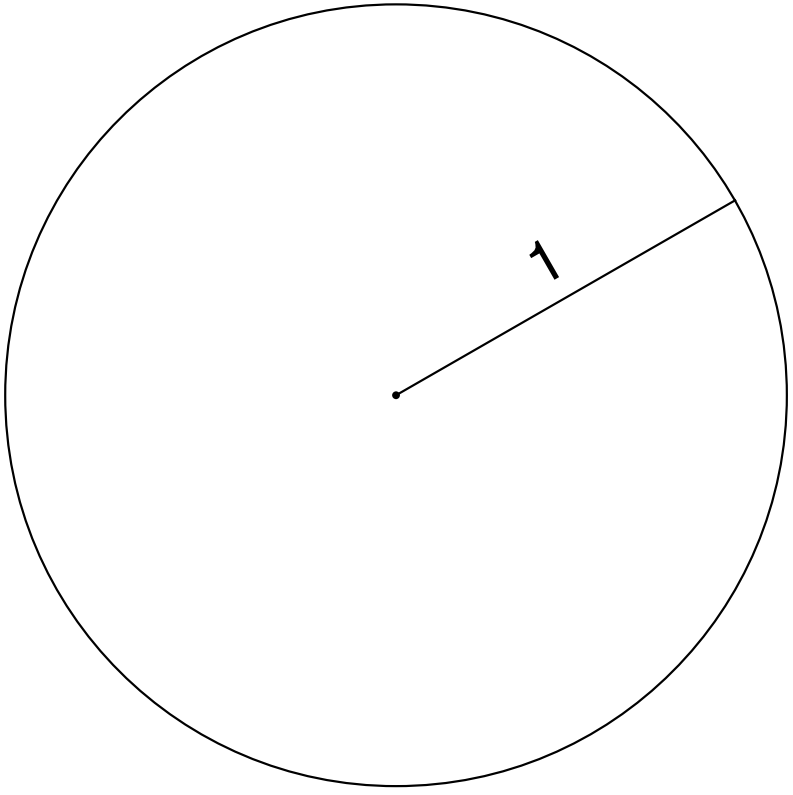
Uniform distribution

Pick a random point that's *equally likely to be anywhere* on $[0, 10]$.

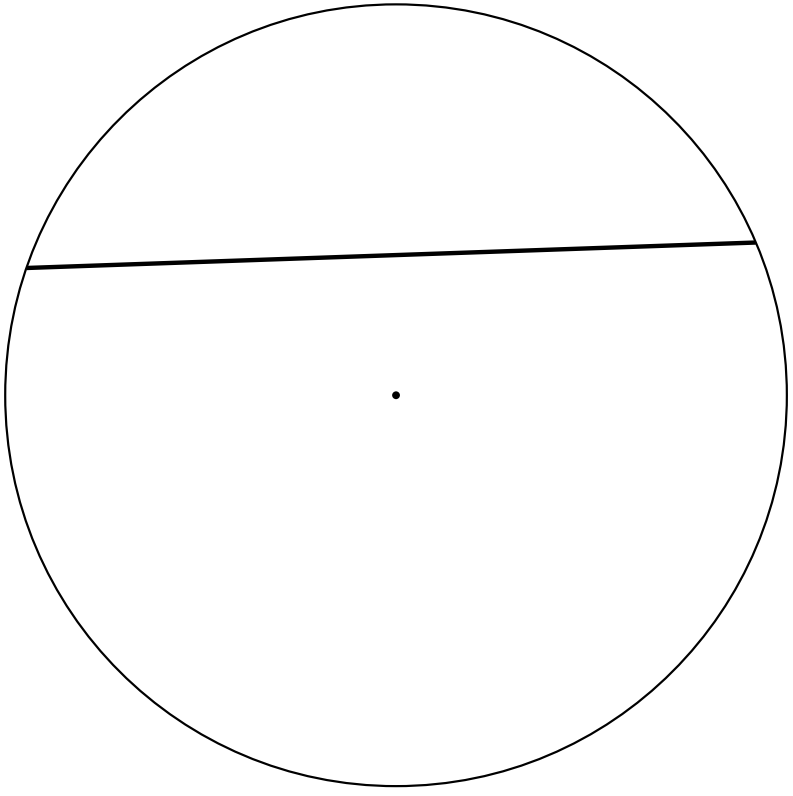


$$\mathbb{P}\{\text{—}\} = \frac{\text{—}}{\text{—}} = \frac{2}{10} = \frac{1}{5}.$$

Random chords and a question

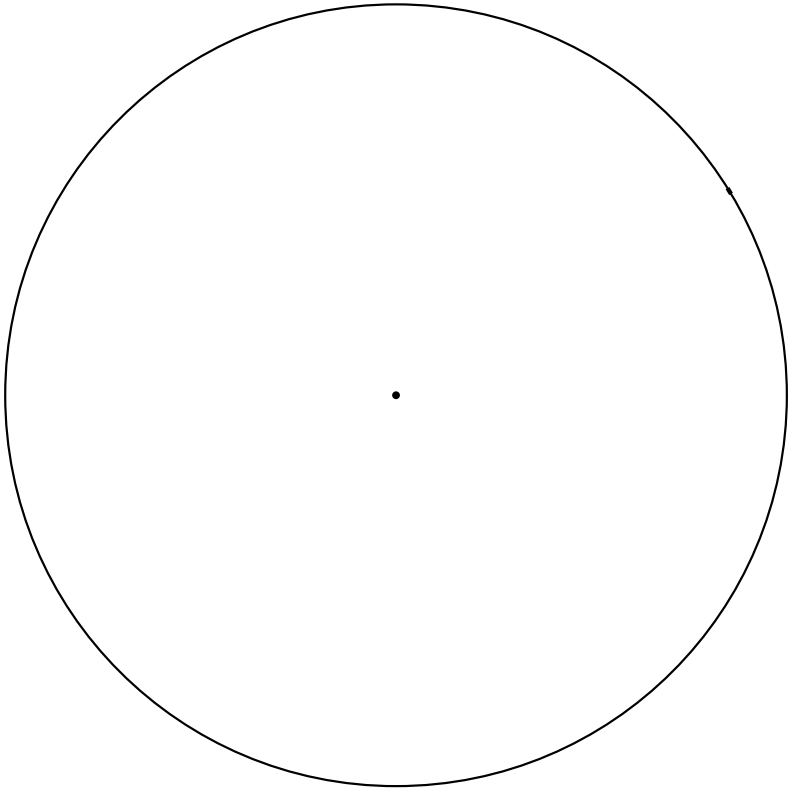


Random chords and a question



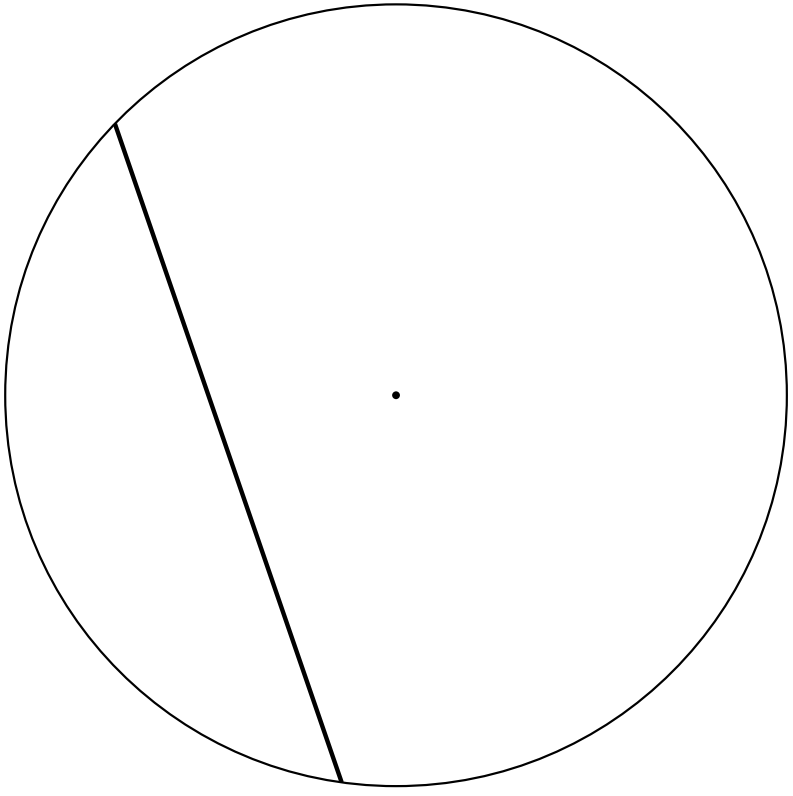
random chord

Random chords and a question



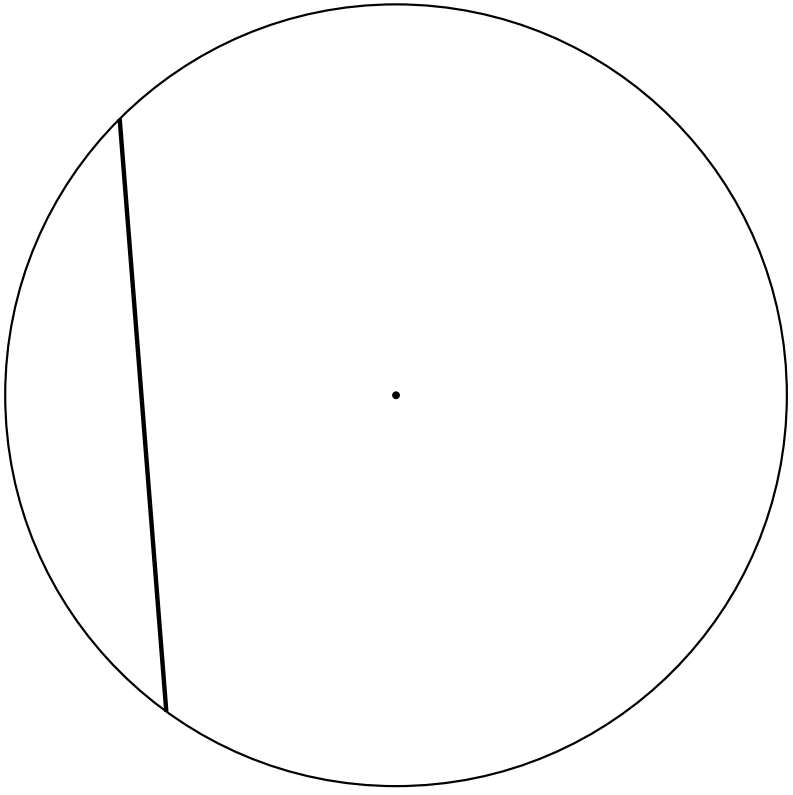
random chord

Random chords and a question



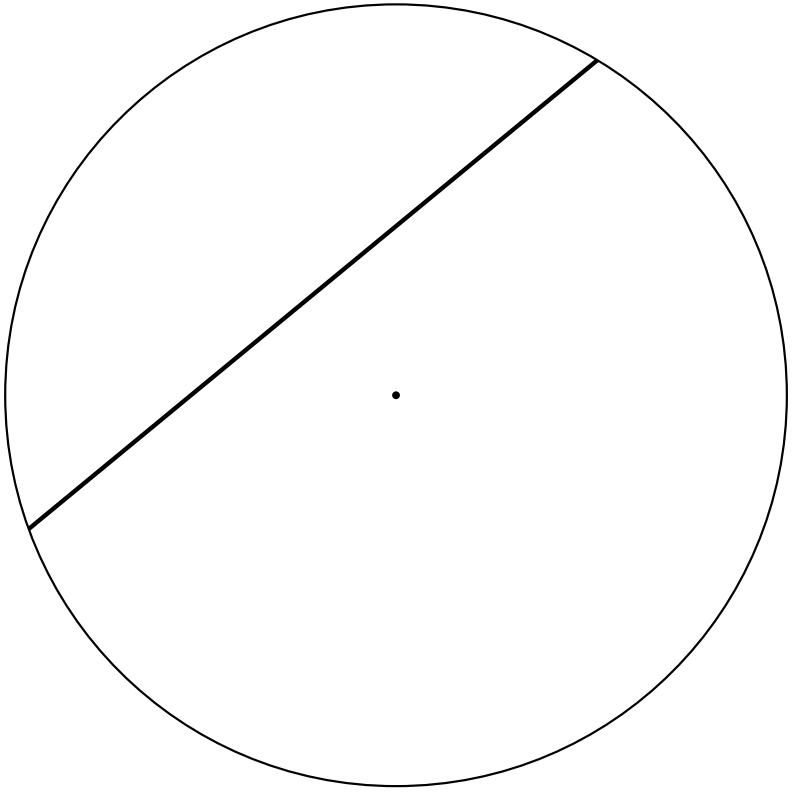
random chord

Random chords and a question



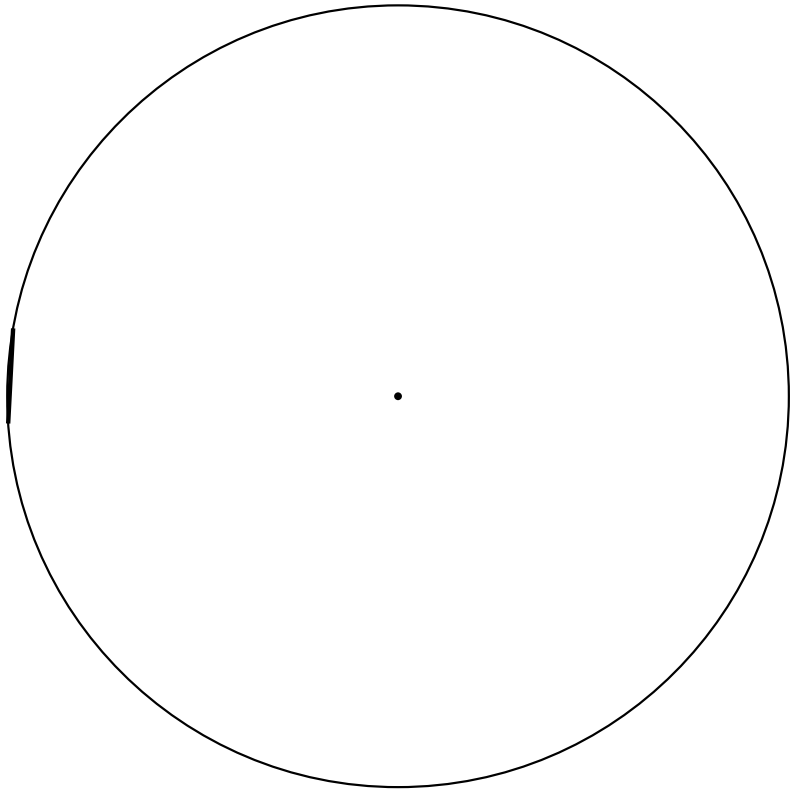
random chord

Random chords and a question



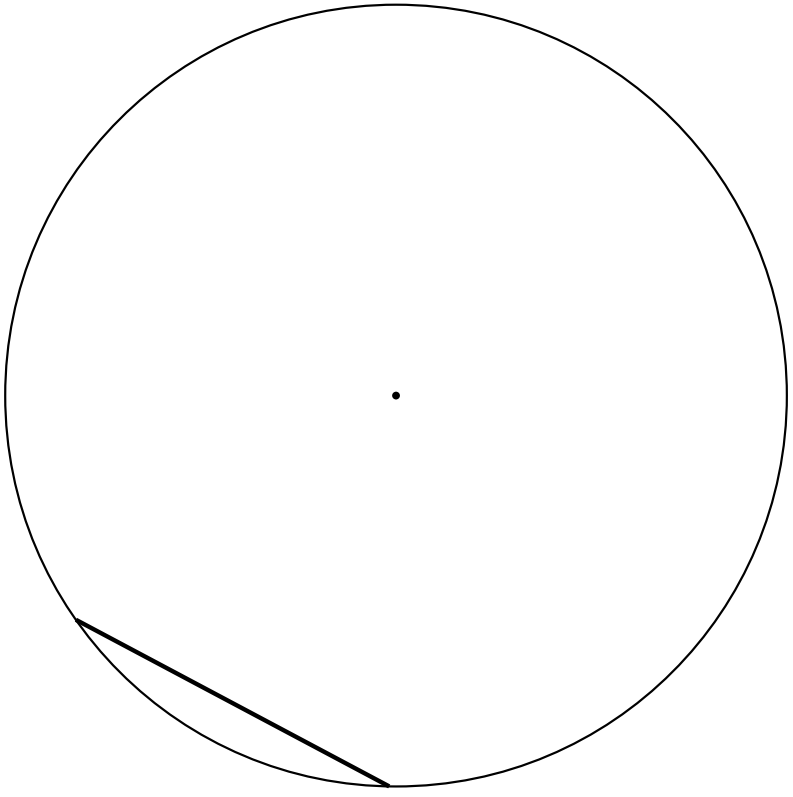
random chord

Random chords and a question



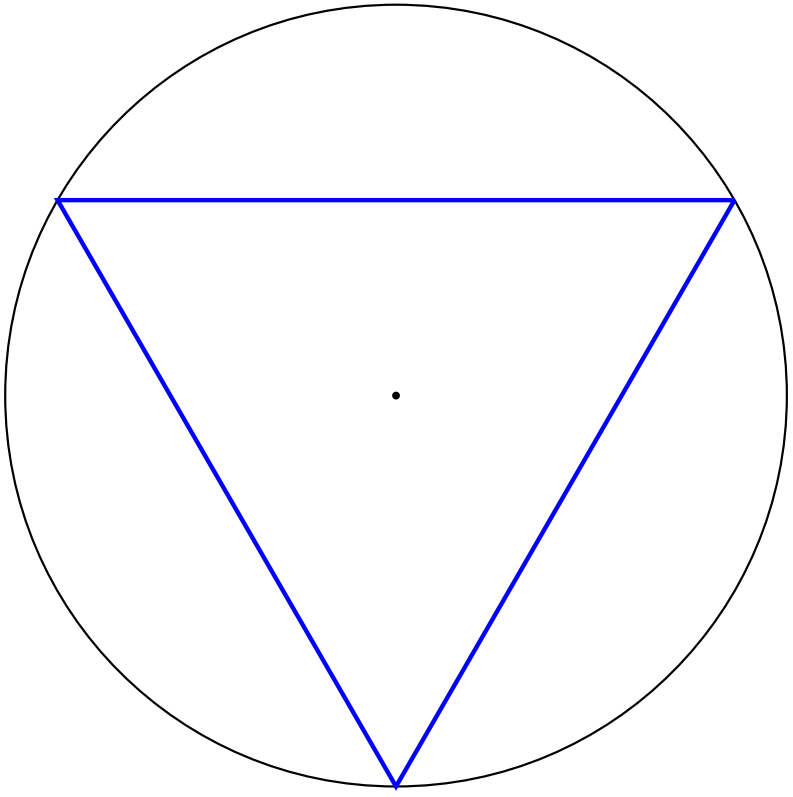
random chord

Random chords and a question



random chord

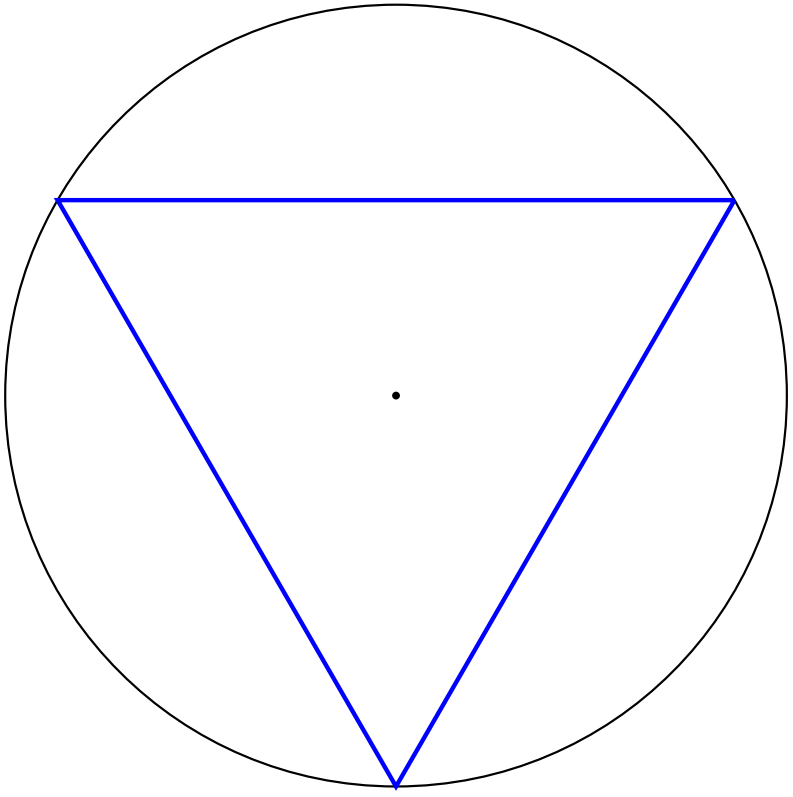
Random chords and a question



random chord

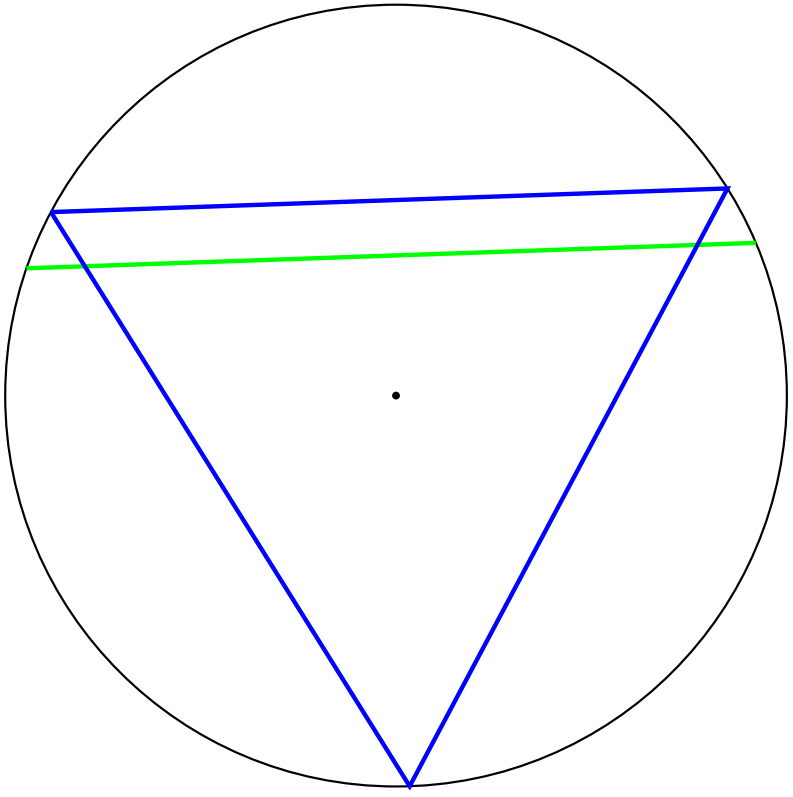


Random chords and a question



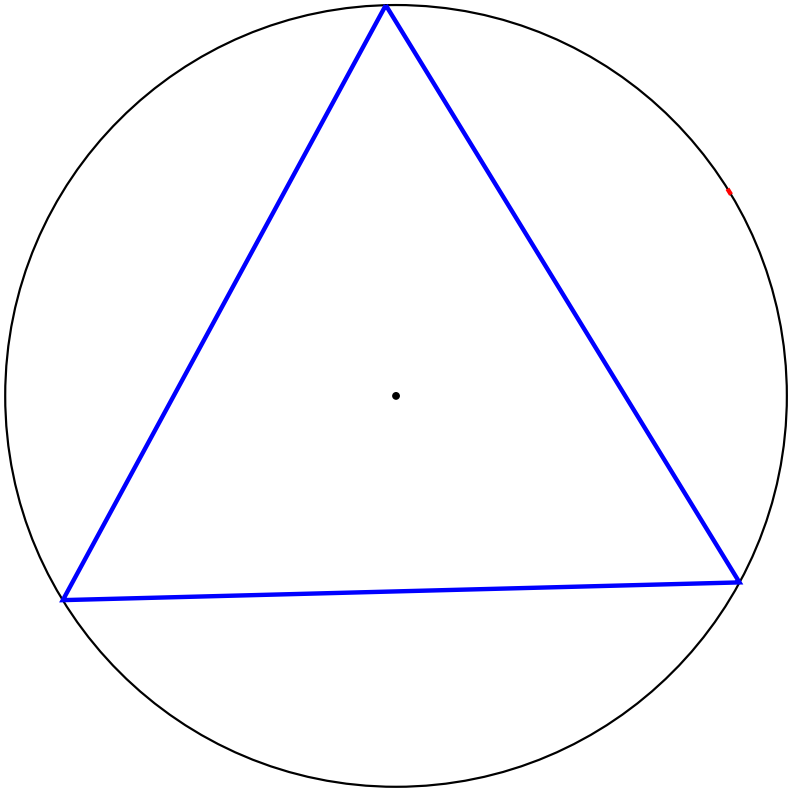
$\mathbb{P}\{$*random chord* is longer than one side of the $\bigtriangledown\}$?

Random chords and a question



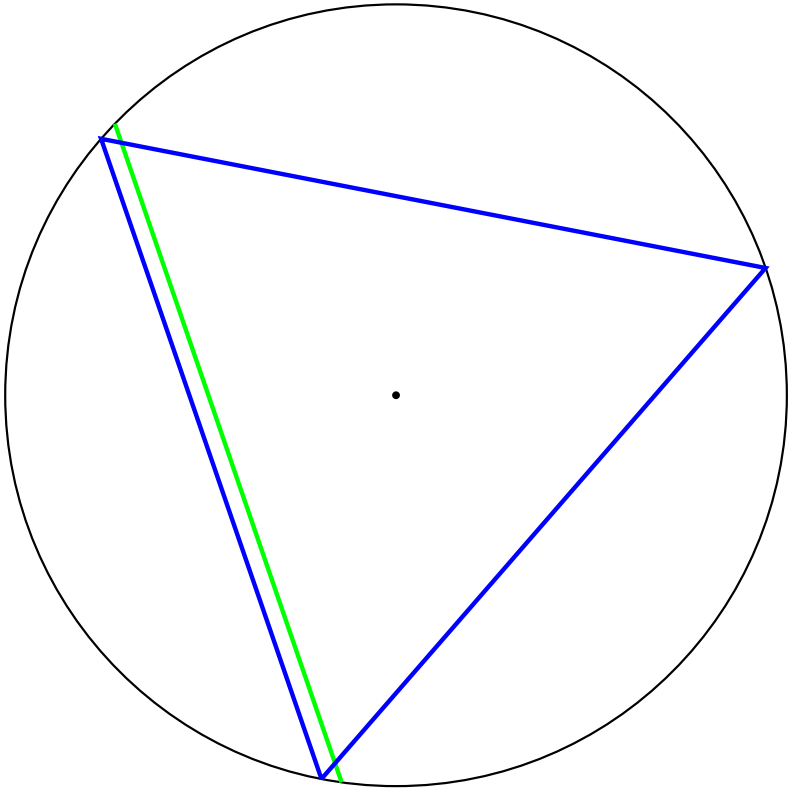
$\mathbb{P}\{$*random chord* is longer than one side of the $\bigtriangledown\}$?

Random chords and a question



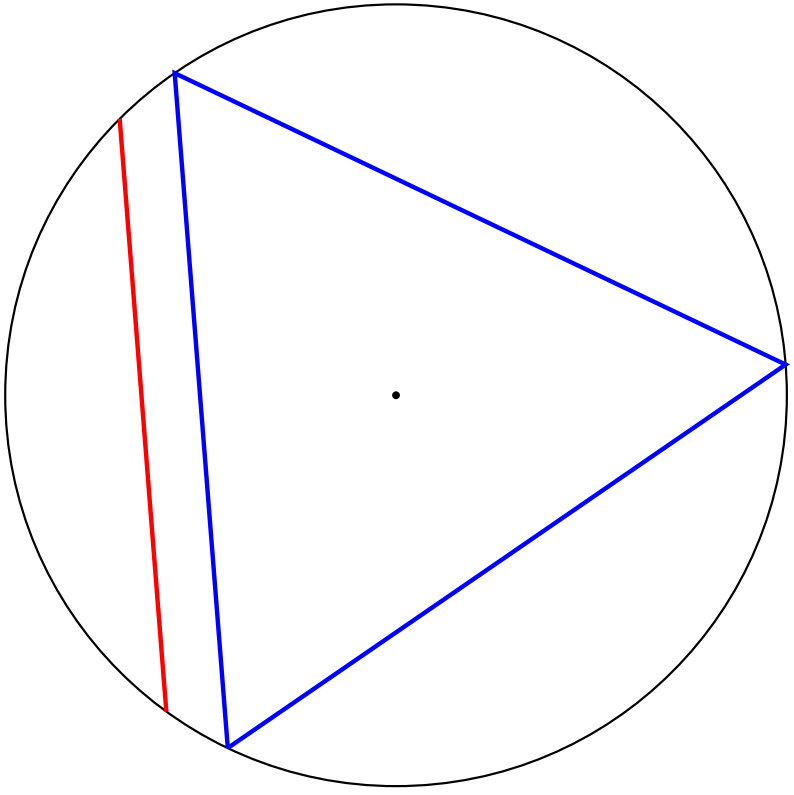
$\mathbb{P}\{$*random chord* is longer than one side of the $\bigtriangledown\}$?

Random chords and a question



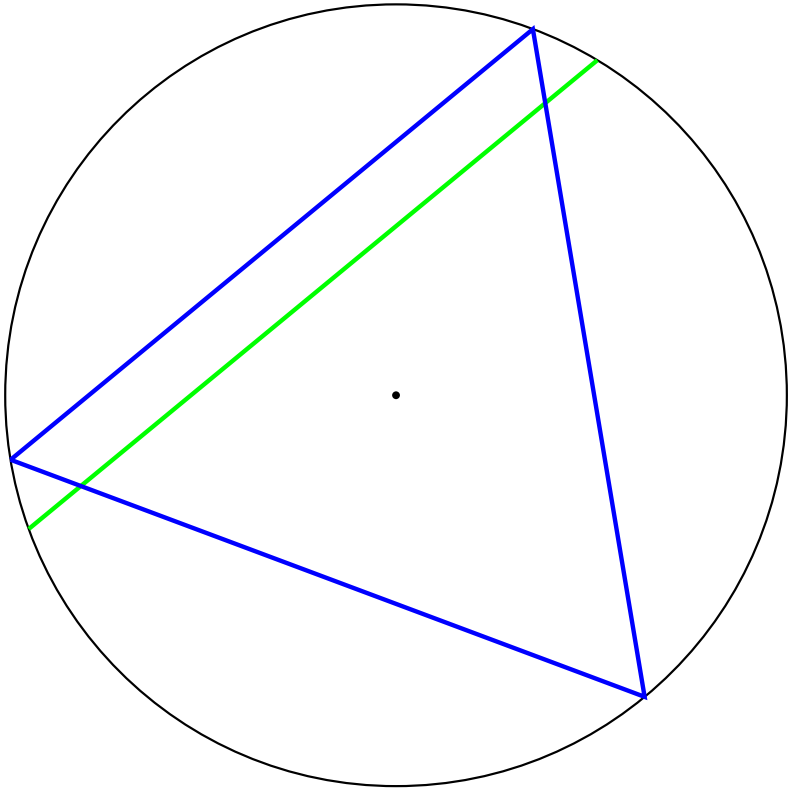
$\mathbb{P}\{$*random chord* is longer than one side of the $\bigtriangledown\}$?

Random chords and a question



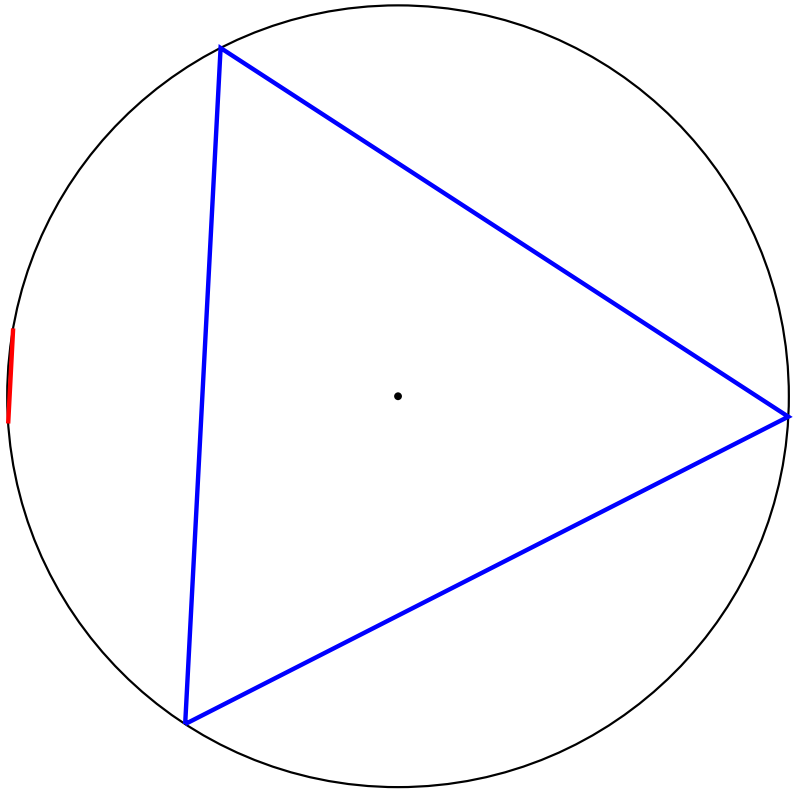
$\mathbb{P}\{$*random chord* is longer than one side of the $\bigtriangledown\}$?

Random chords and a question



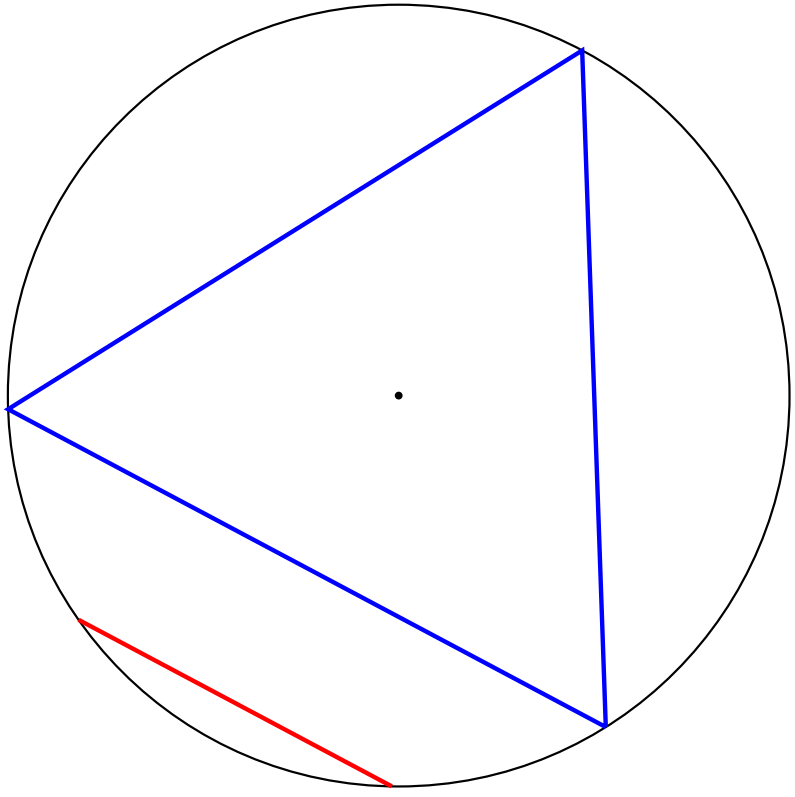
$\mathbb{P}\{$*random chord* is longer than one side of the $\bigtriangledown\}$?

Random chords and a question



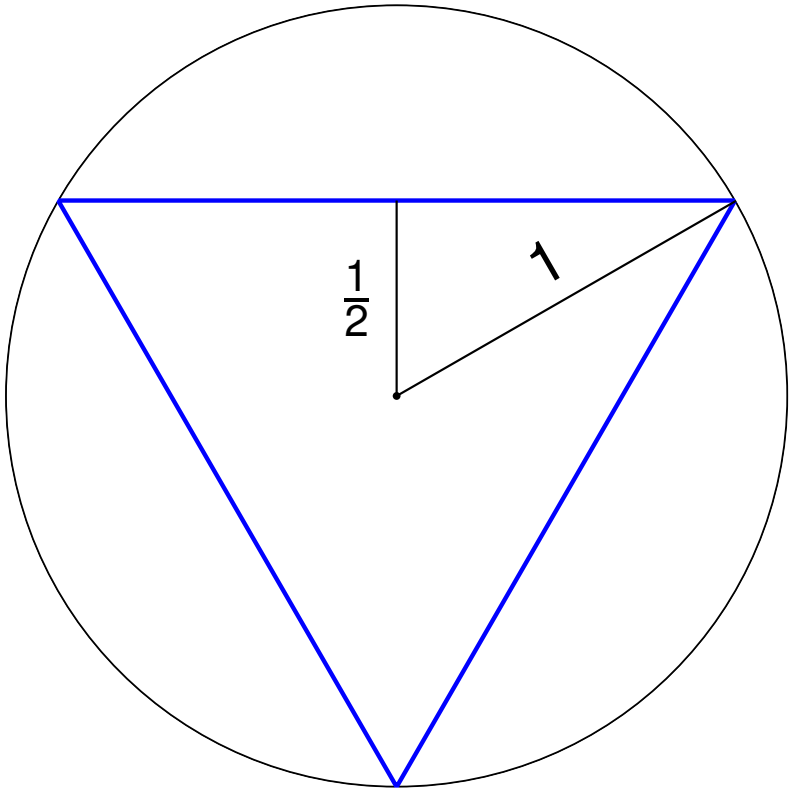
$\mathbb{P}\{$*random chord* is longer than one side of the $\bigtriangledown\}$?

Random chords and a question

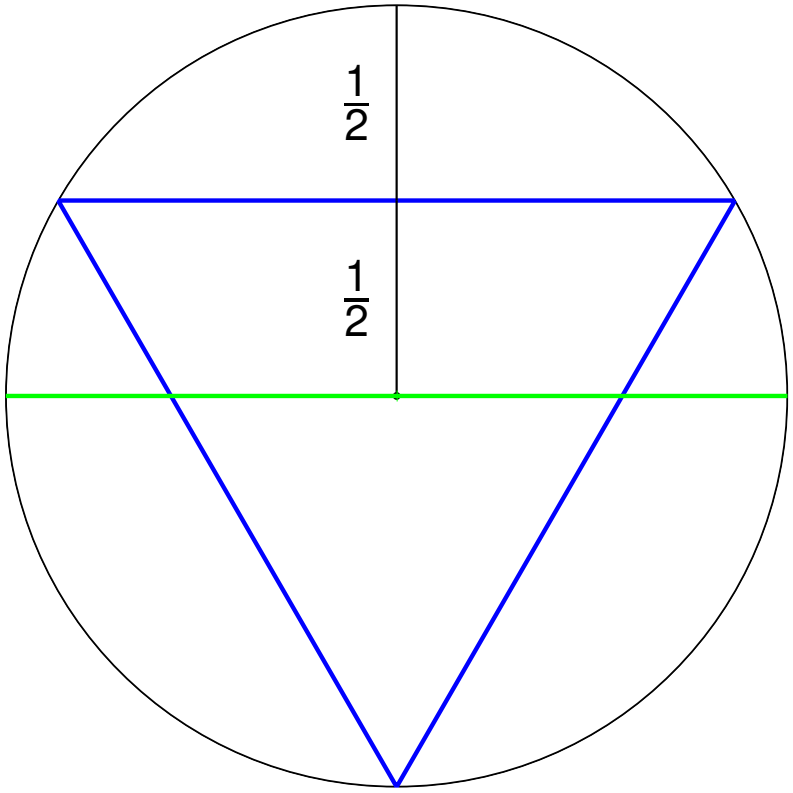


$\mathbb{P}\{$*random chord* is longer than one side of the $\bigtriangledown\}$?

Answer: $\frac{1}{2}$

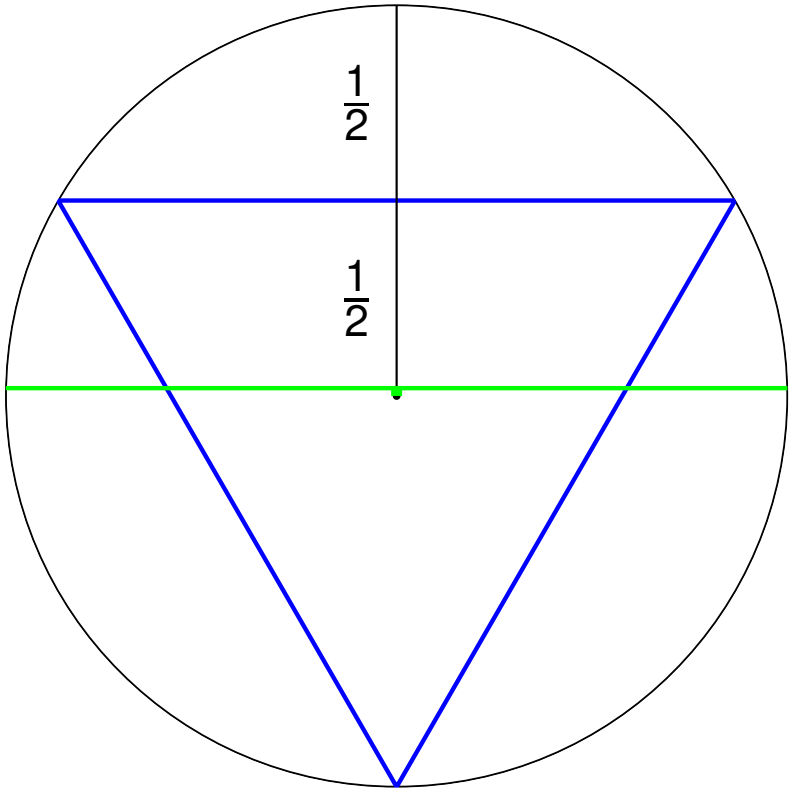


Answer: $\frac{1}{2}$



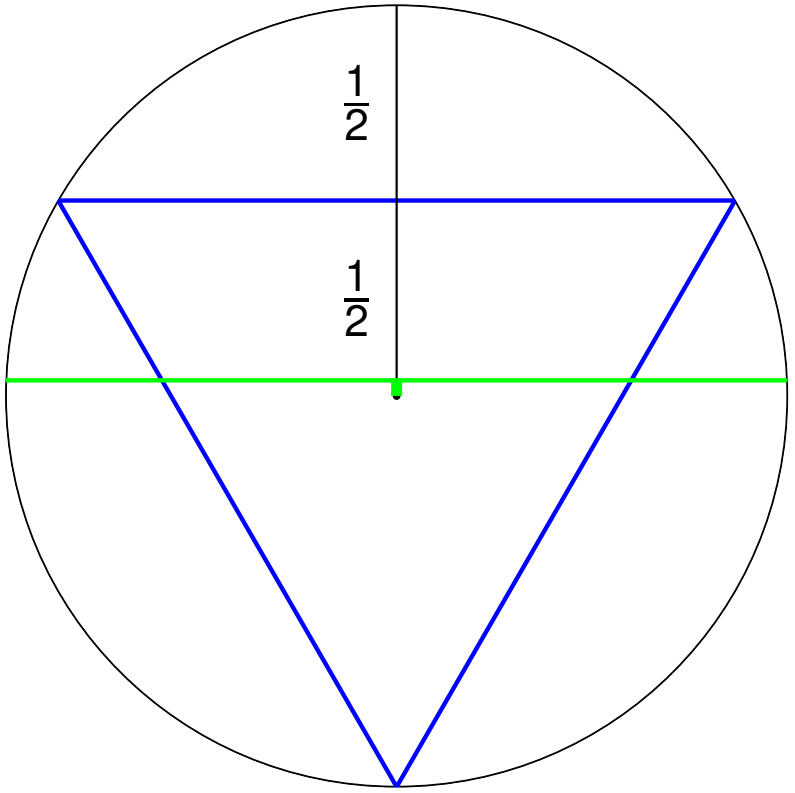
Distance from center:

Answer: $\frac{1}{2}$



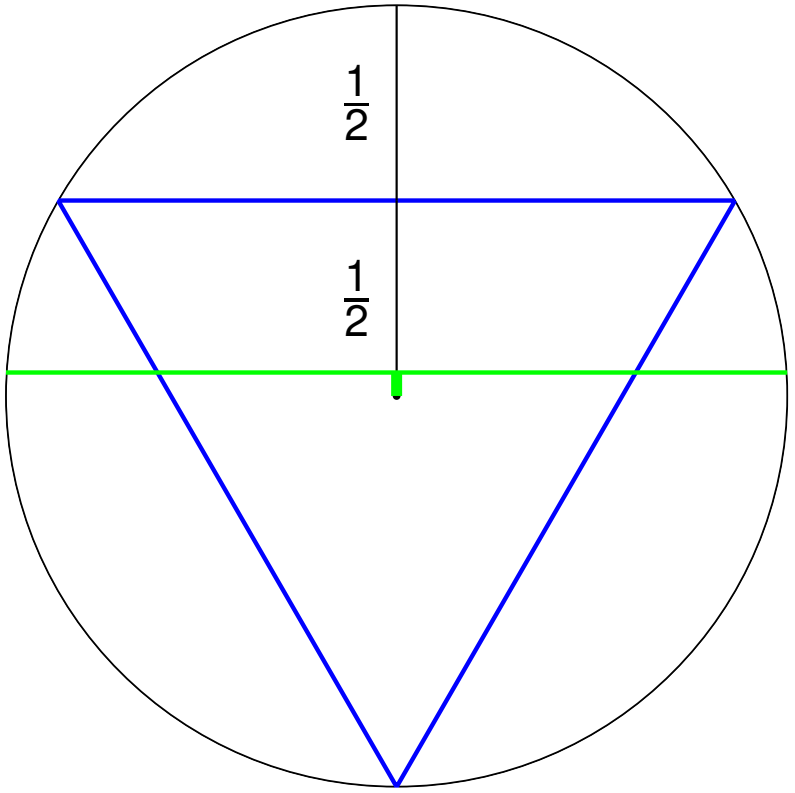
Distance from center:

Answer: $\frac{1}{2}$



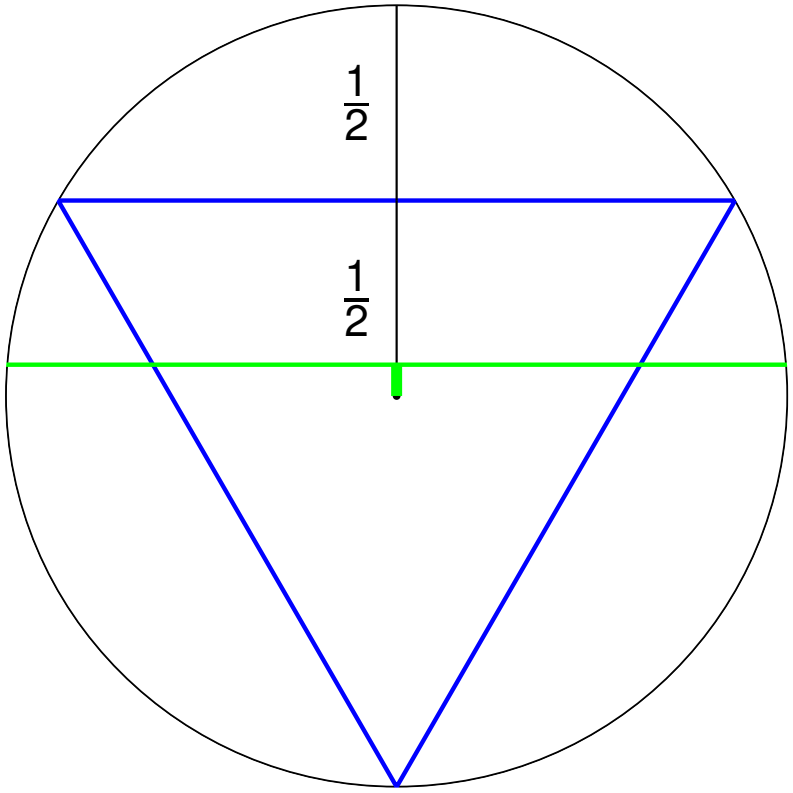
Distance from center:

Answer: $\frac{1}{2}$



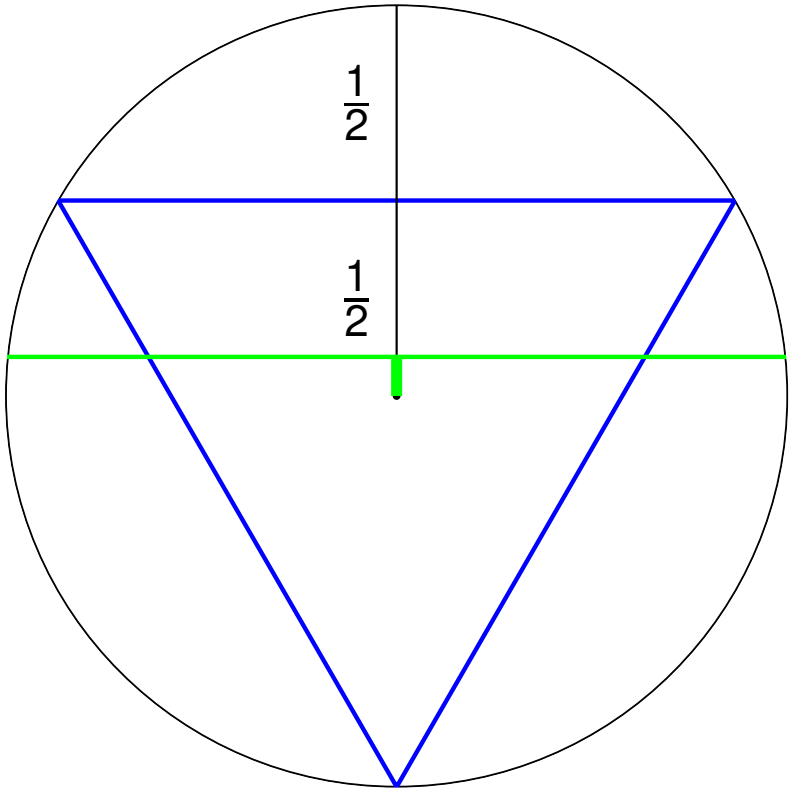
Distance from center:

Answer: $\frac{1}{2}$



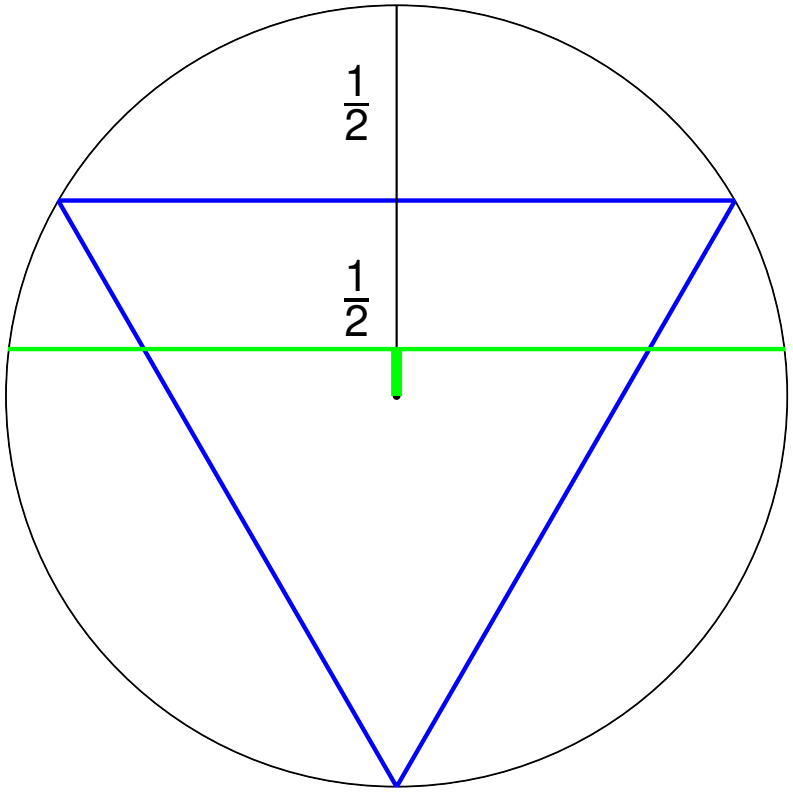
Distance from center:

Answer: $\frac{1}{2}$



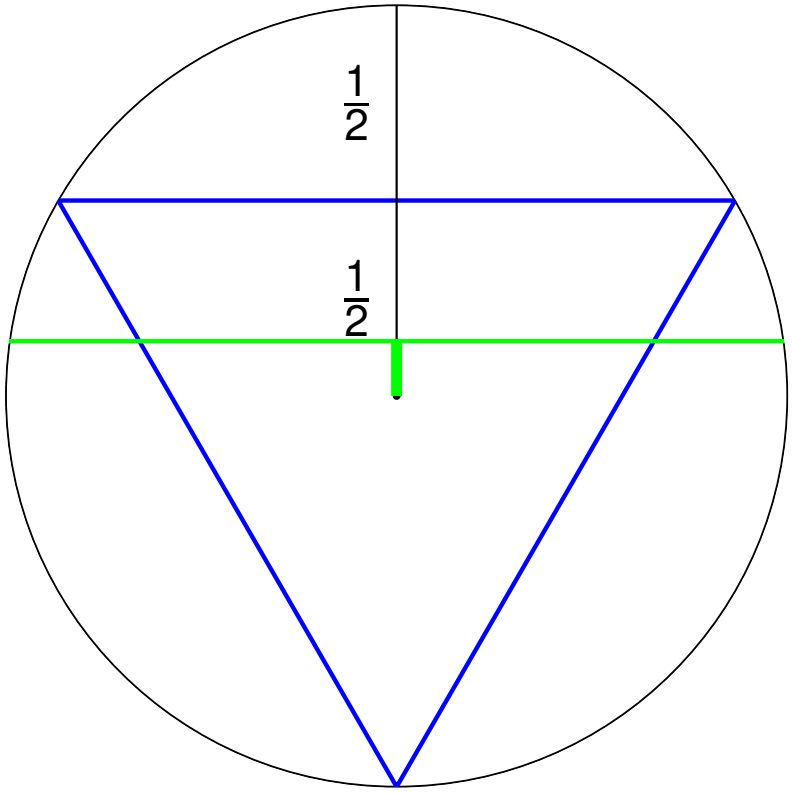
Distance from center:

Answer: $\frac{1}{2}$



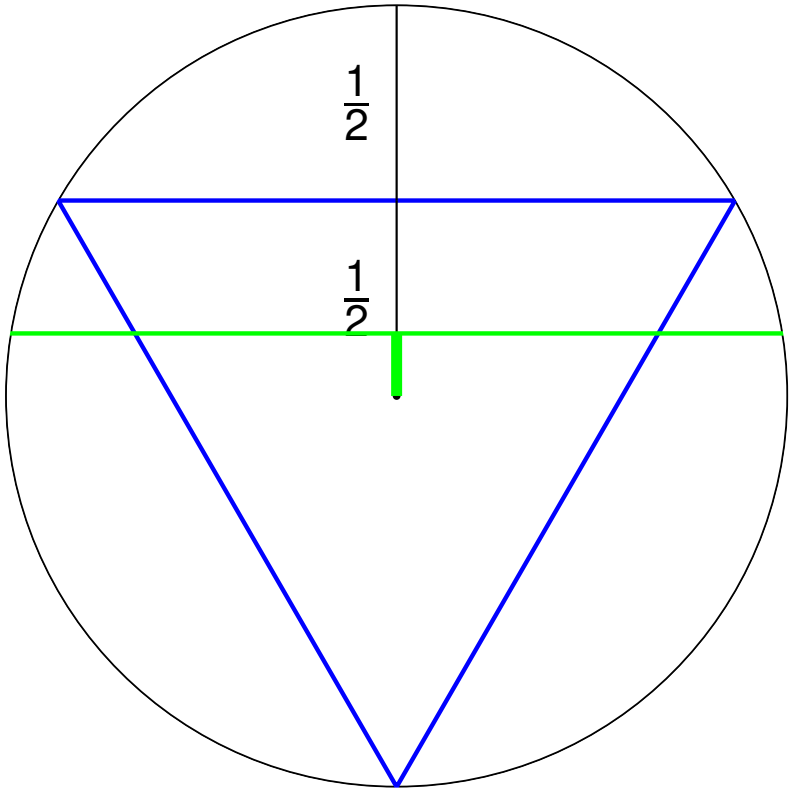
Distance from center:

Answer: $\frac{1}{2}$



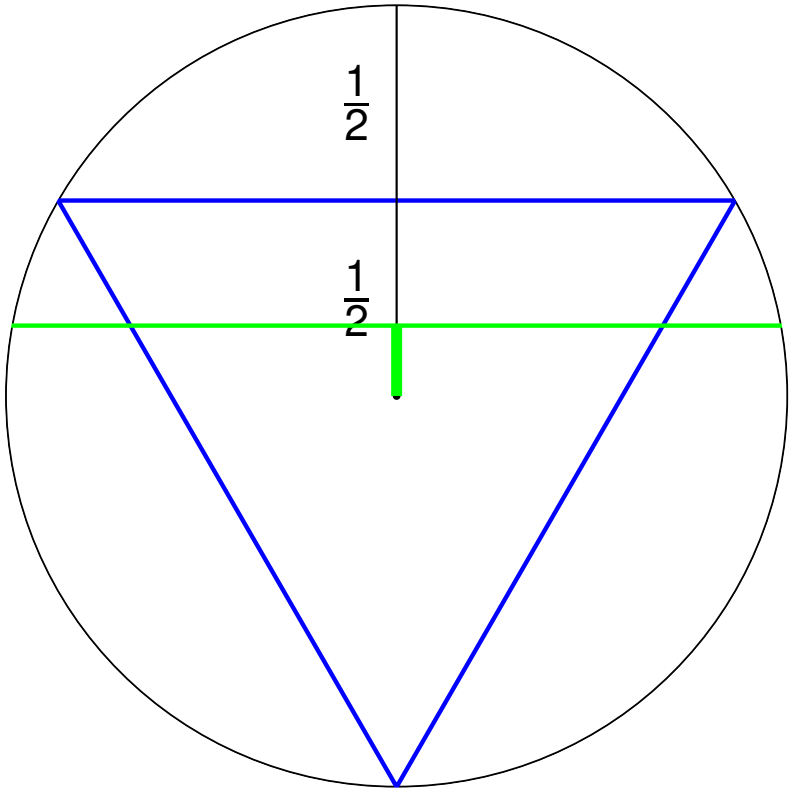
Distance from center:

Answer: $\frac{1}{2}$



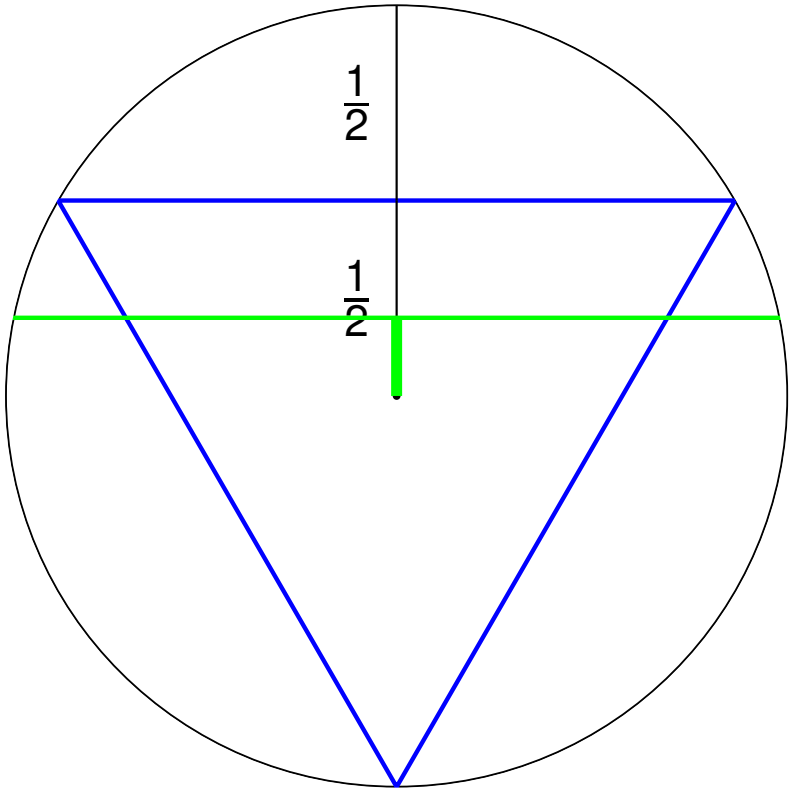
Distance from center:

Answer: $\frac{1}{2}$



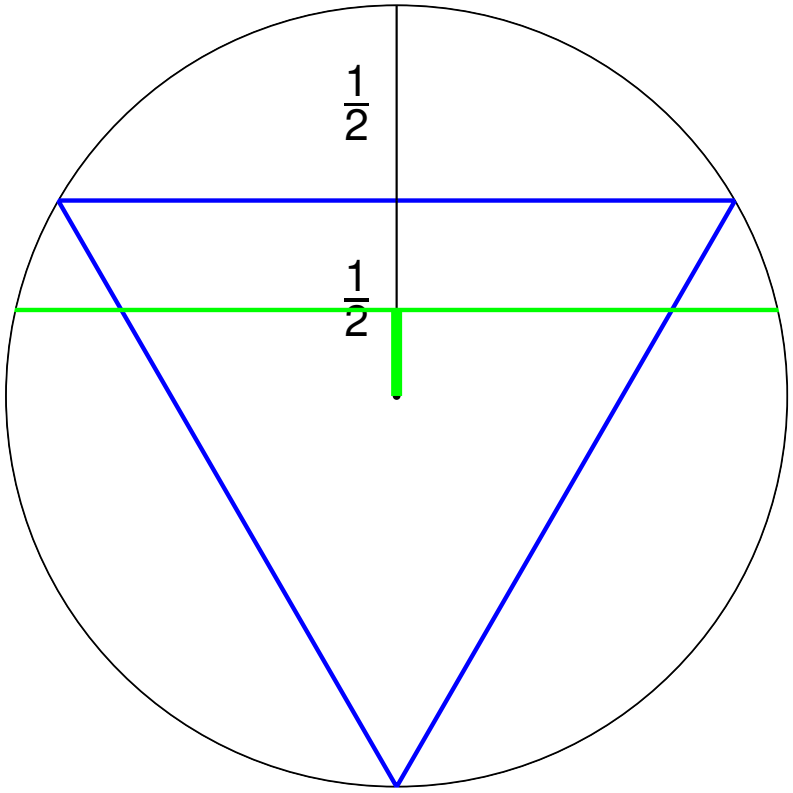
Distance from center:

Answer: $\frac{1}{2}$



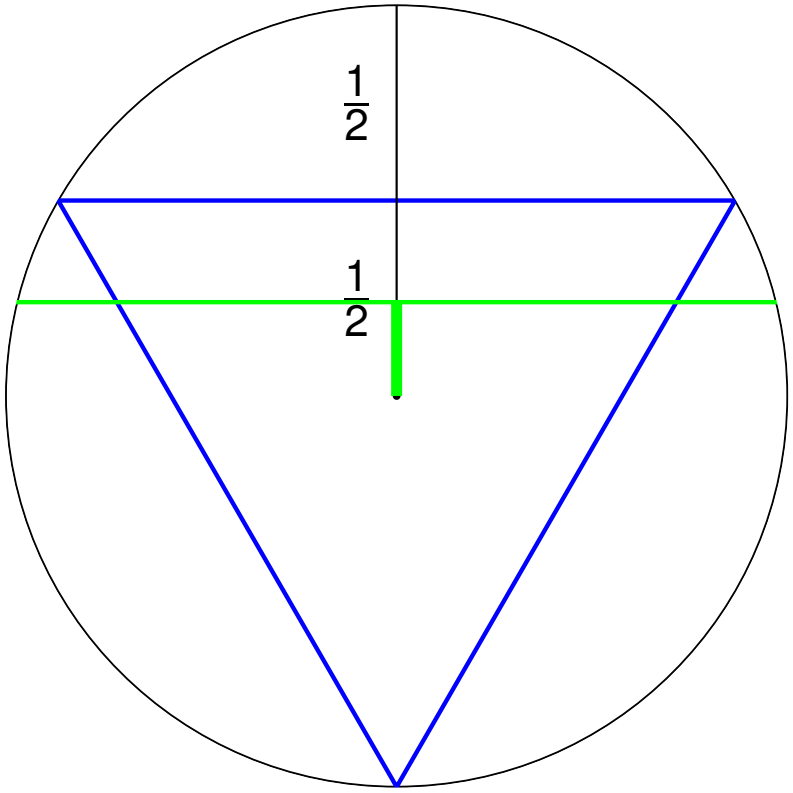
Distance from center:

Answer: $\frac{1}{2}$



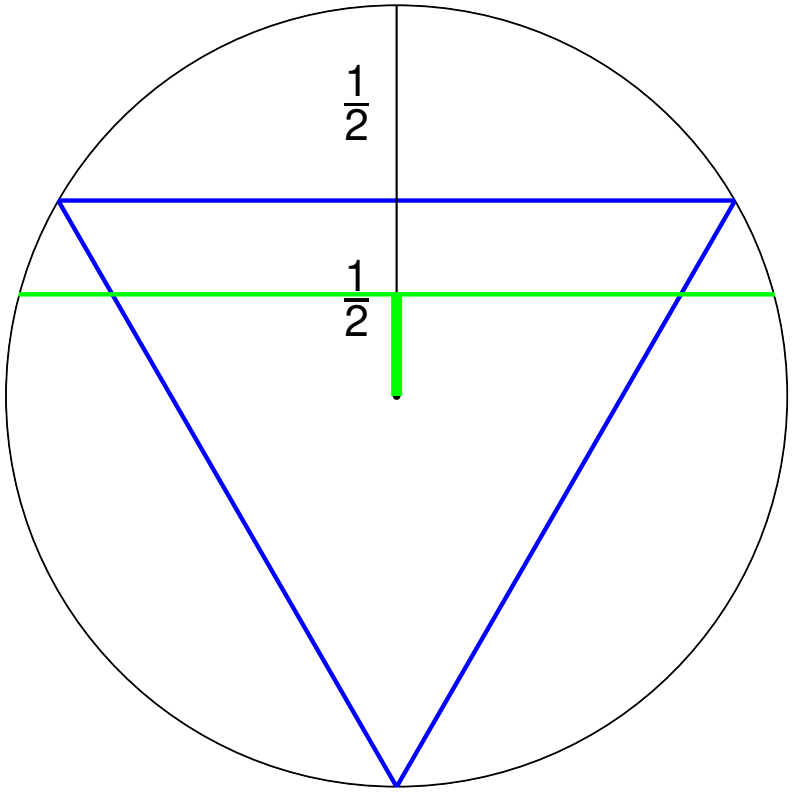
Distance from center:

Answer: $\frac{1}{2}$



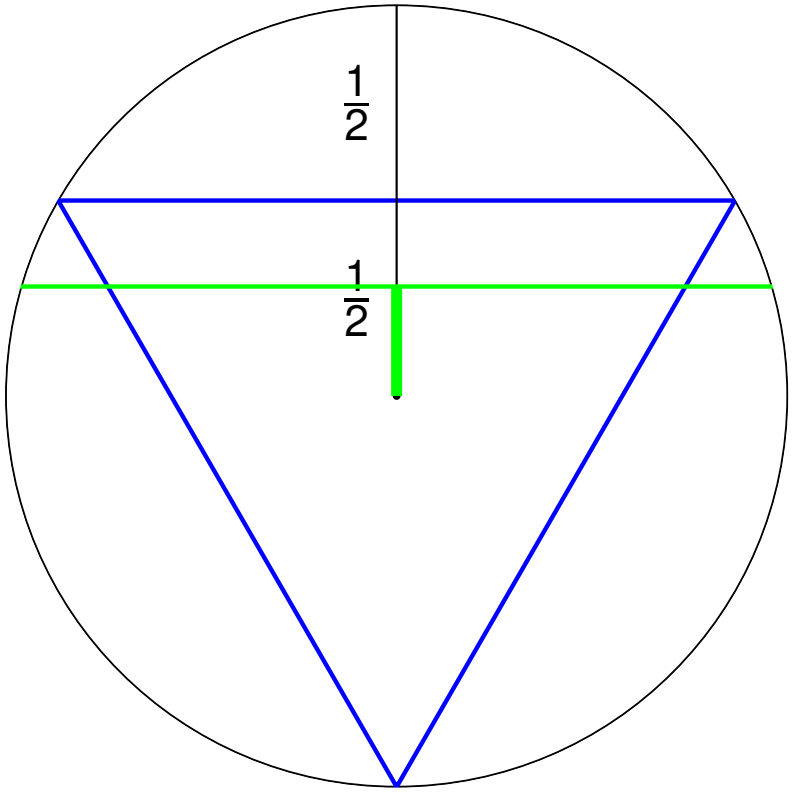
Distance from center:

Answer: $\frac{1}{2}$



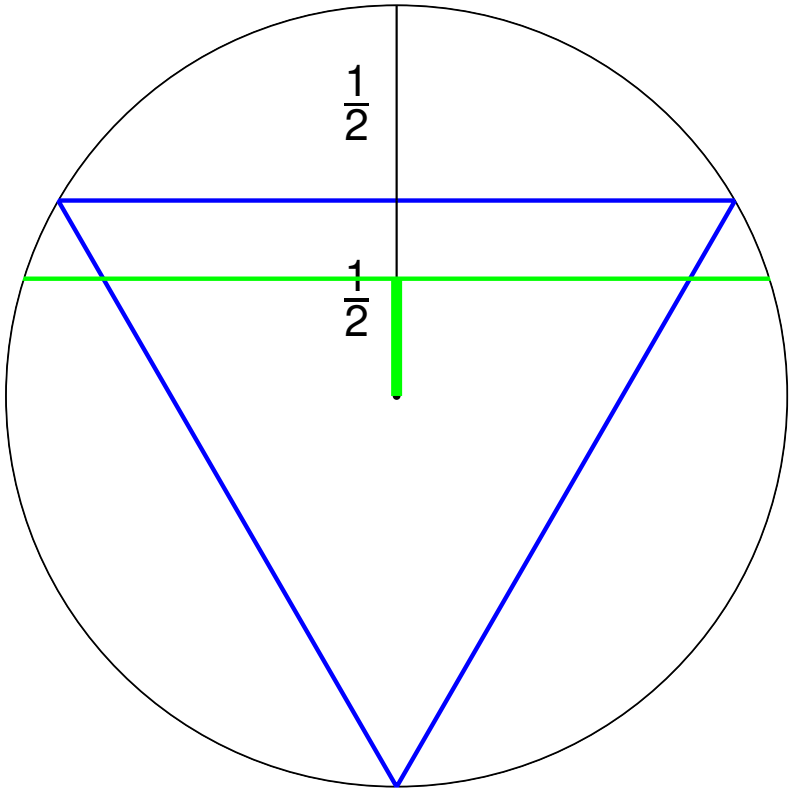
Distance from center:

Answer: $\frac{1}{2}$



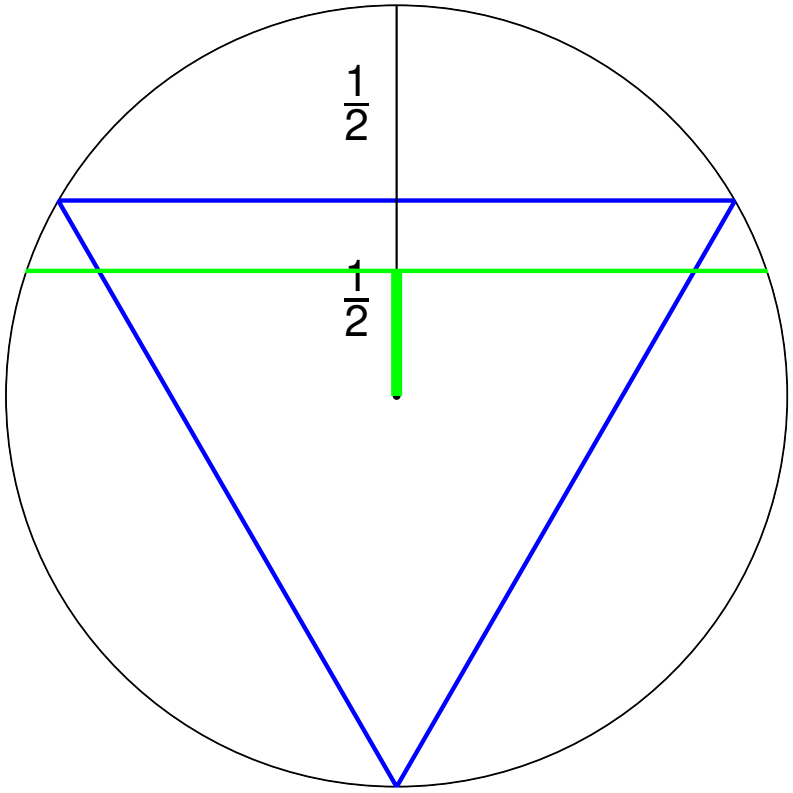
Distance from center:

Answer: $\frac{1}{2}$



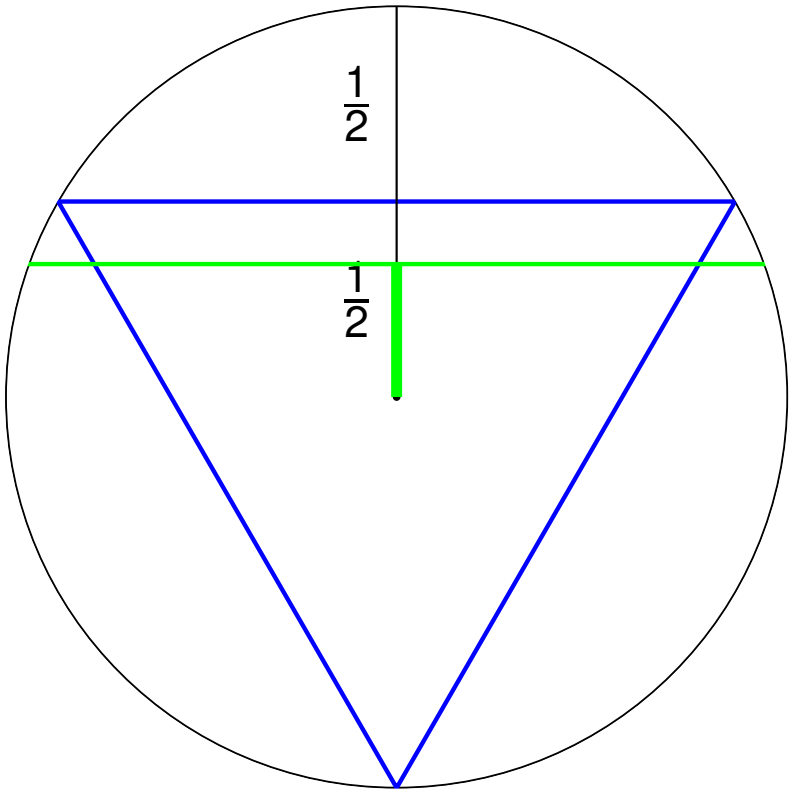
Distance from center:

Answer: $\frac{1}{2}$



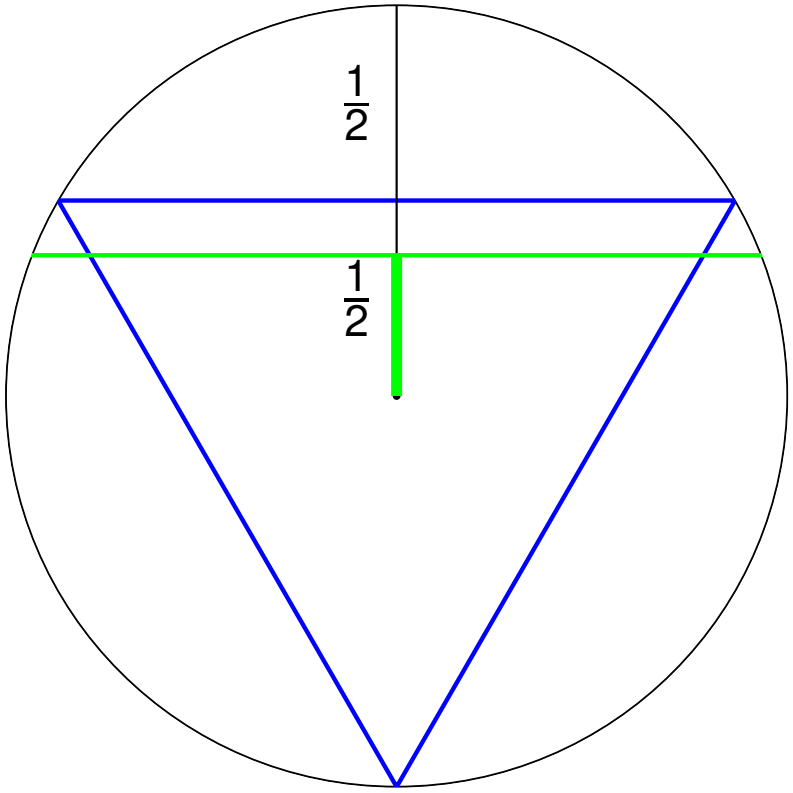
Distance from center:

Answer: $\frac{1}{2}$



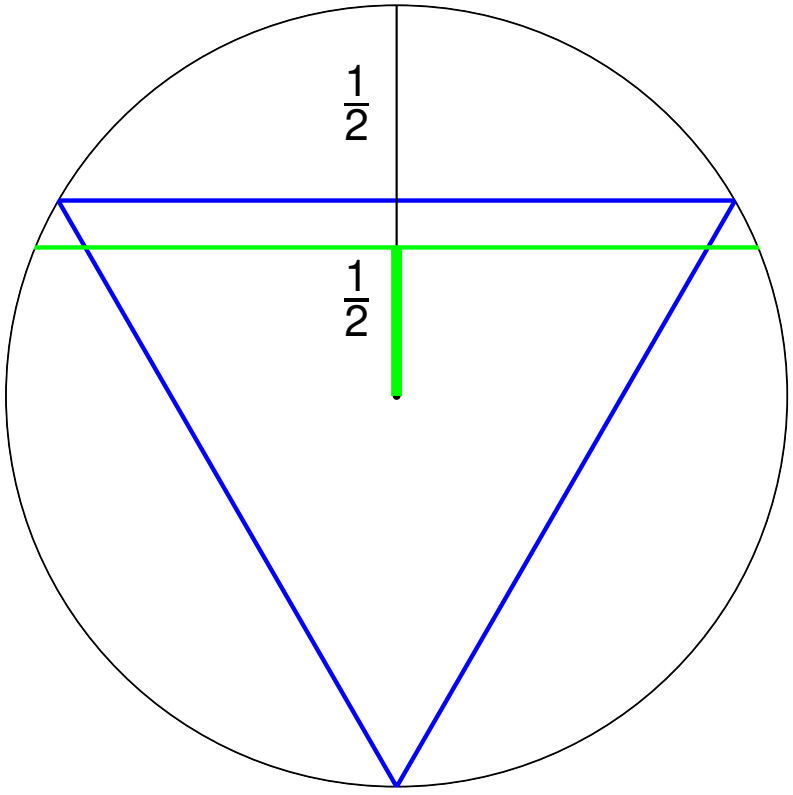
Distance from center:

Answer: $\frac{1}{2}$



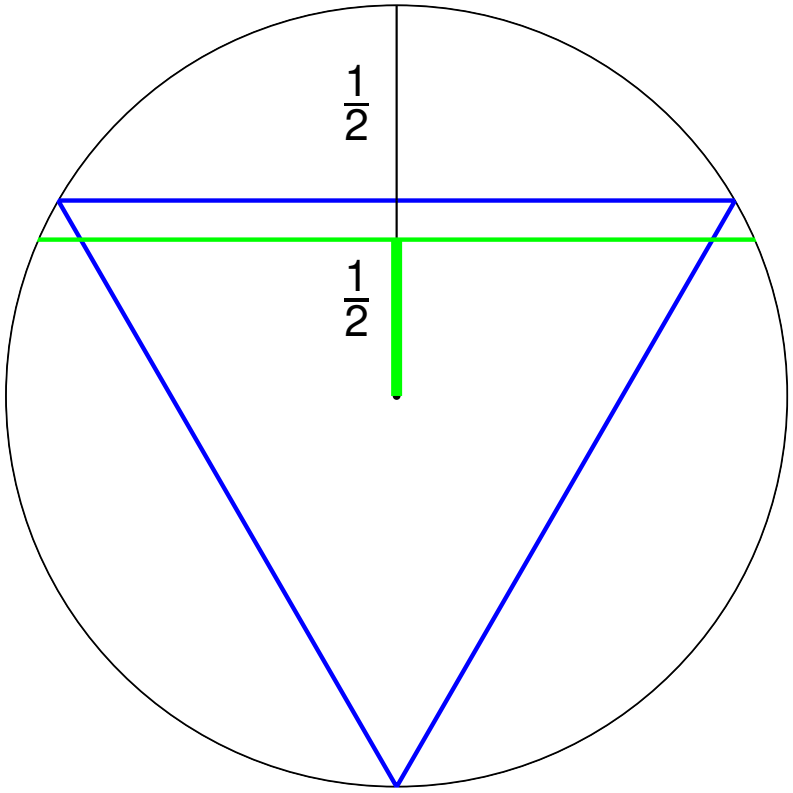
Distance from center:

Answer: $\frac{1}{2}$



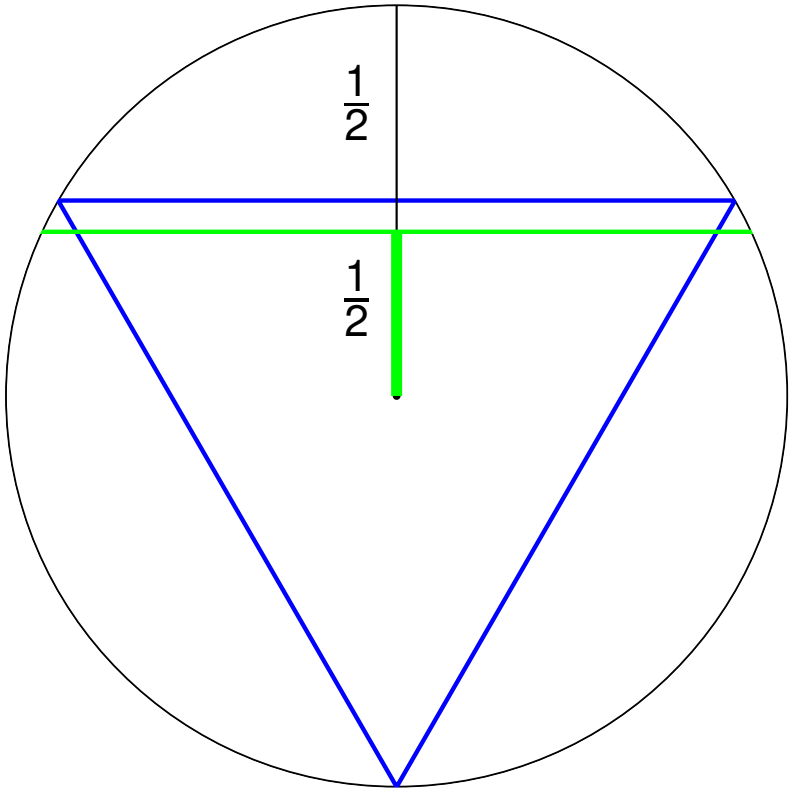
Distance from center:

Answer: $\frac{1}{2}$



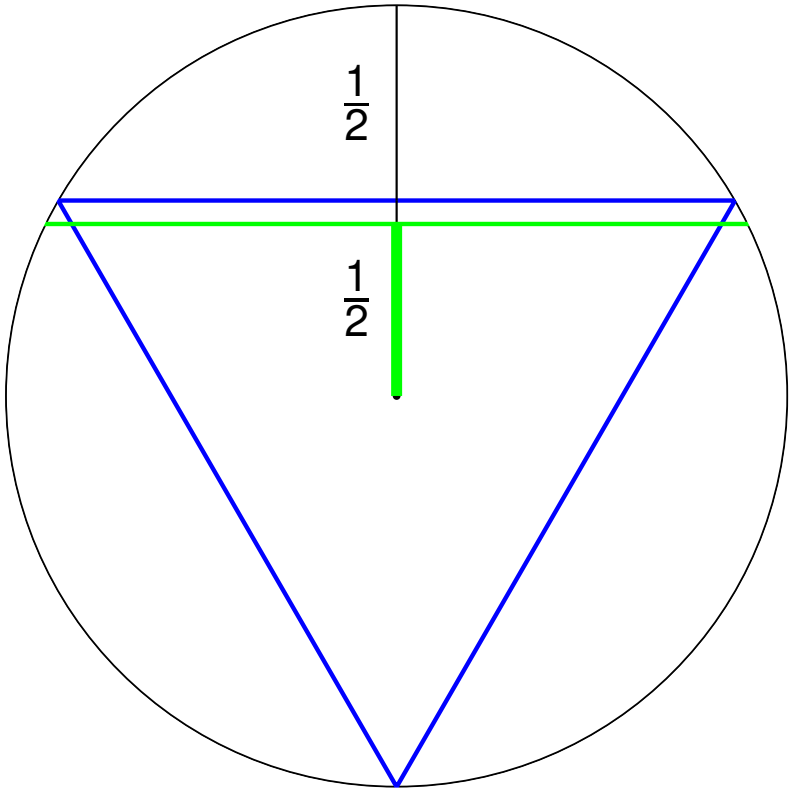
Distance from center:

Answer: $\frac{1}{2}$



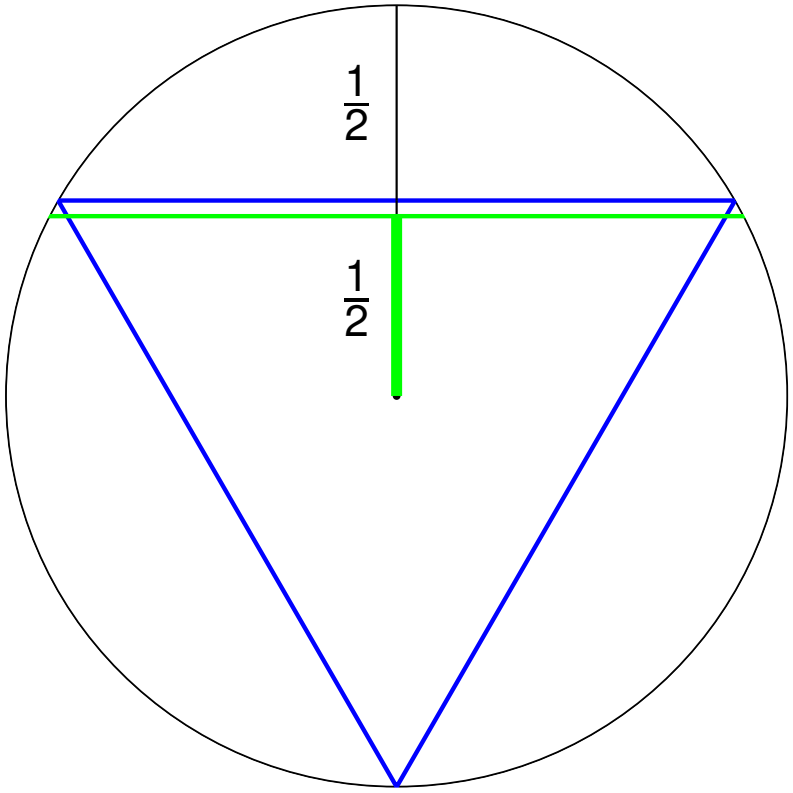
Distance from center:

Answer: $\frac{1}{2}$



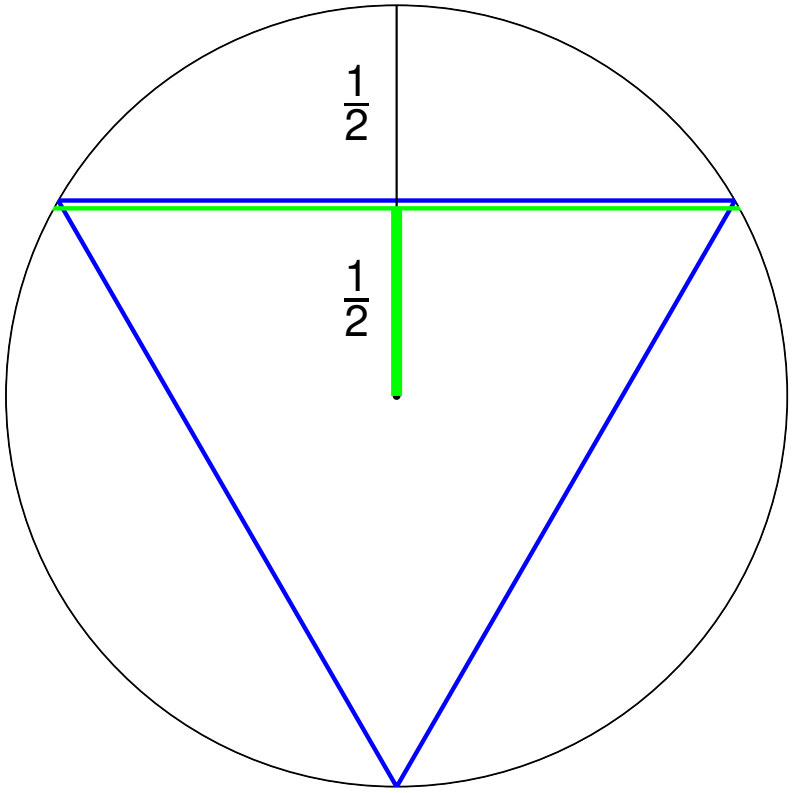
Distance from center:

Answer: $\frac{1}{2}$



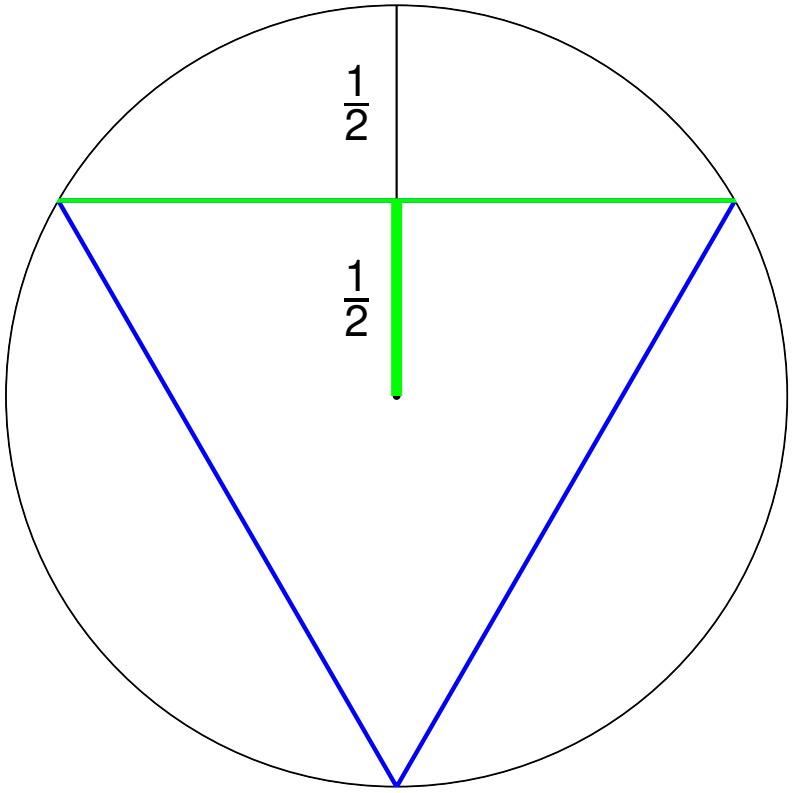
Distance from center:

Answer: $\frac{1}{2}$



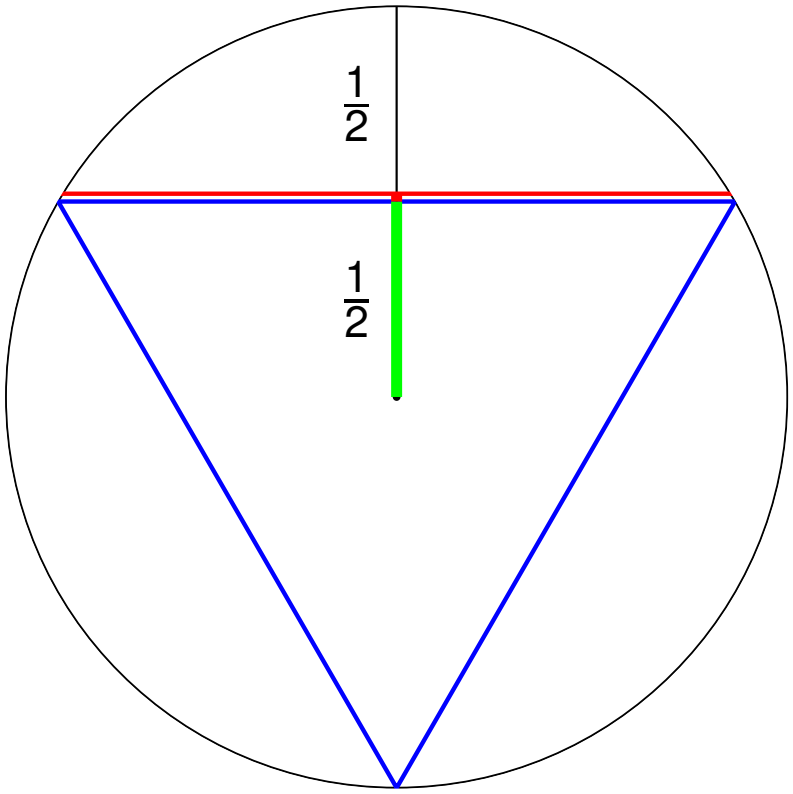
Distance from center:

Answer: $\frac{1}{2}$



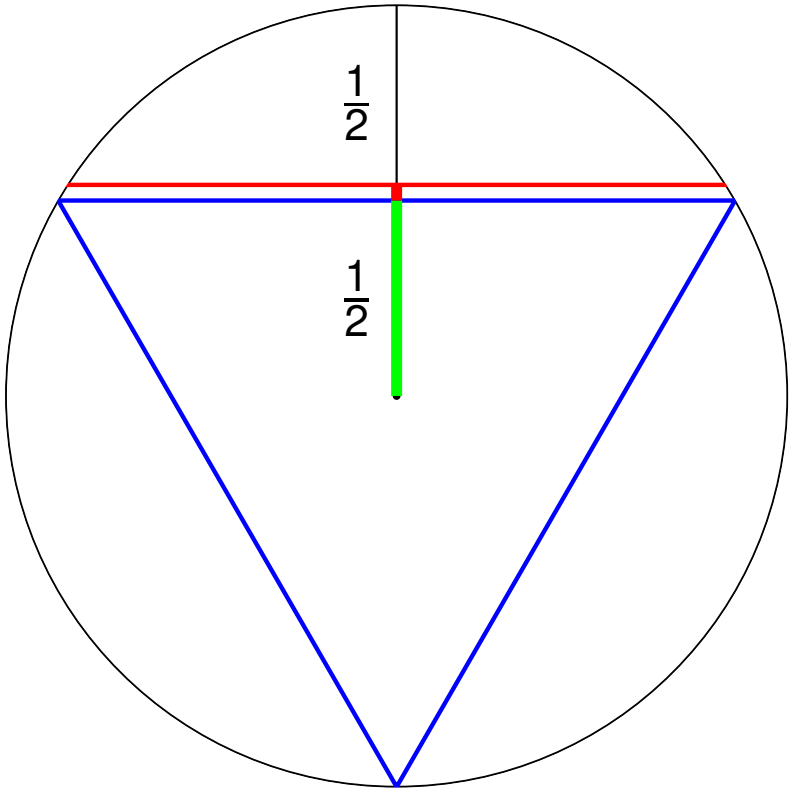
Distance from center:

Answer: $\frac{1}{2}$



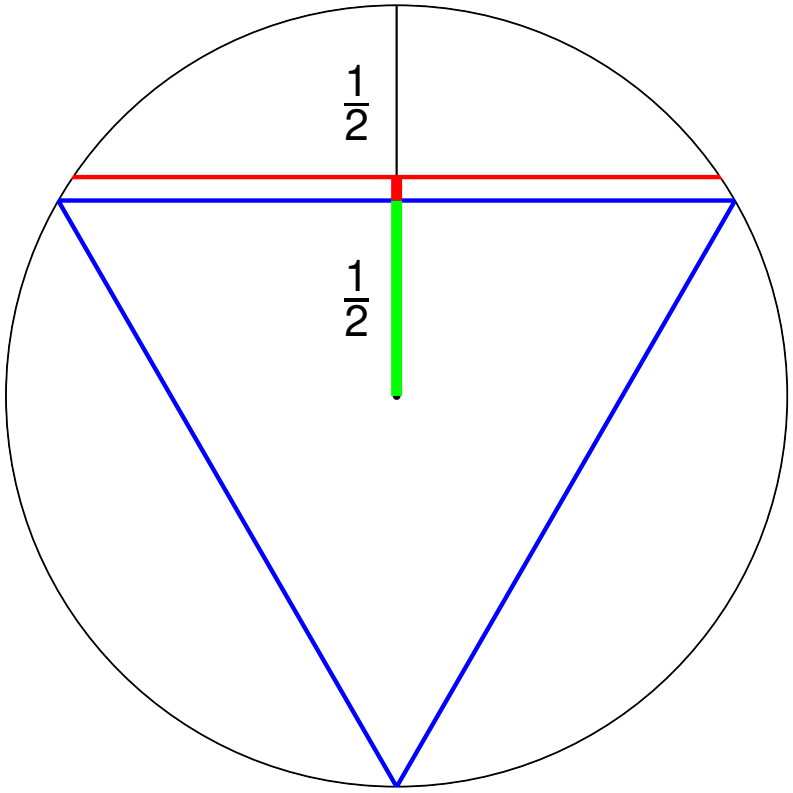
Distance from center:

Answer: $\frac{1}{2}$



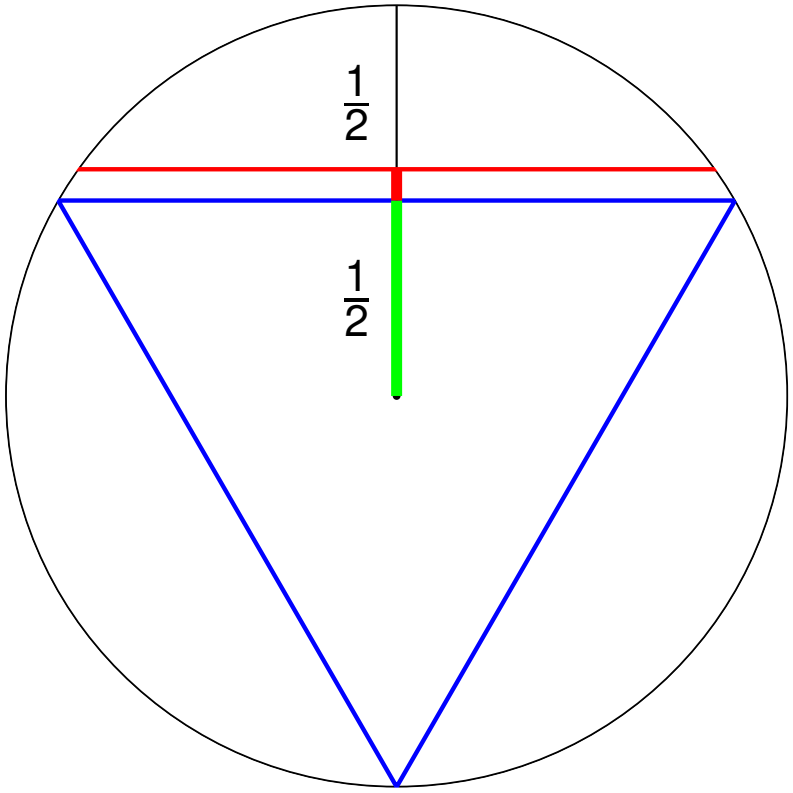
Distance from center:

Answer: $\frac{1}{2}$



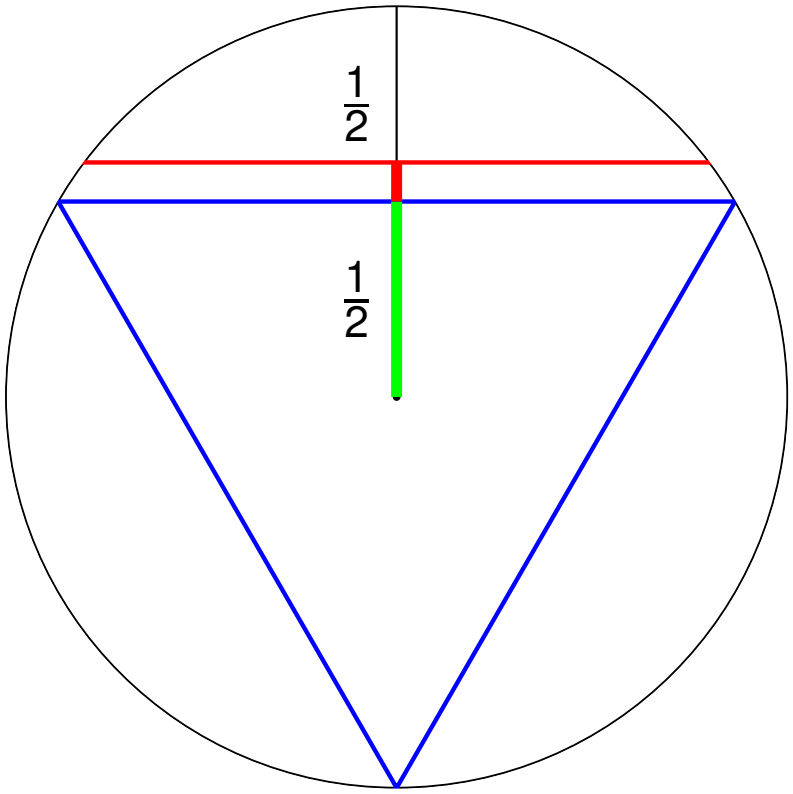
Distance from center:

Answer: $\frac{1}{2}$



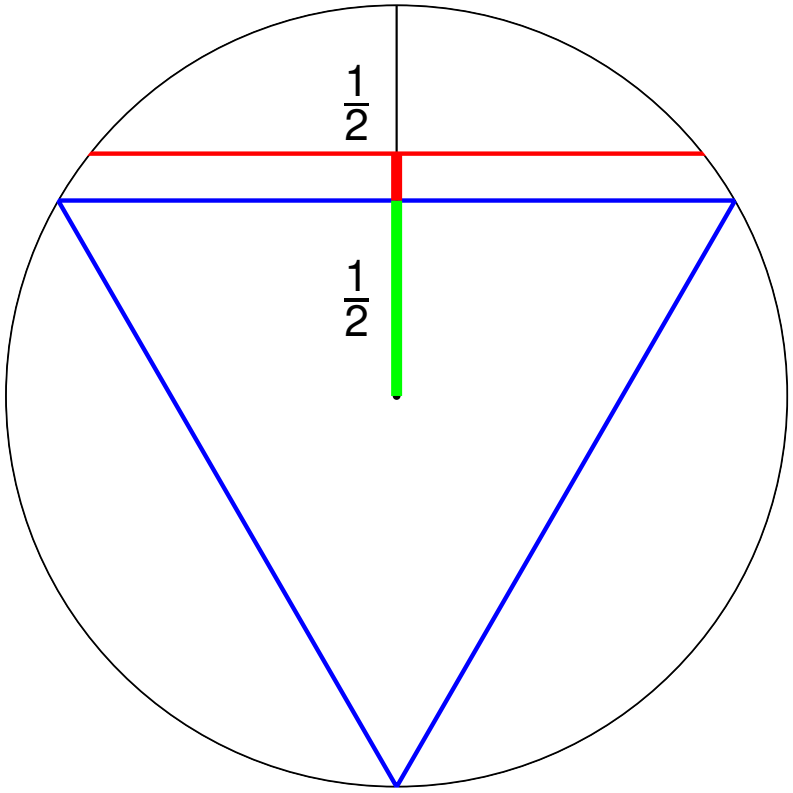
Distance from center:

Answer: $\frac{1}{2}$



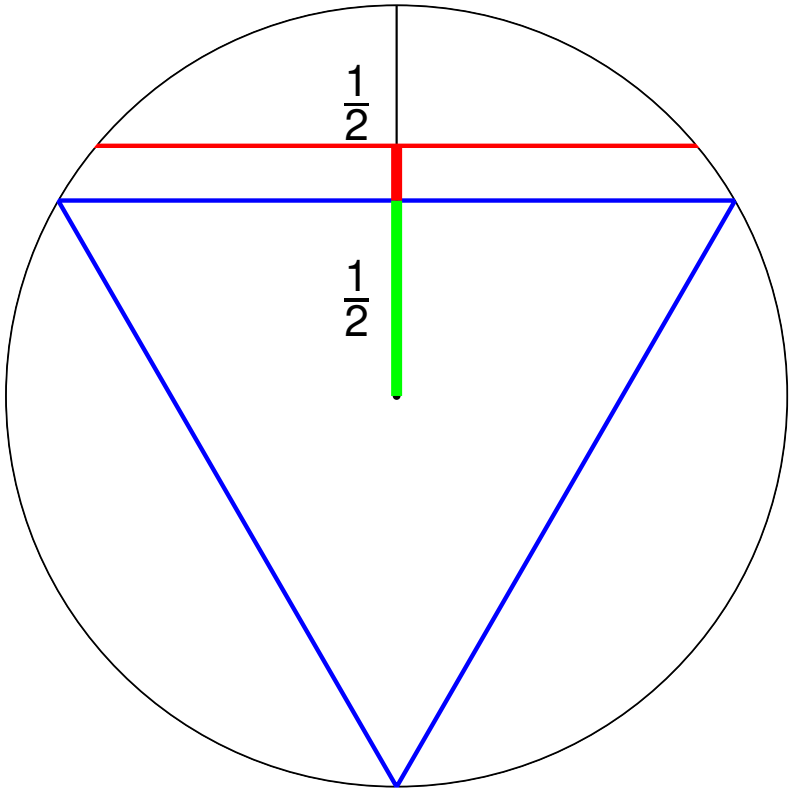
Distance from center:

Answer: $\frac{1}{2}$



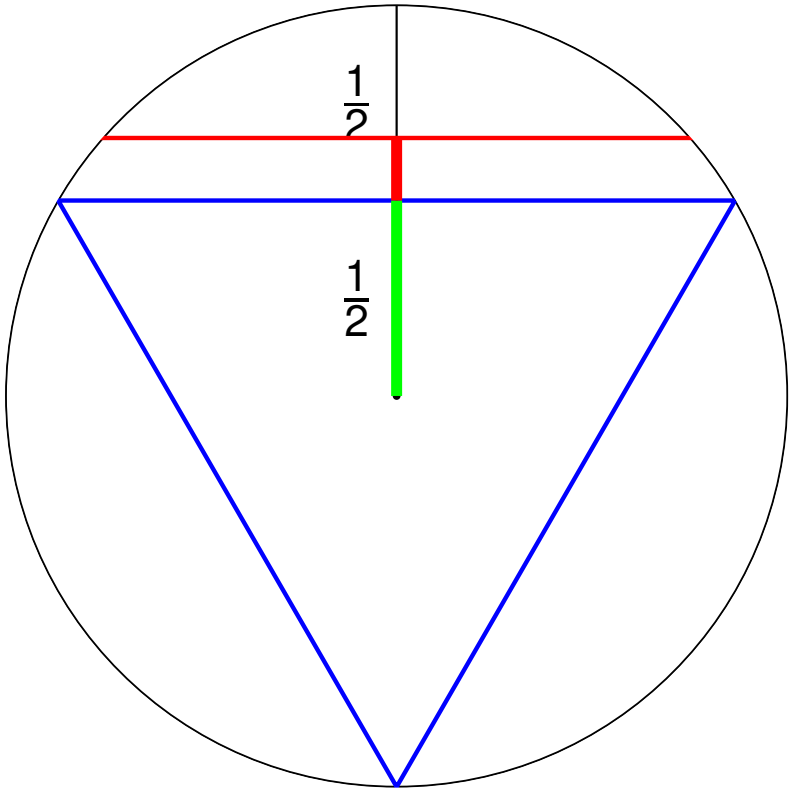
Distance from center:

Answer: $\frac{1}{2}$



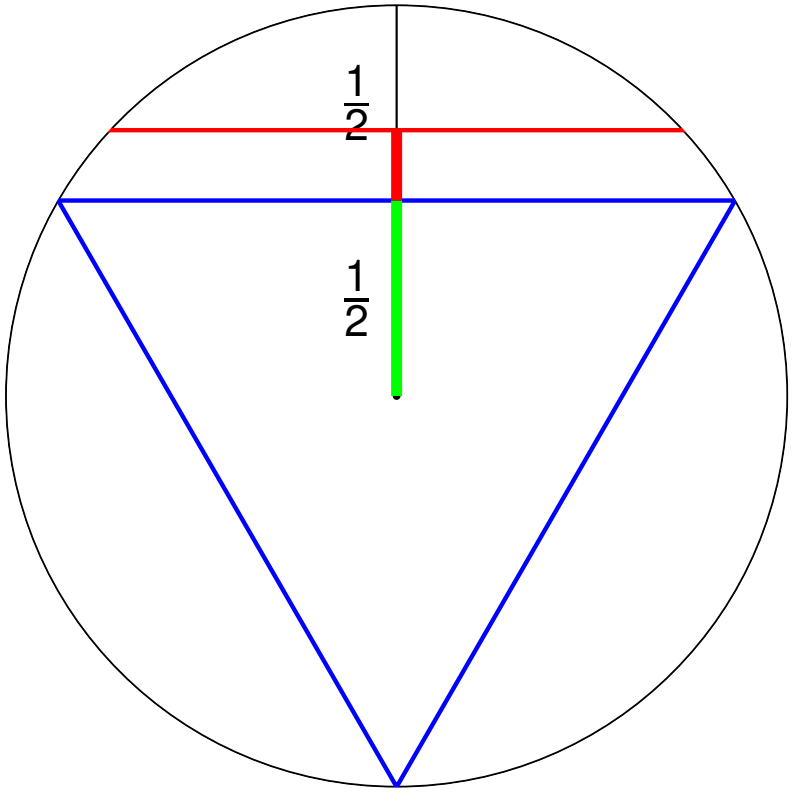
Distance from center:

Answer: $\frac{1}{2}$



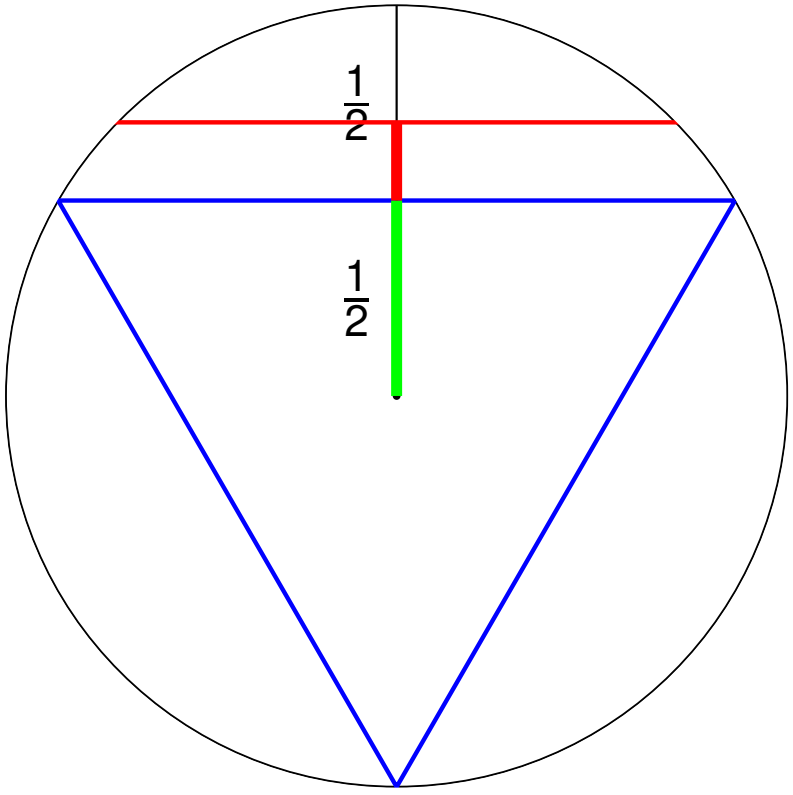
Distance from center:

Answer: $\frac{1}{2}$



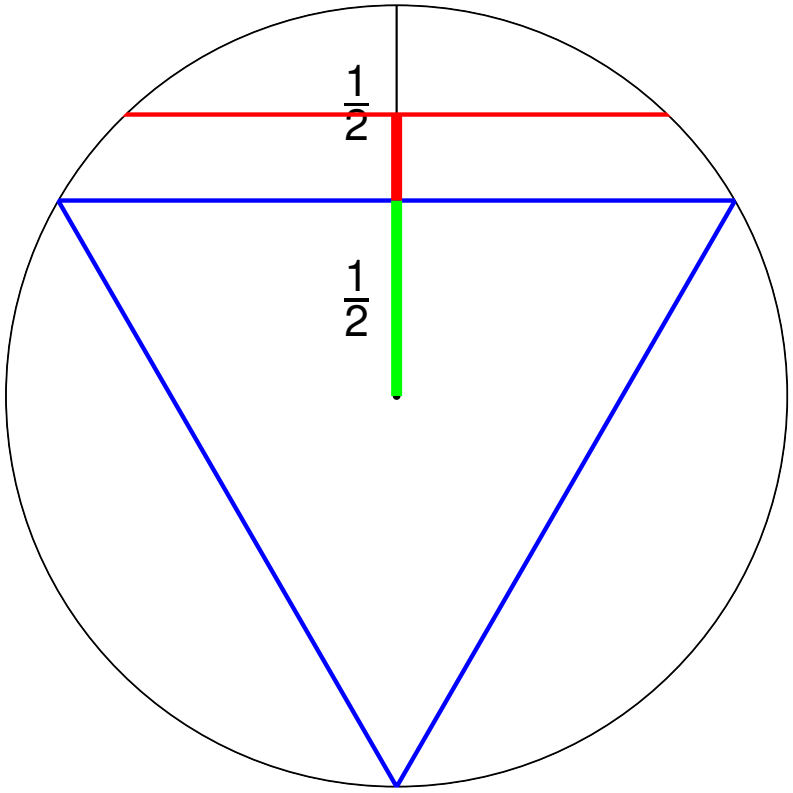
Distance from center:

Answer: $\frac{1}{2}$



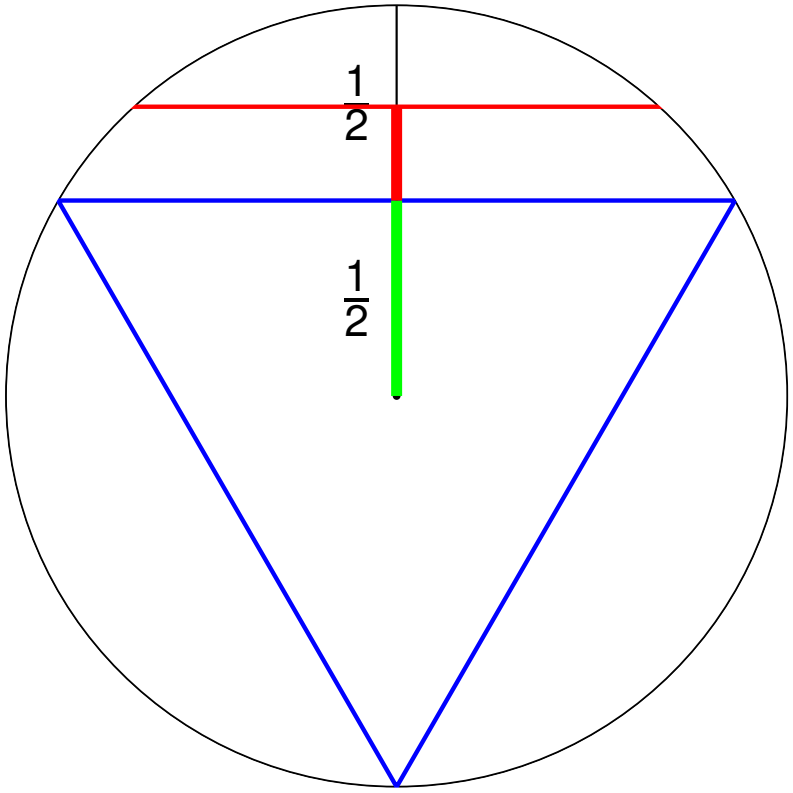
Distance from center:

Answer: $\frac{1}{2}$



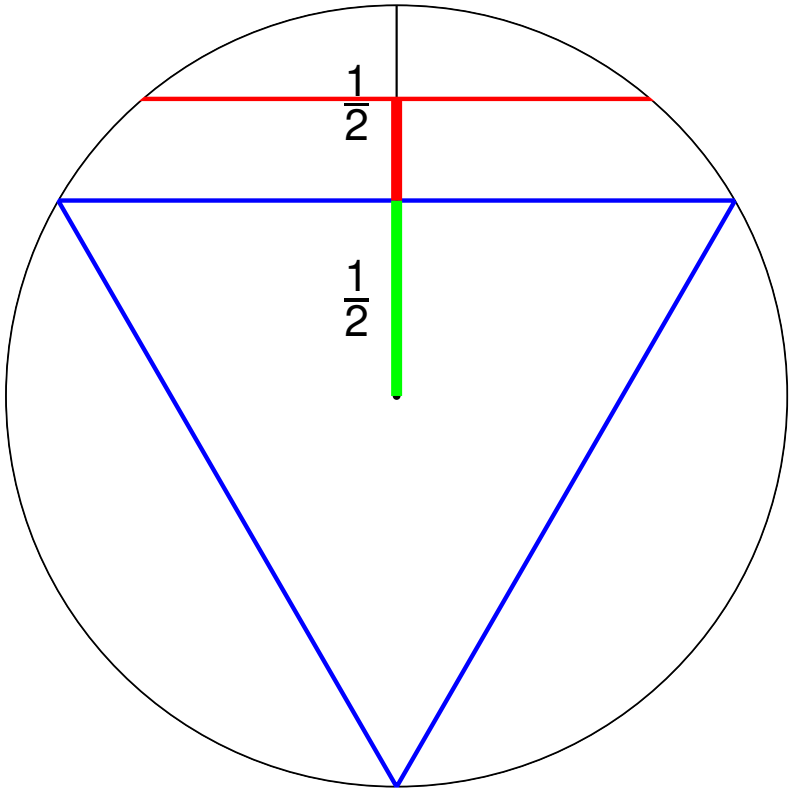
Distance from center:

Answer: $\frac{1}{2}$



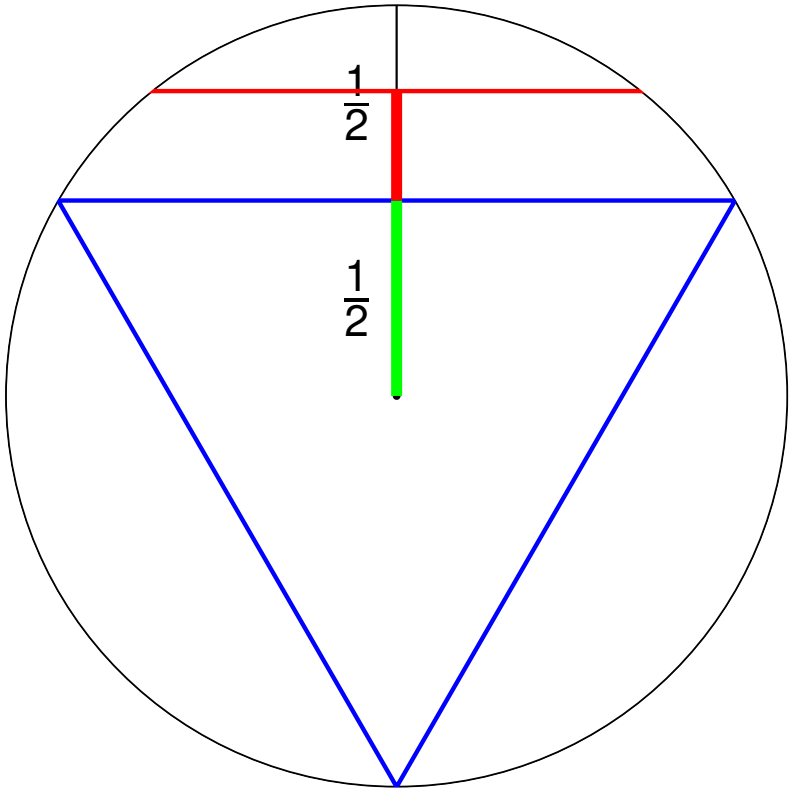
Distance from center:

Answer: $\frac{1}{2}$



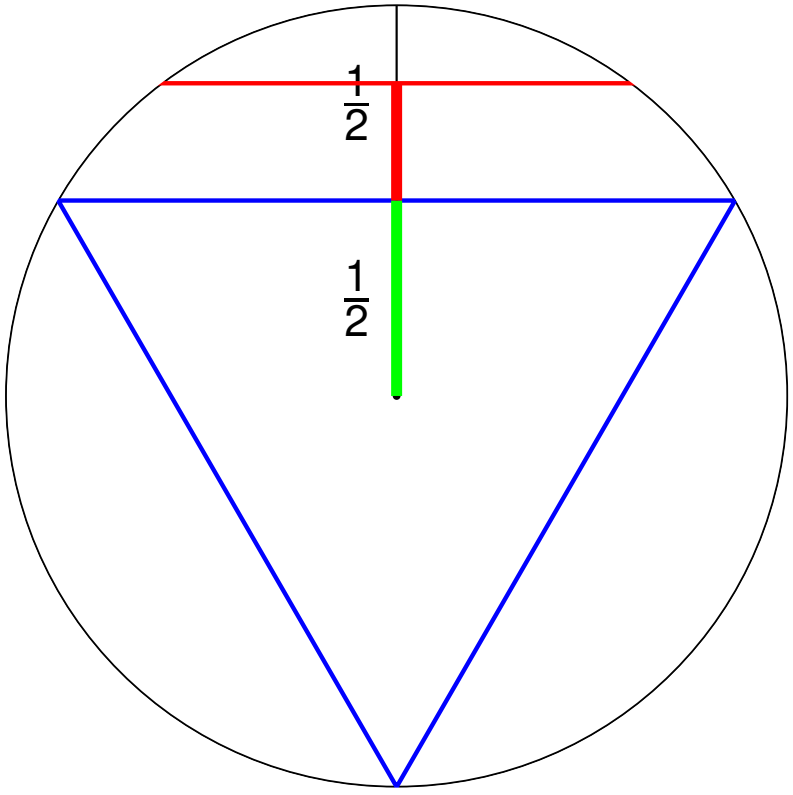
Distance from center:

Answer: $\frac{1}{2}$



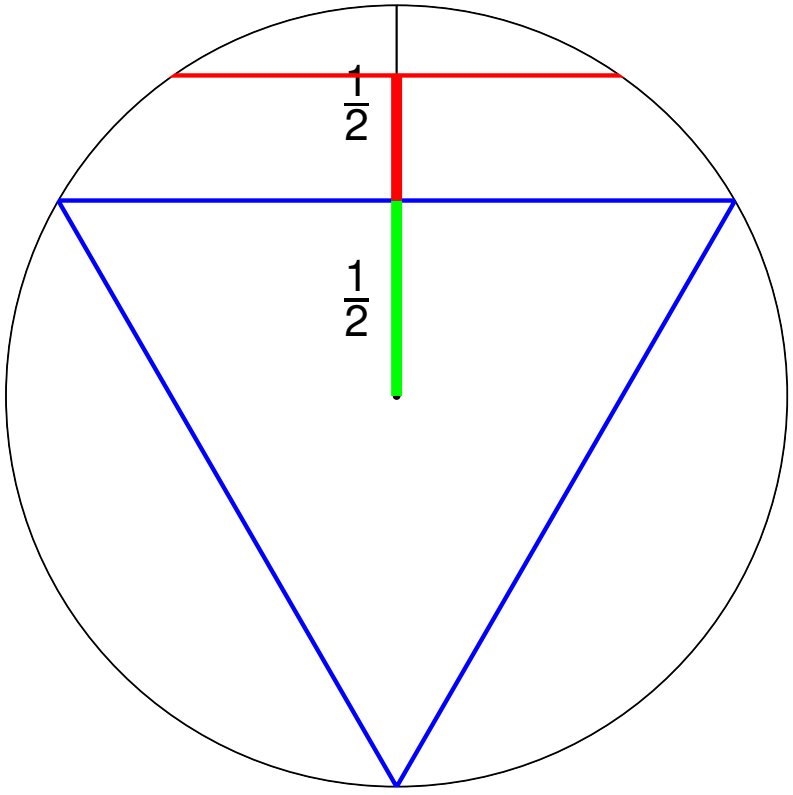
Distance from center:

Answer: $\frac{1}{2}$



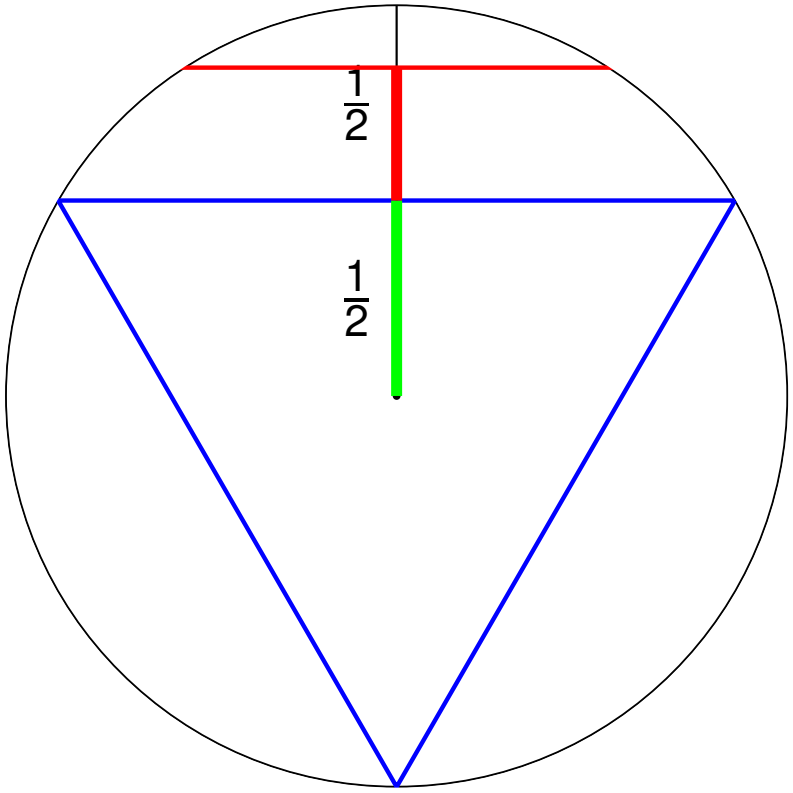
Distance from center:

Answer: $\frac{1}{2}$



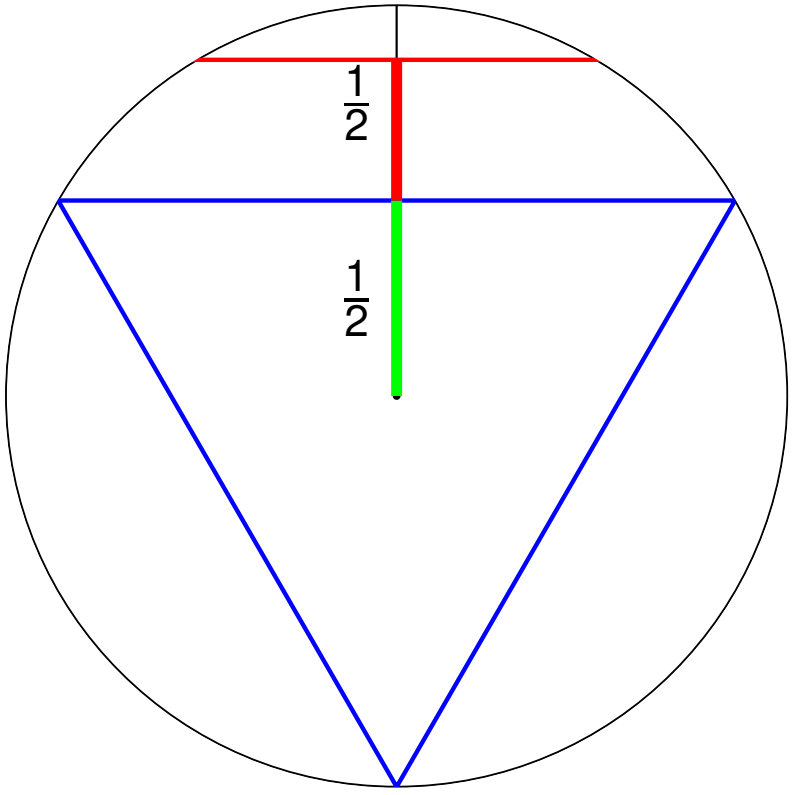
Distance from center:

Answer: $\frac{1}{2}$



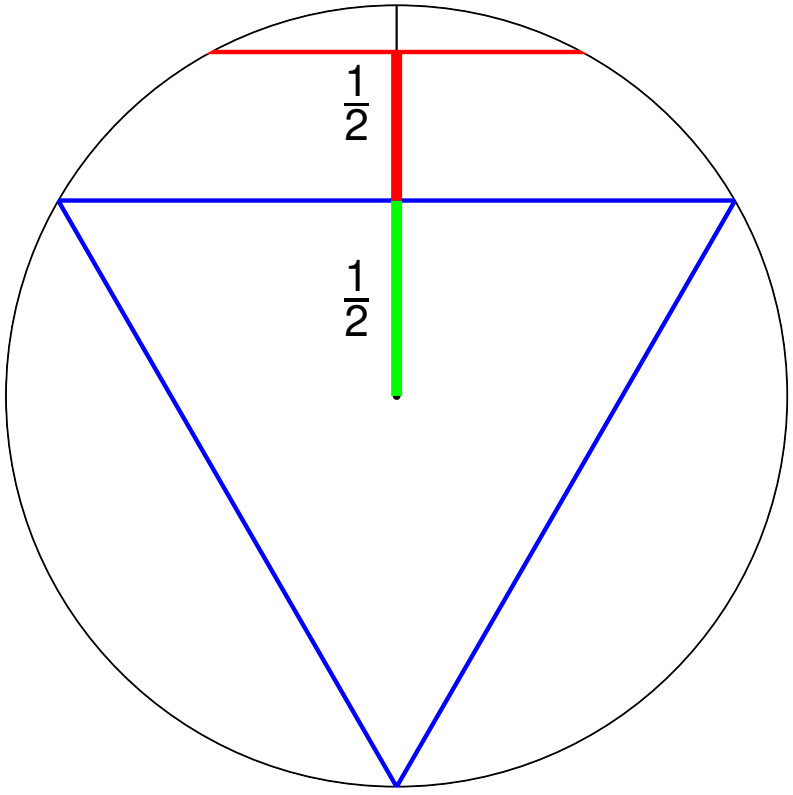
Distance from center:

Answer: $\frac{1}{2}$



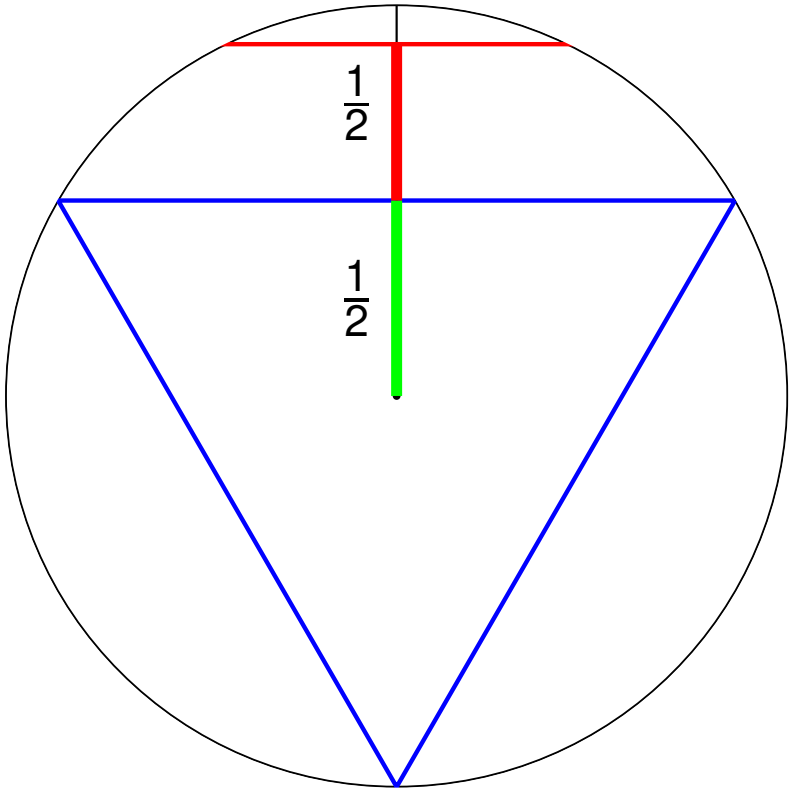
Distance from center:

Answer: $\frac{1}{2}$



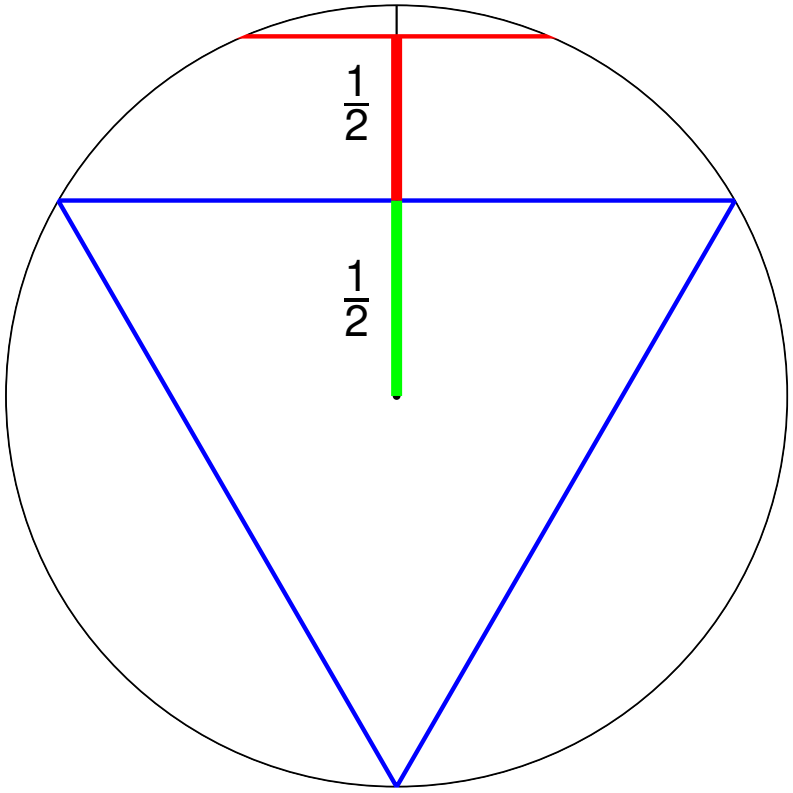
Distance from center:

Answer: $\frac{1}{2}$



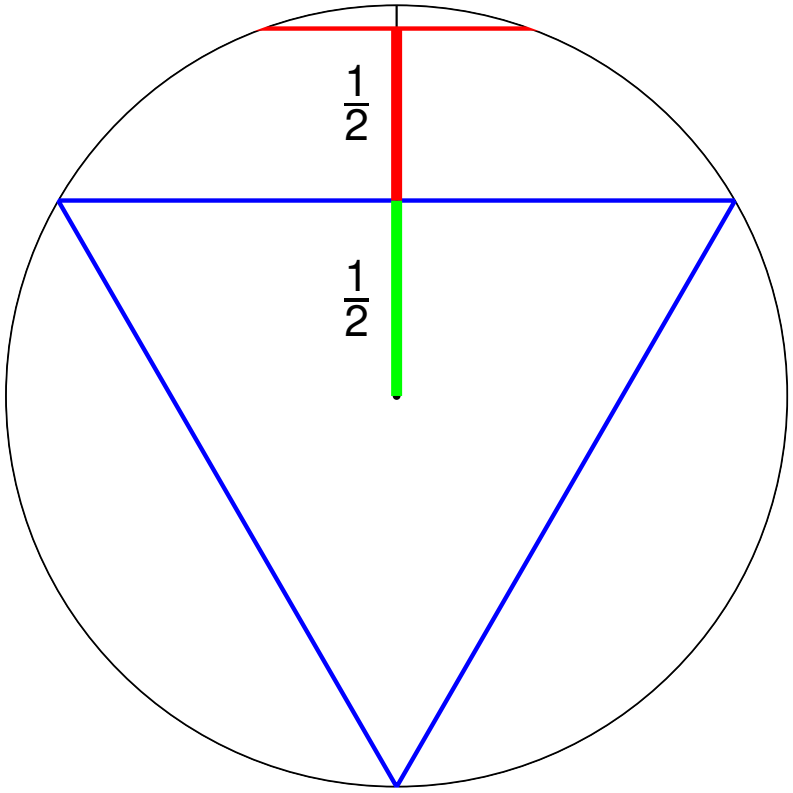
Distance from center:

Answer: $\frac{1}{2}$



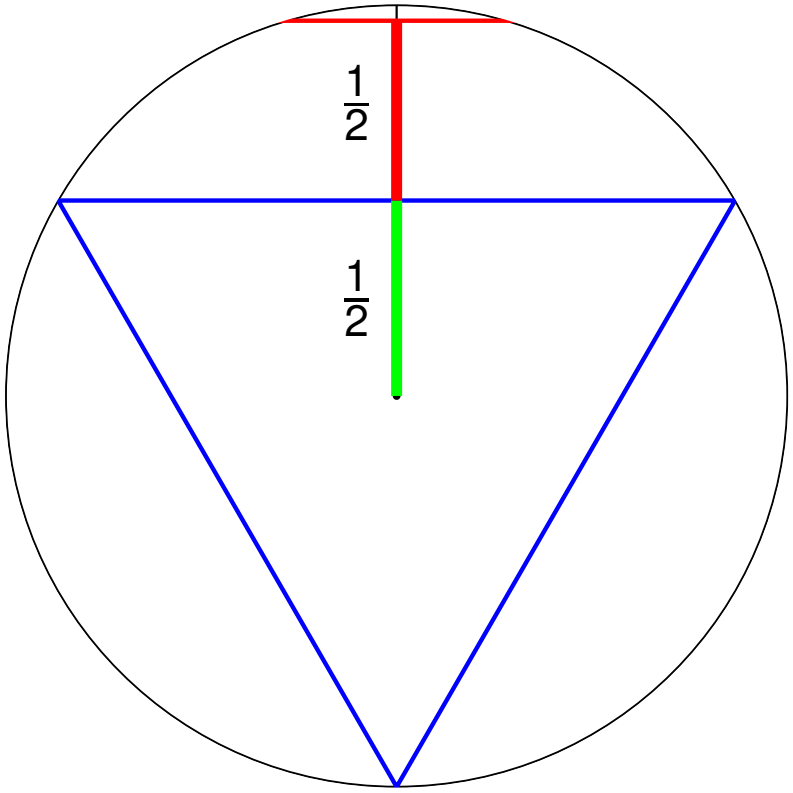
Distance from center:

Answer: $\frac{1}{2}$



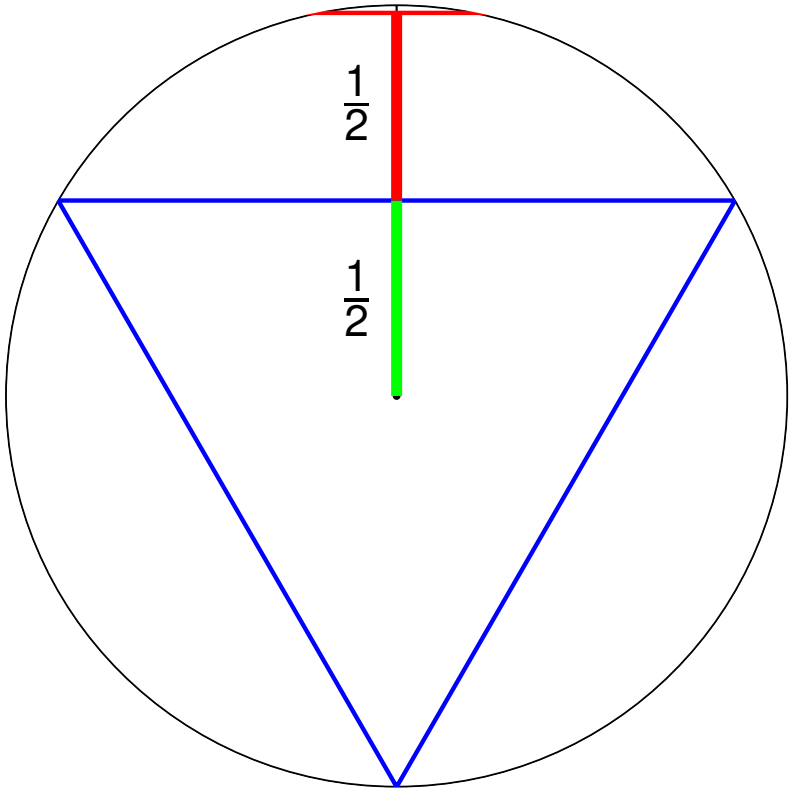
Distance from center:

Answer: $\frac{1}{2}$



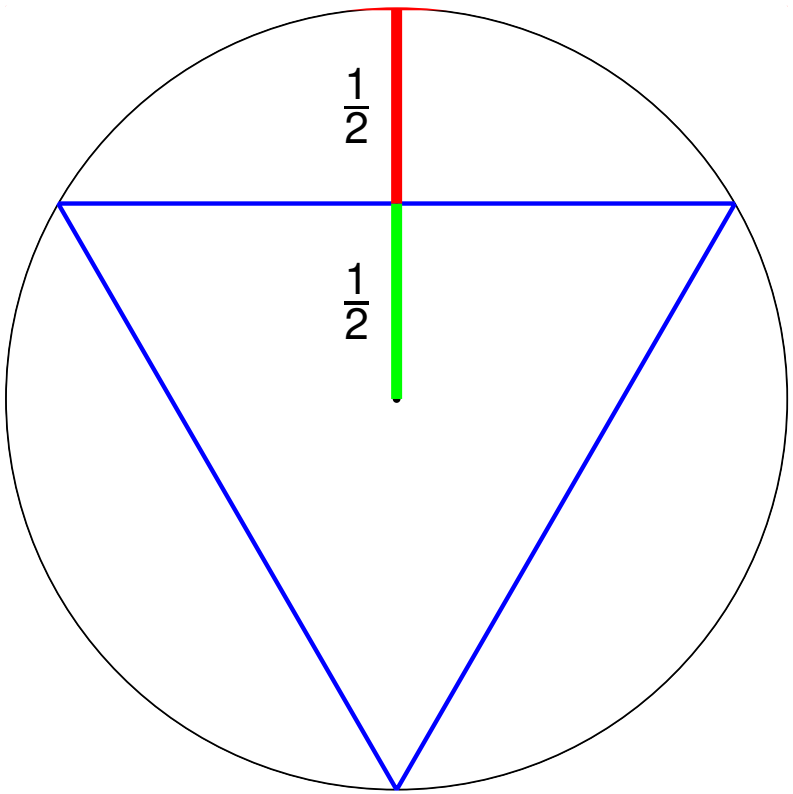
Distance from center:

Answer: $\frac{1}{2}$



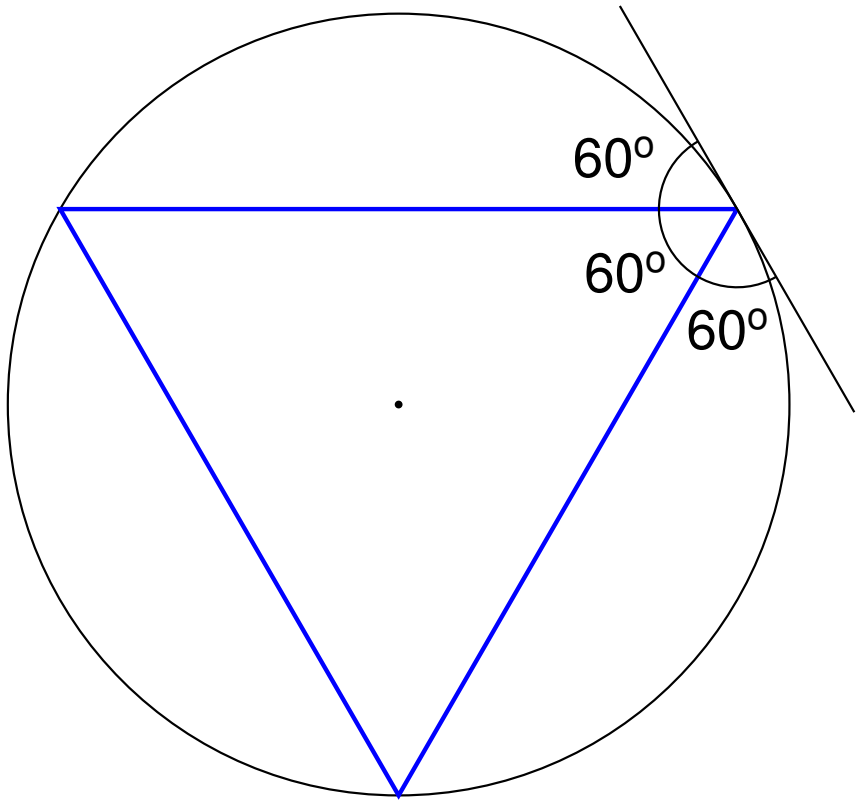
Distance from center:

Answer: $\frac{1}{2}$

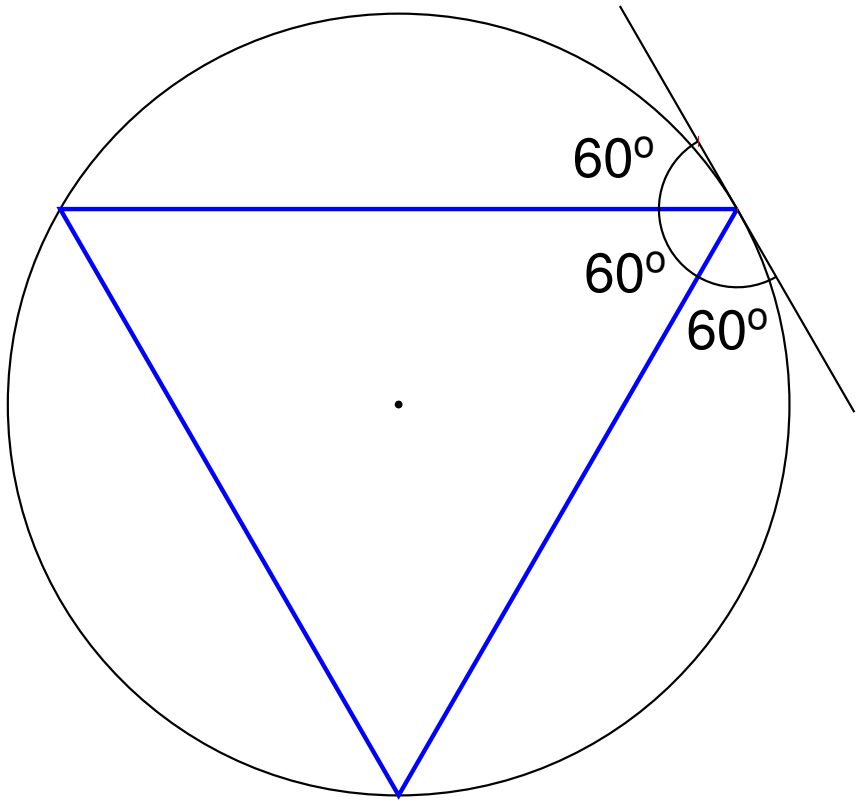


Distance from center: $\frac{\frac{1}{2}}{\frac{1}{2}+\frac{1}{2}} = \frac{1}{2}$.

Wait. . . it's $\frac{1}{3}$!

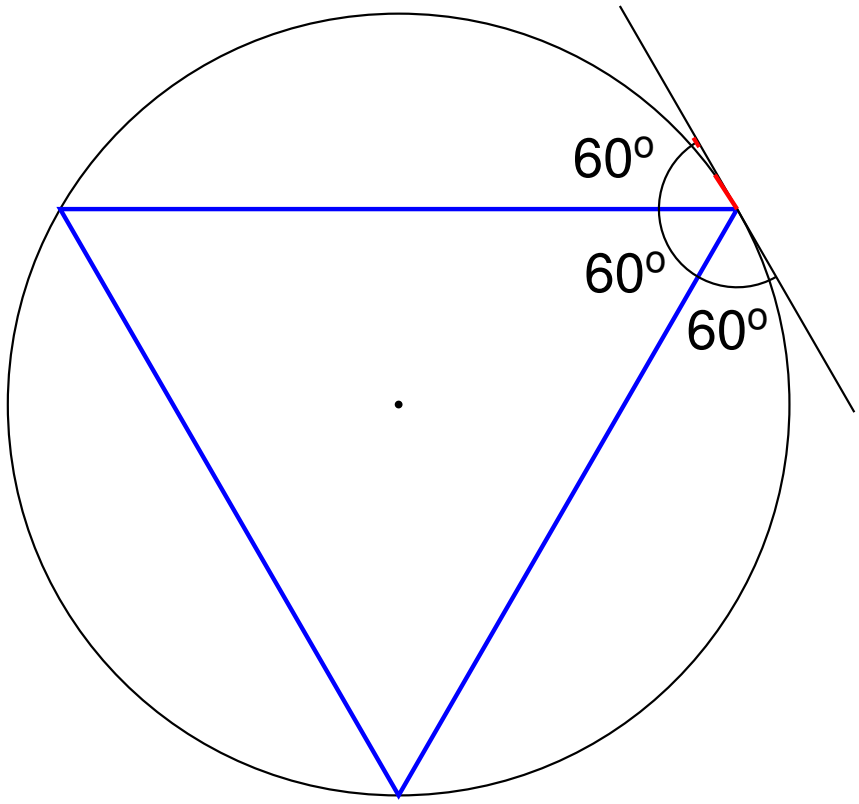


Wait. . . it's $\frac{1}{3}$!



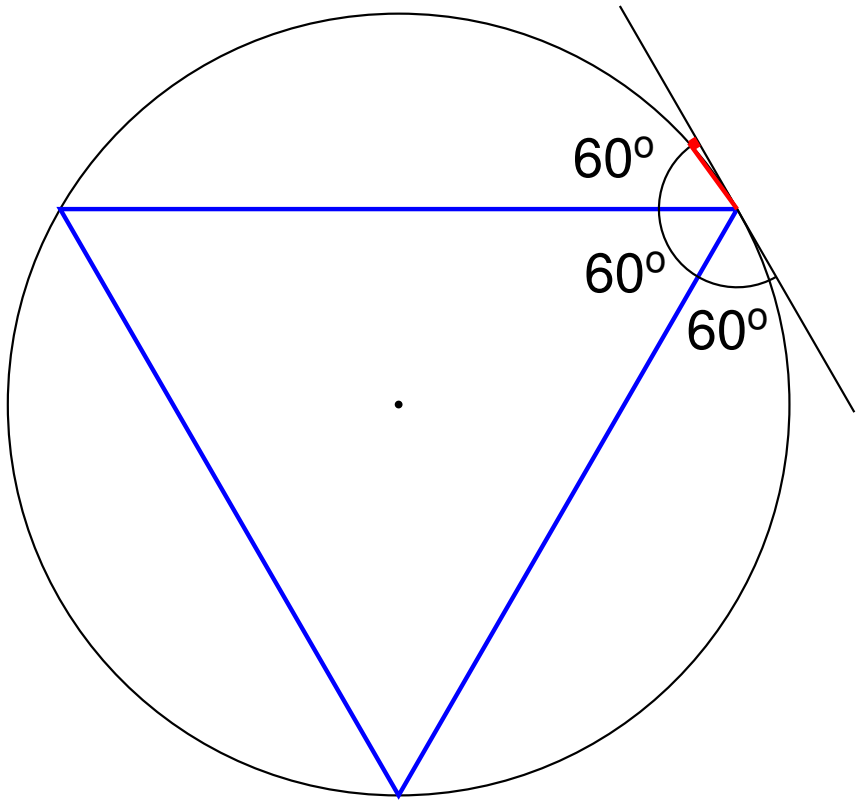
Angle:

Wait. . . it's $\frac{1}{3}$!



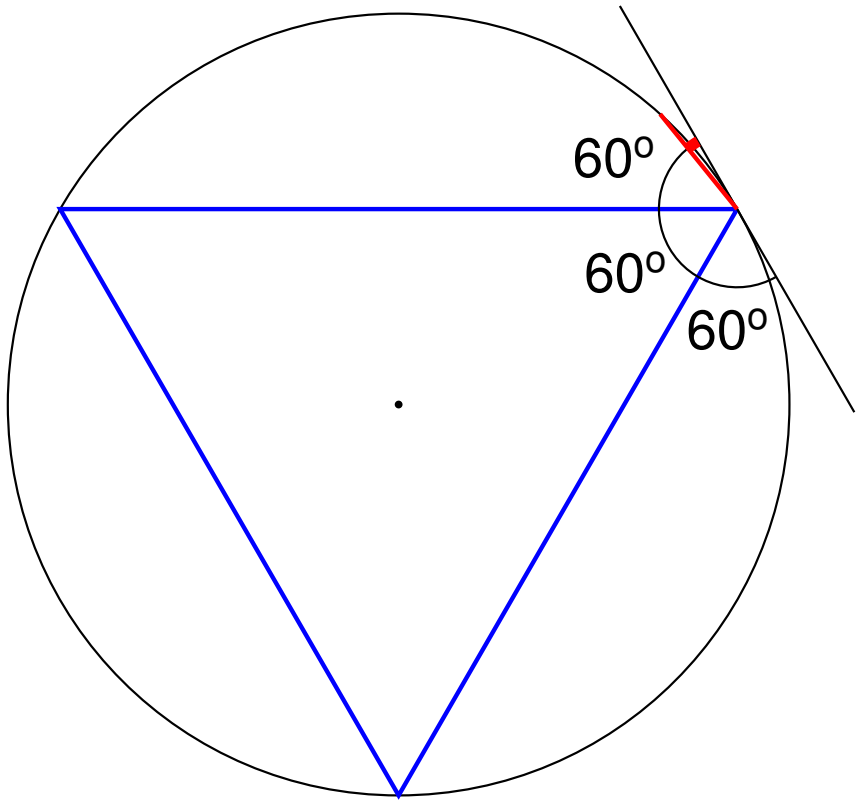
Angle:

Wait. . . it's $\frac{1}{3}$!



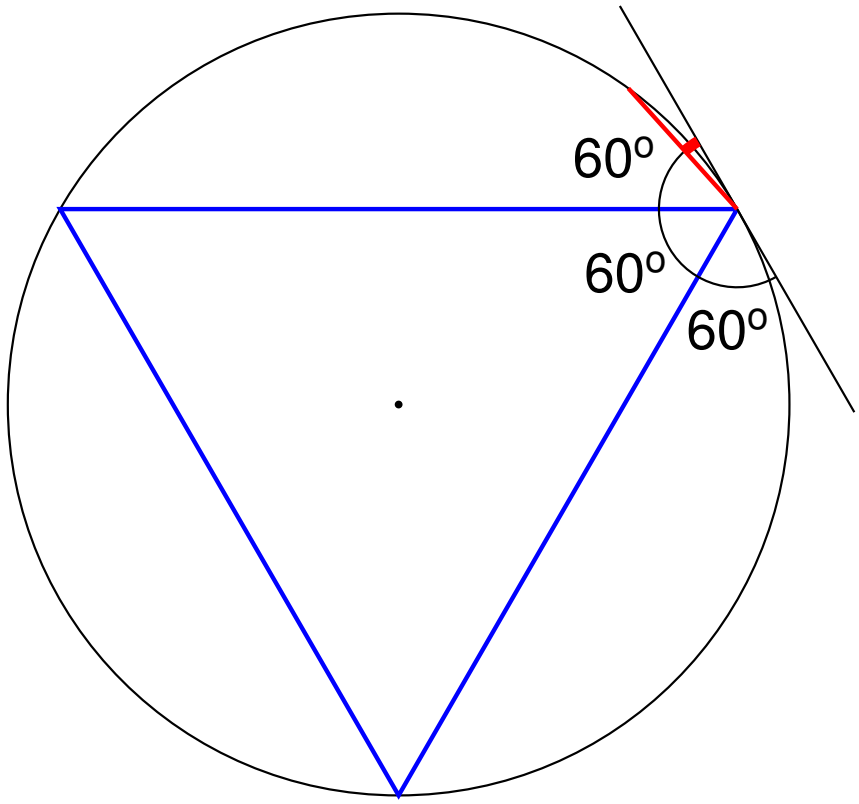
Angle:

Wait. . . it's $\frac{1}{3}$!



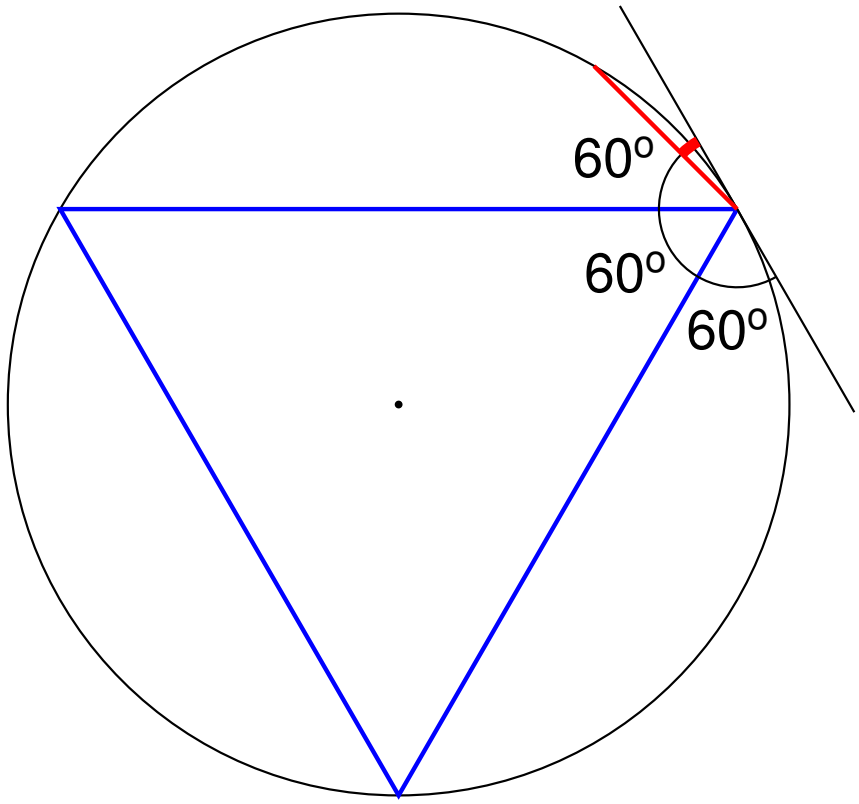
Angle:

Wait. . . it's $\frac{1}{3}$!



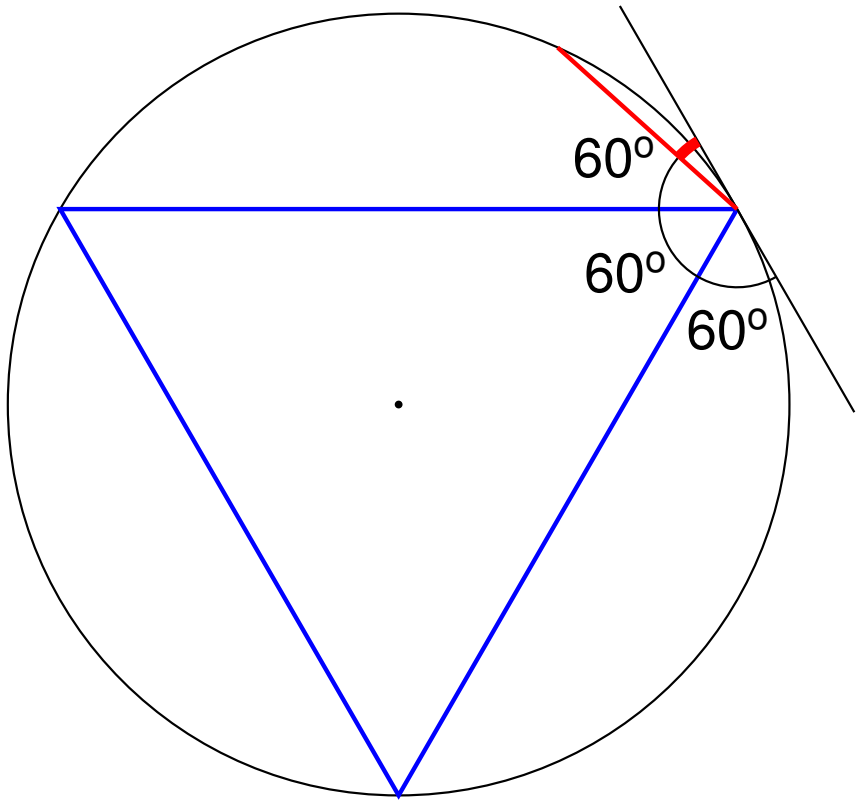
Angle:

Wait. . . it's $\frac{1}{3}$!



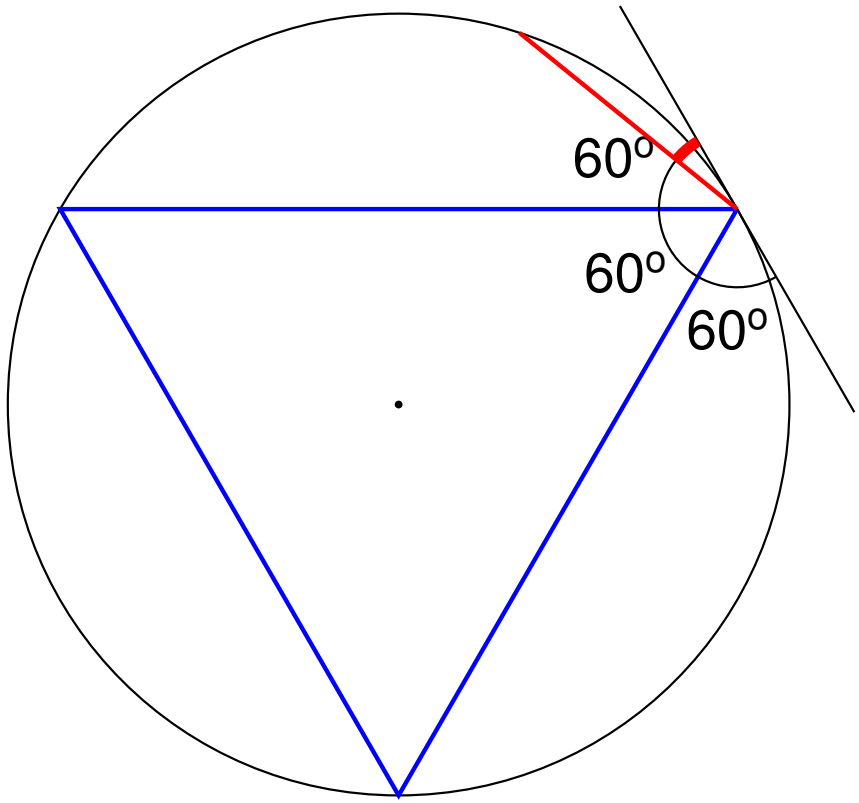
Angle:

Wait. . . it's $\frac{1}{3}$!



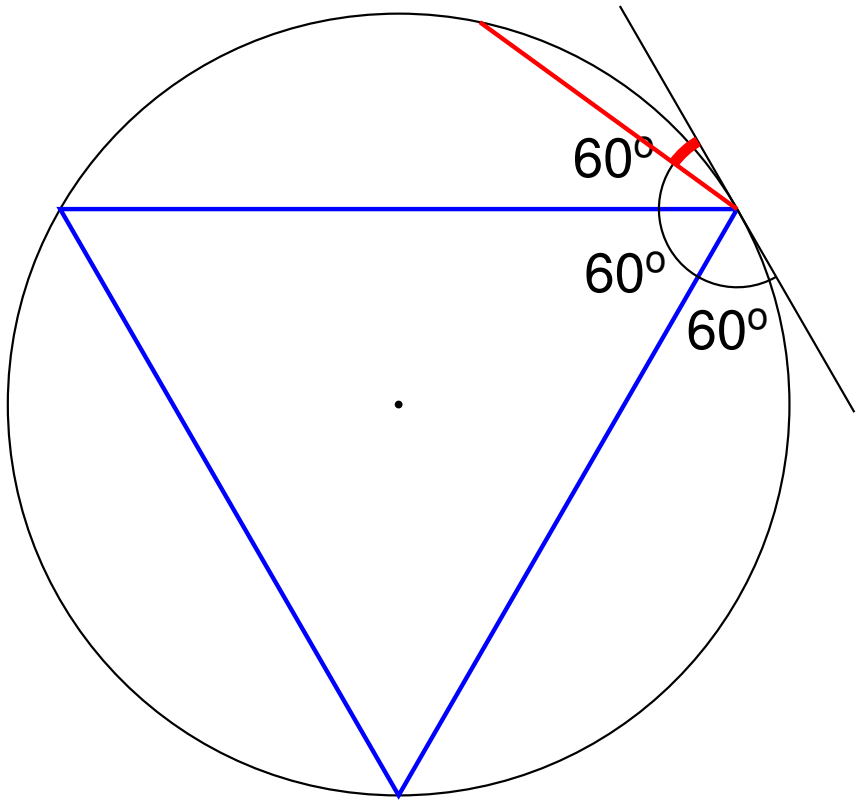
Angle:

Wait. . . it's $\frac{1}{3}$!



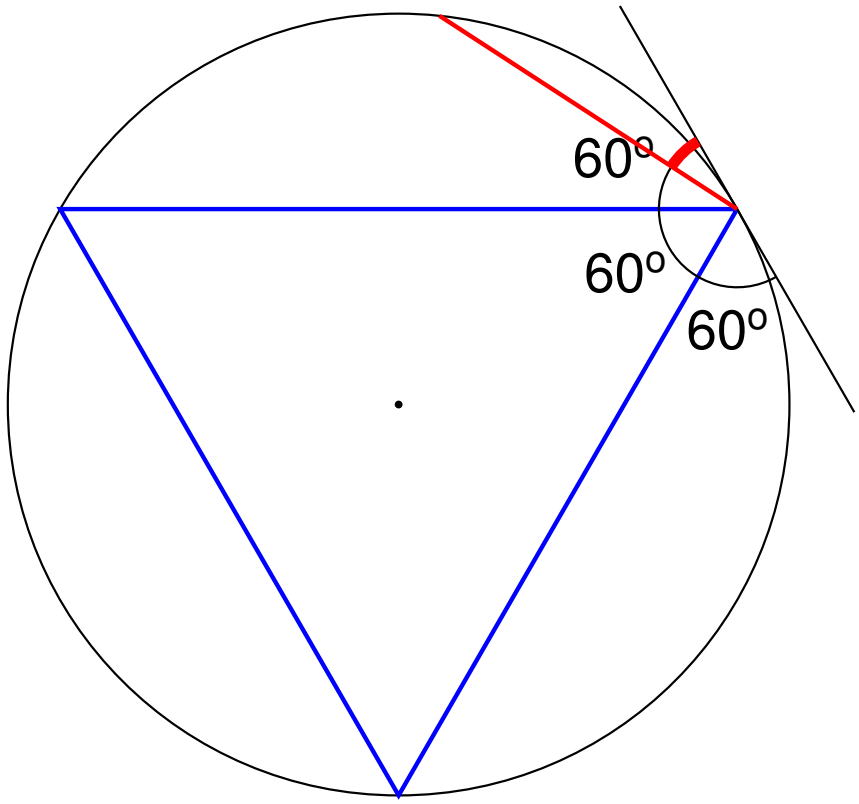
Angle:

Wait. . . it's $\frac{1}{3}$!



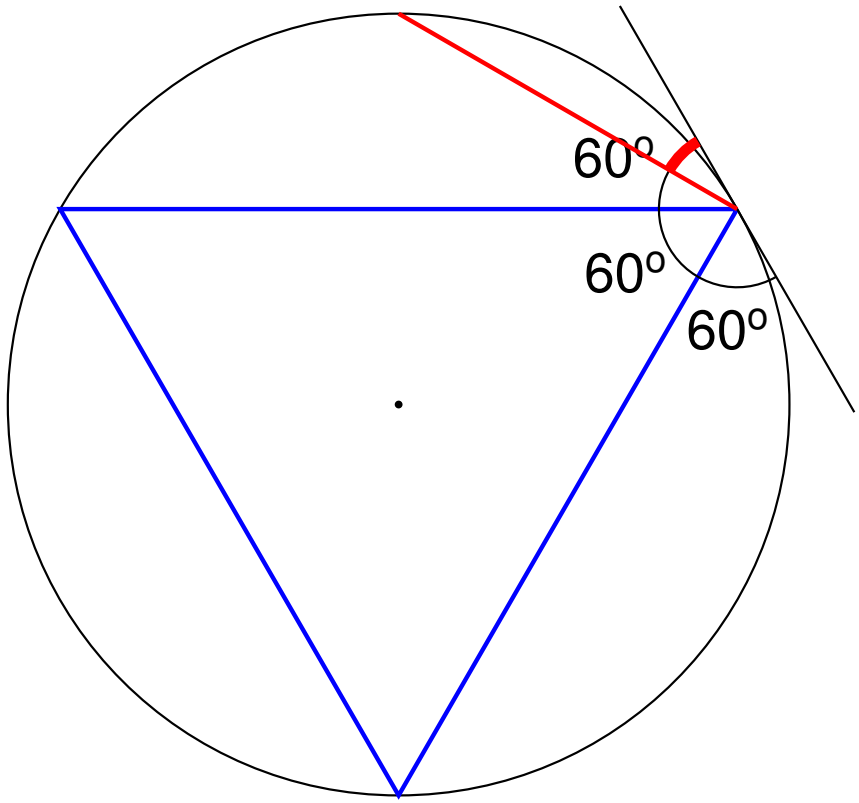
Angle:

Wait. . . it's $\frac{1}{3}$!



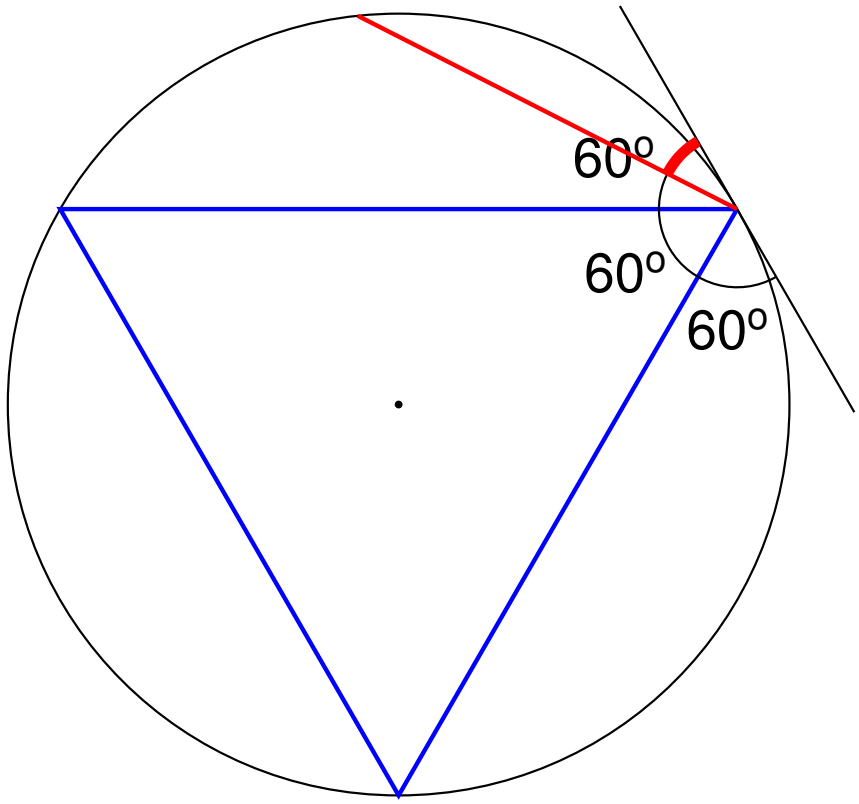
Angle:

Wait. . . it's $\frac{1}{3}$!



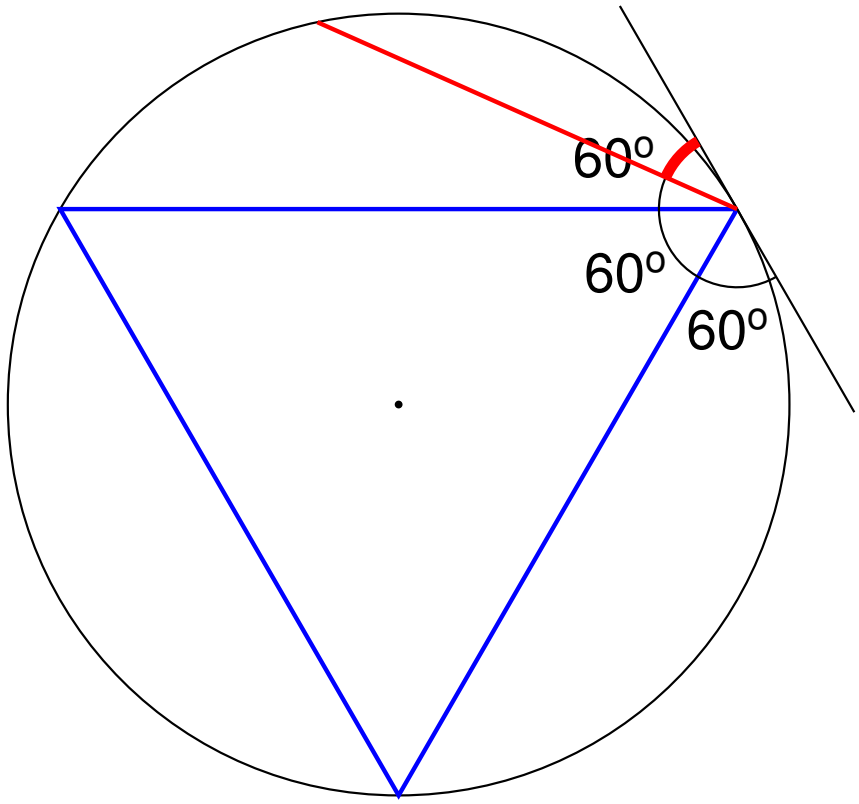
Angle:

Wait. . . it's $\frac{1}{3}$!



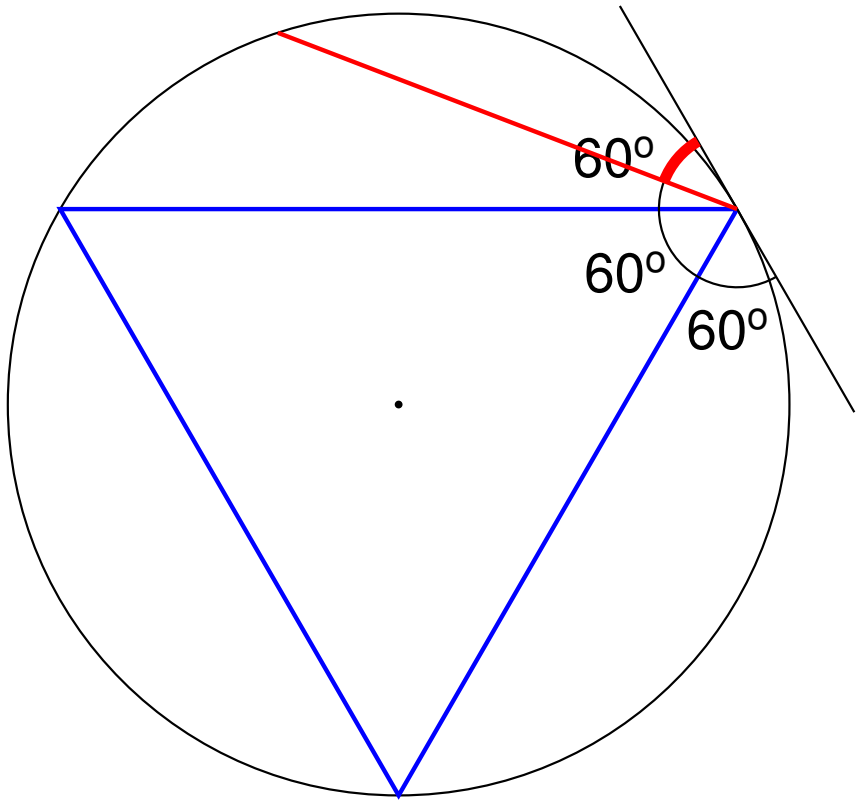
Angle:

Wait. . . it's $\frac{1}{3}$!



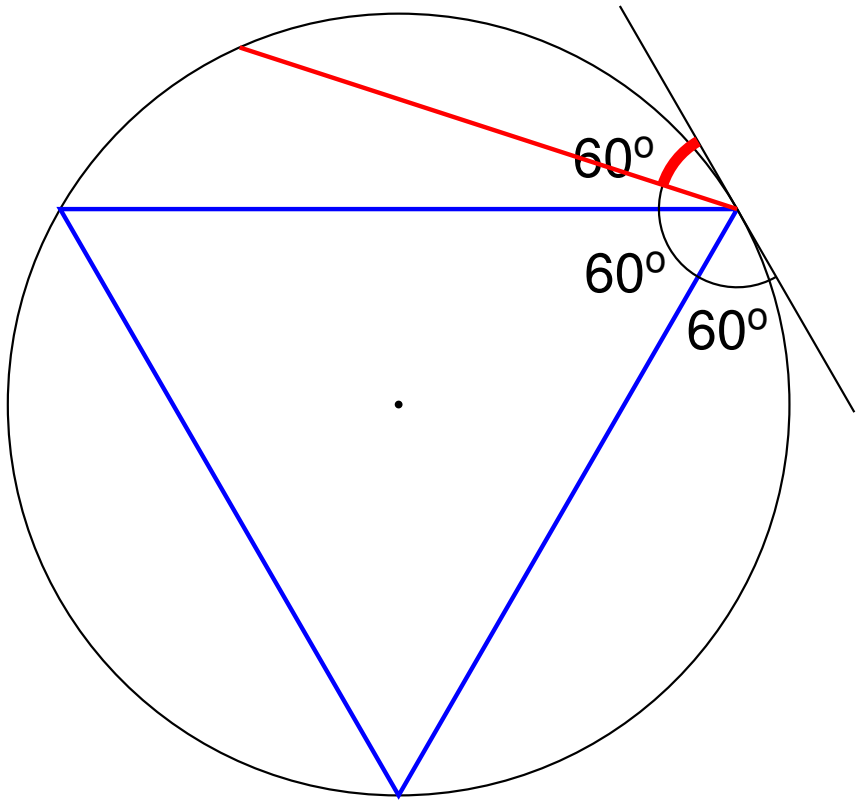
Angle:

Wait. . . it's $\frac{1}{3}$!



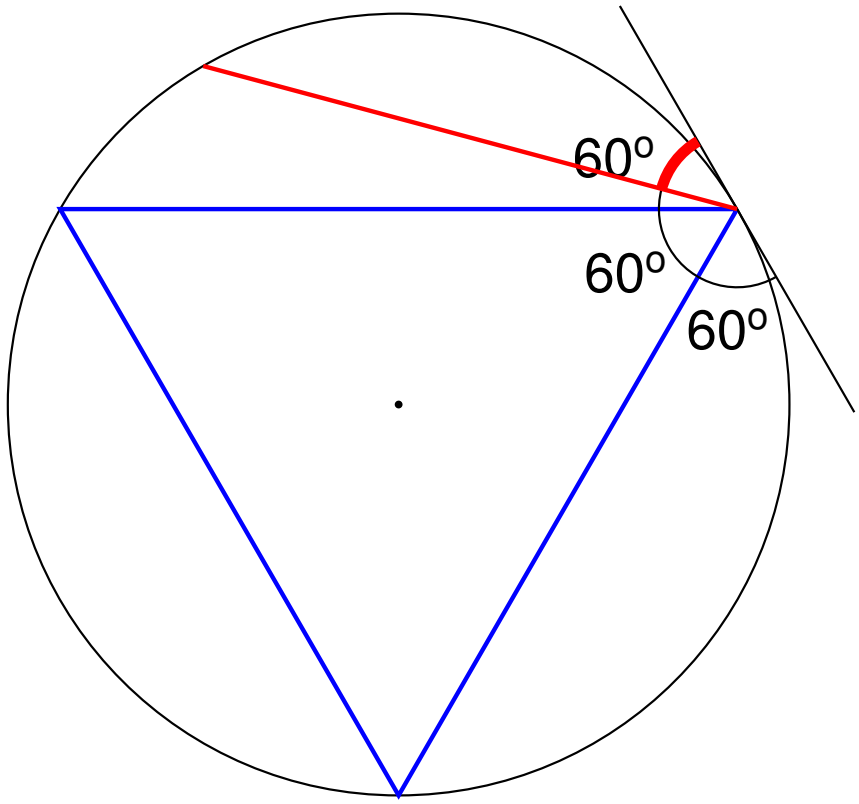
Angle:

Wait. . . it's $\frac{1}{3}$!



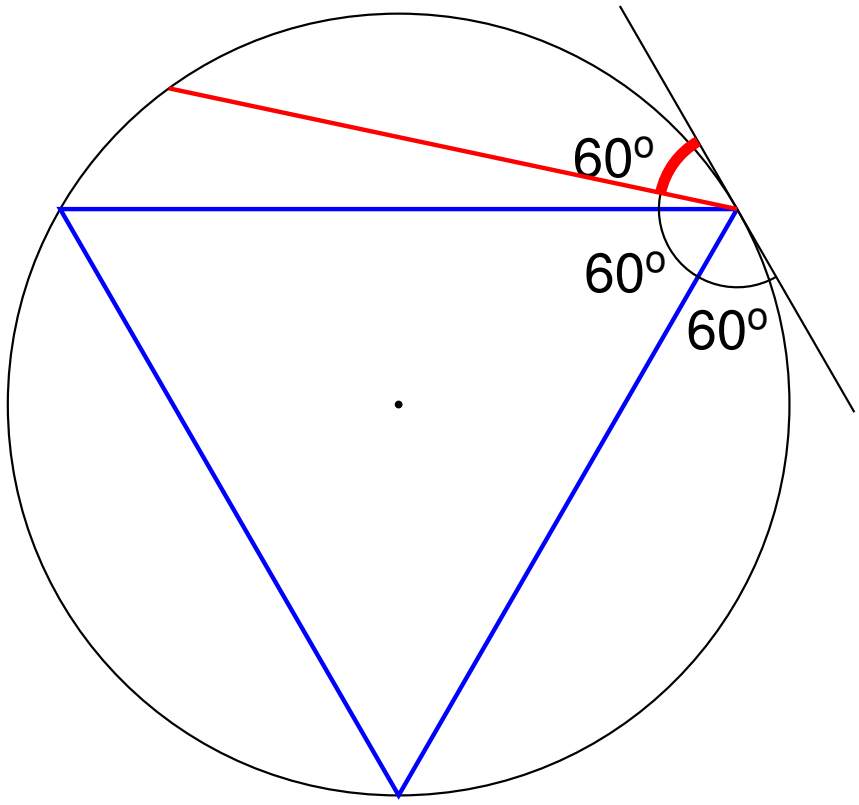
Angle:

Wait. . . it's $\frac{1}{3}$!



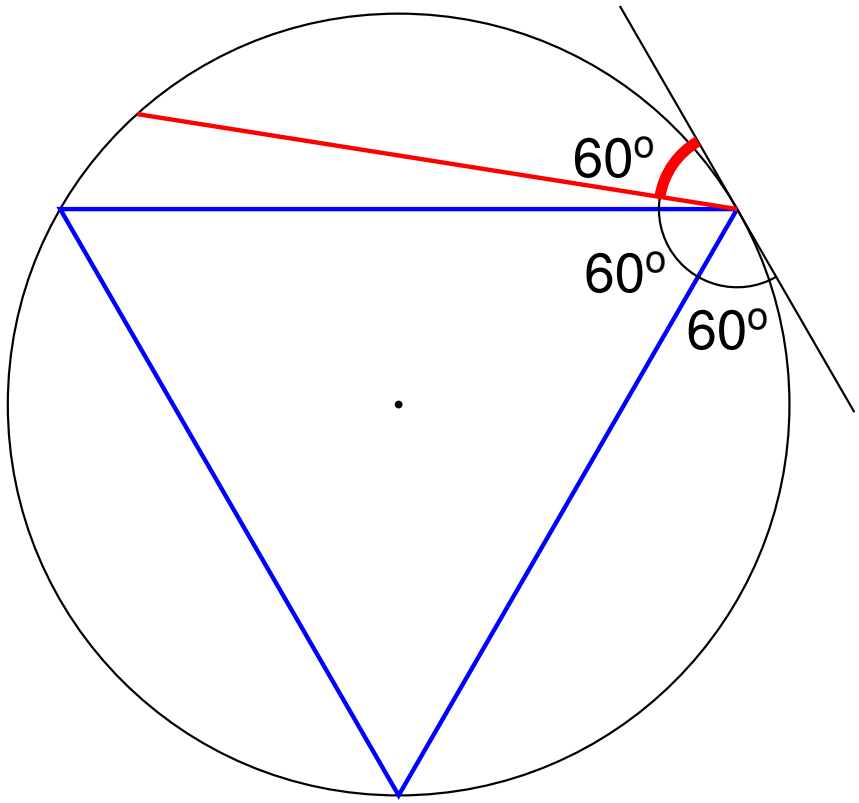
Angle:

Wait. . . it's $\frac{1}{3}$!



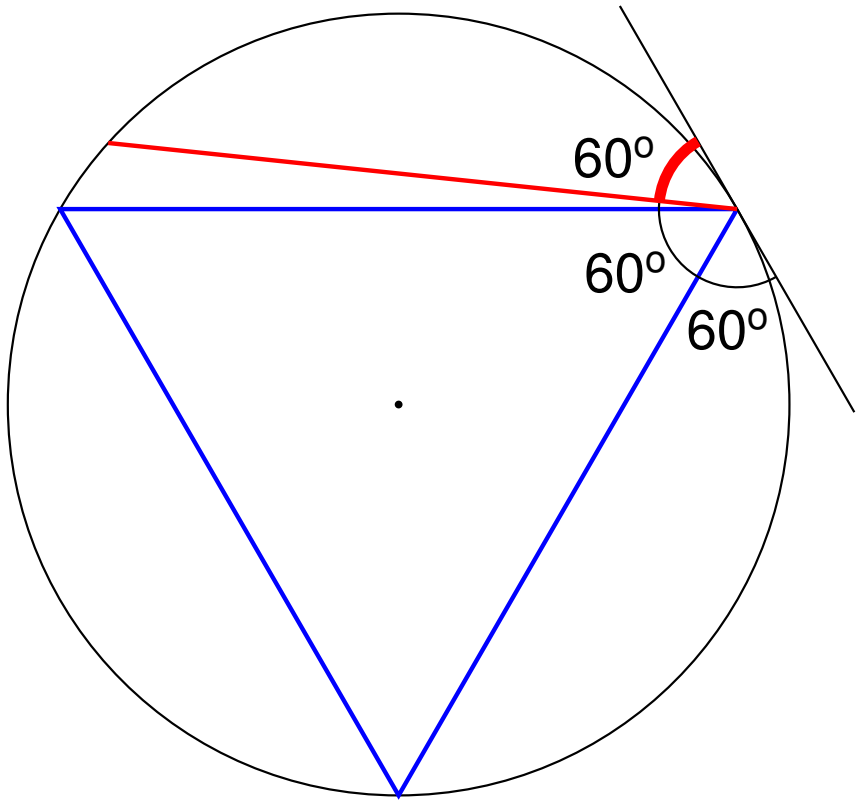
Angle:

Wait. . . it's $\frac{1}{3}$!



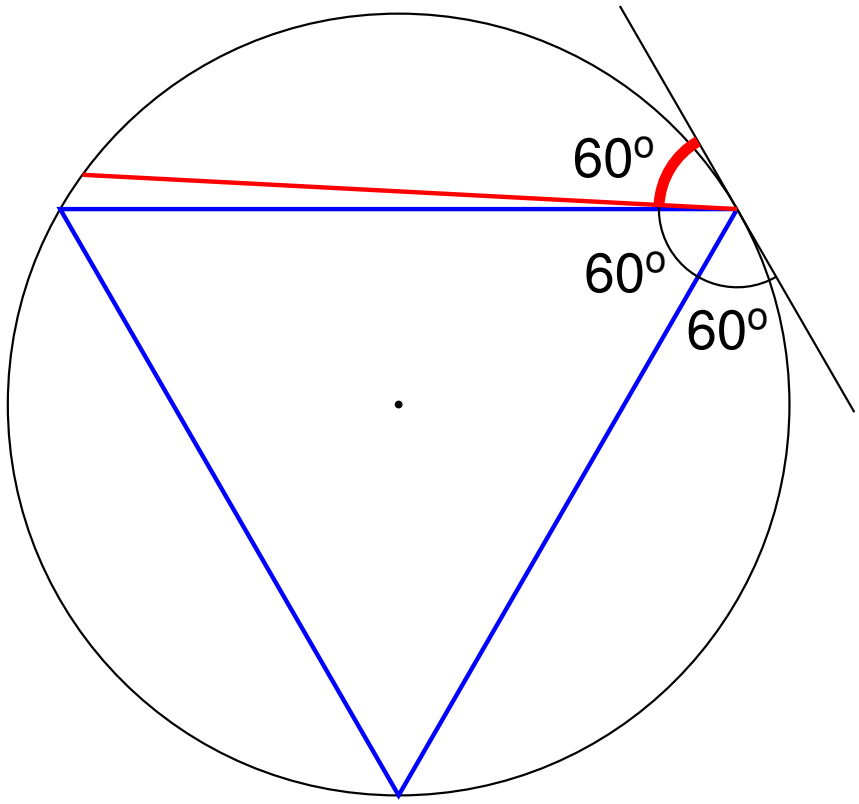
Angle:

Wait. . . it's $\frac{1}{3}$!



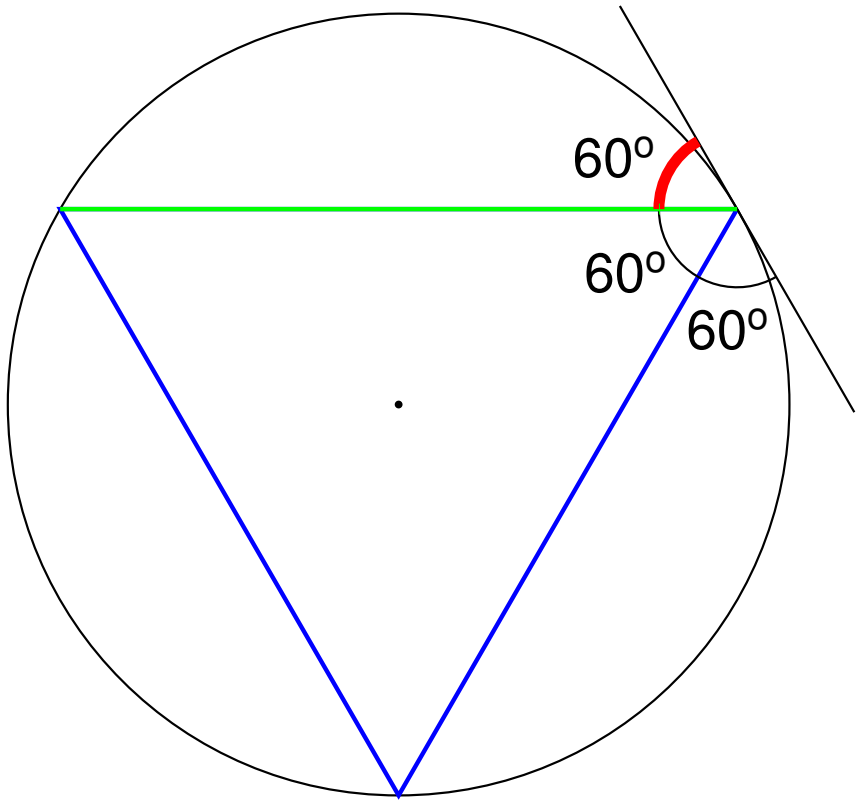
Angle:

Wait. . . it's $\frac{1}{3}$!



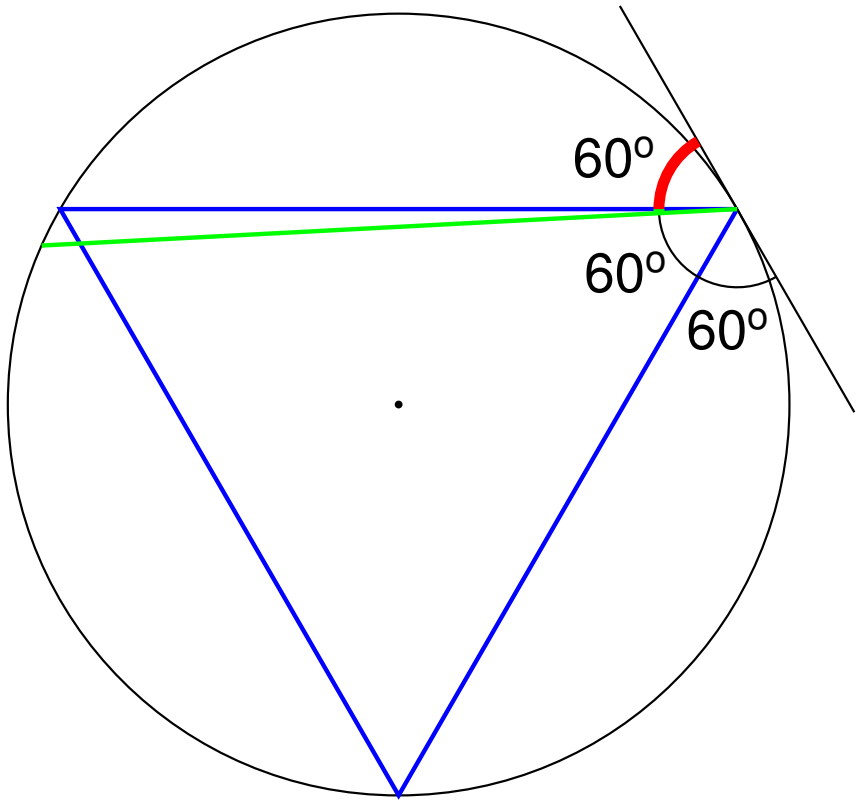
Angle:

Wait. . . it's $\frac{1}{3}$!



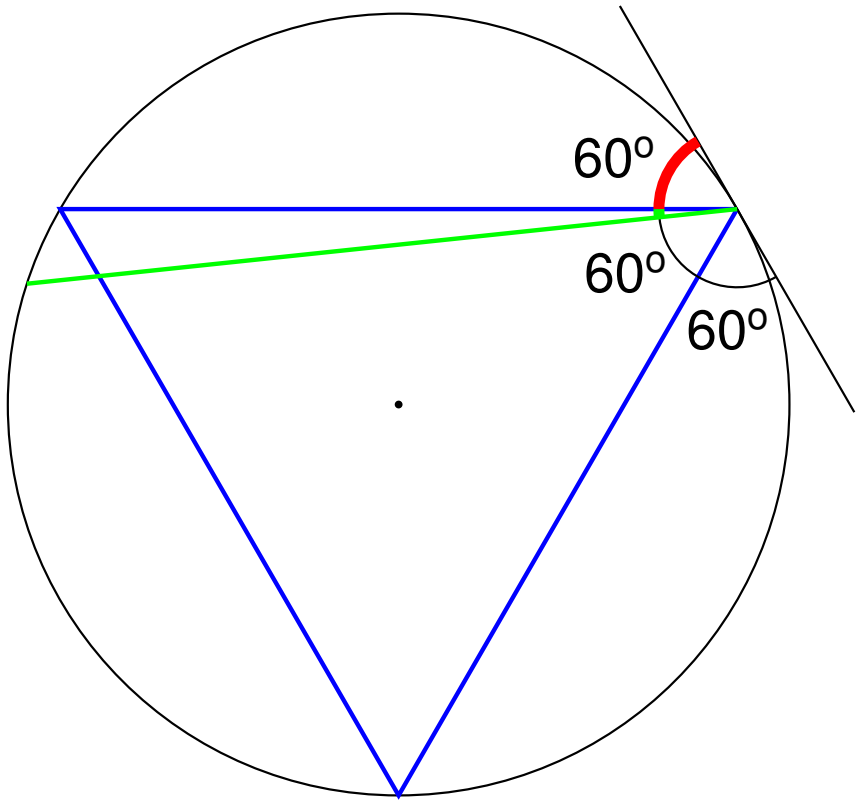
Angle:

Wait. . . it's $\frac{1}{3}$!



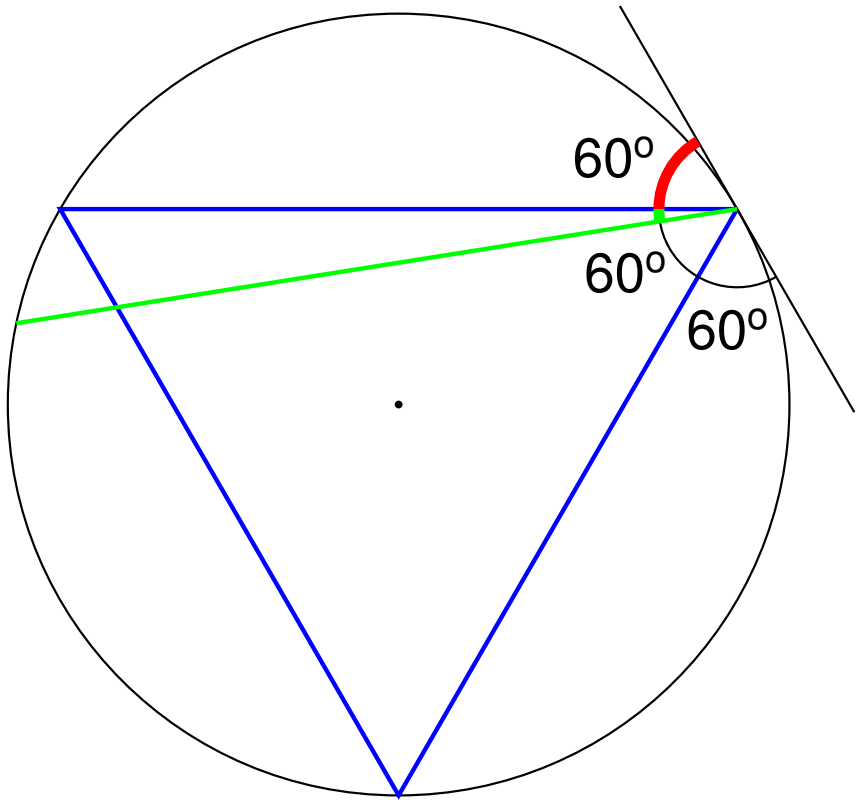
Angle:

Wait. . . it's $\frac{1}{3}$!



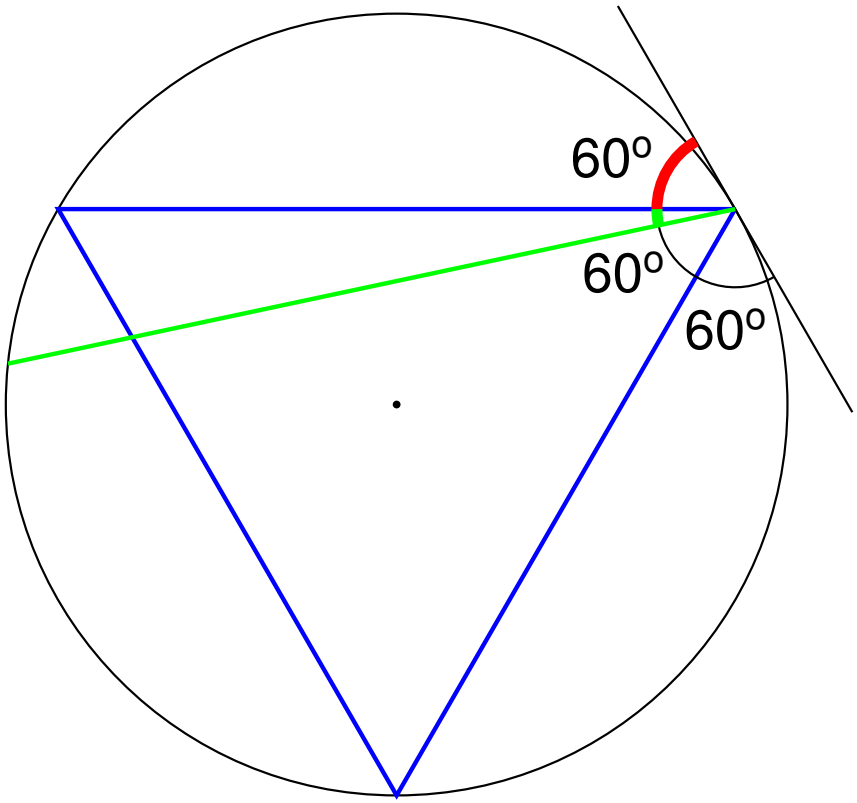
Angle:

Wait. . . it's $\frac{1}{3}$!



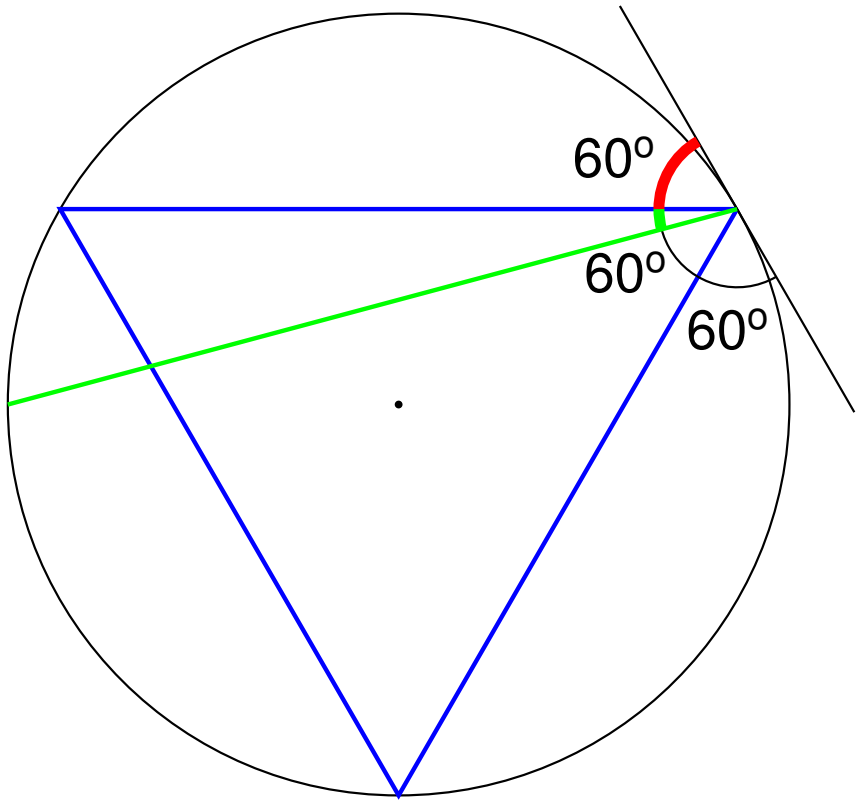
Angle:

Wait. . . it's $\frac{1}{3}$!



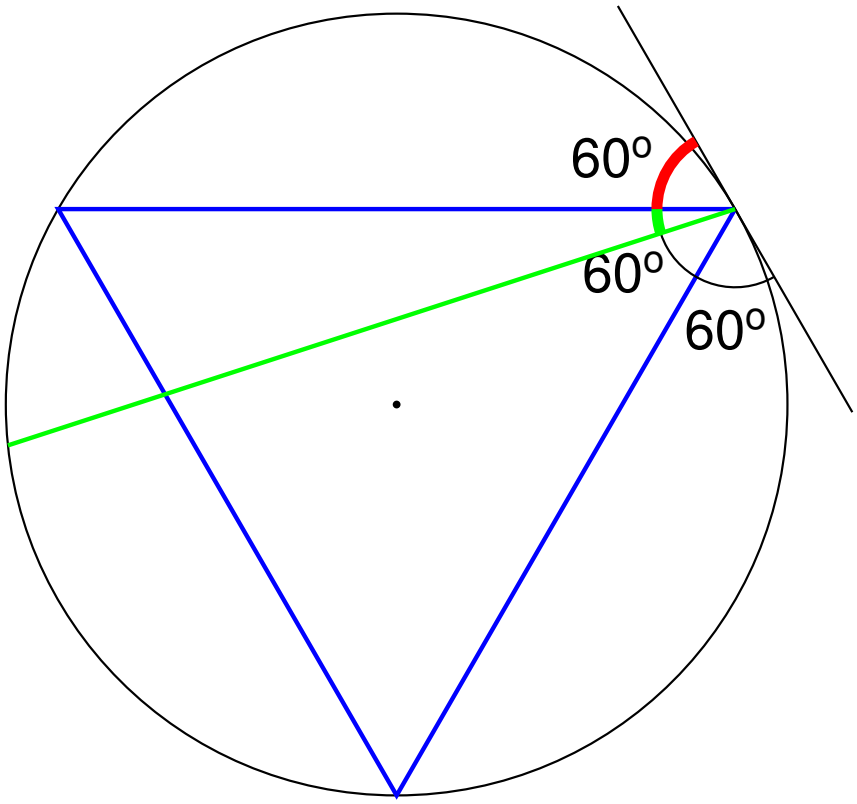
Angle:

Wait. . . it's $\frac{1}{3}$!



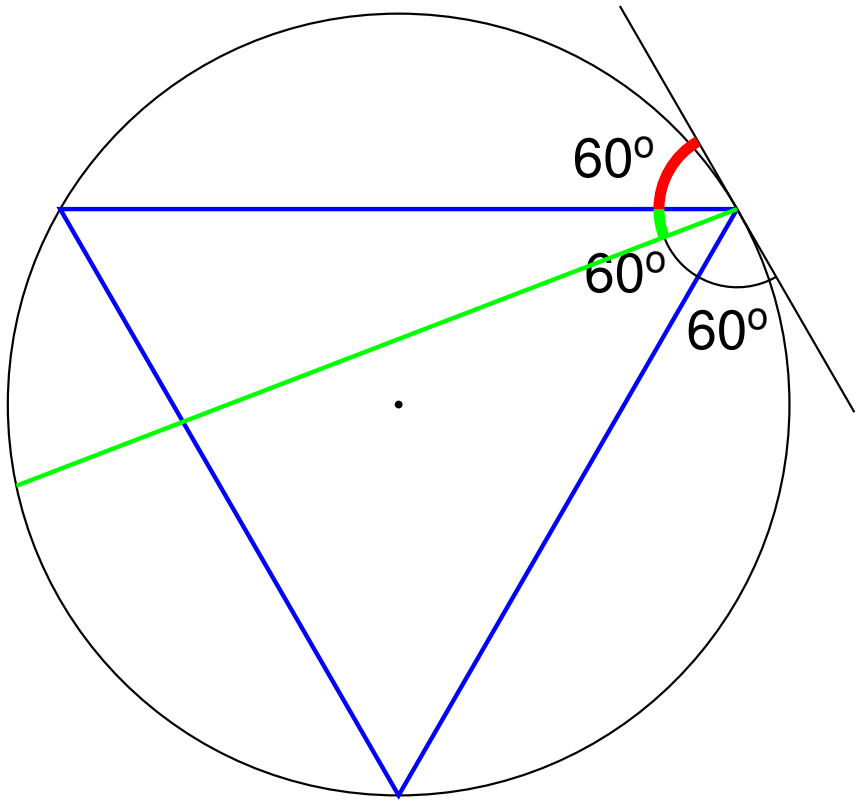
Angle:

Wait. . . it's $\frac{1}{3}$!



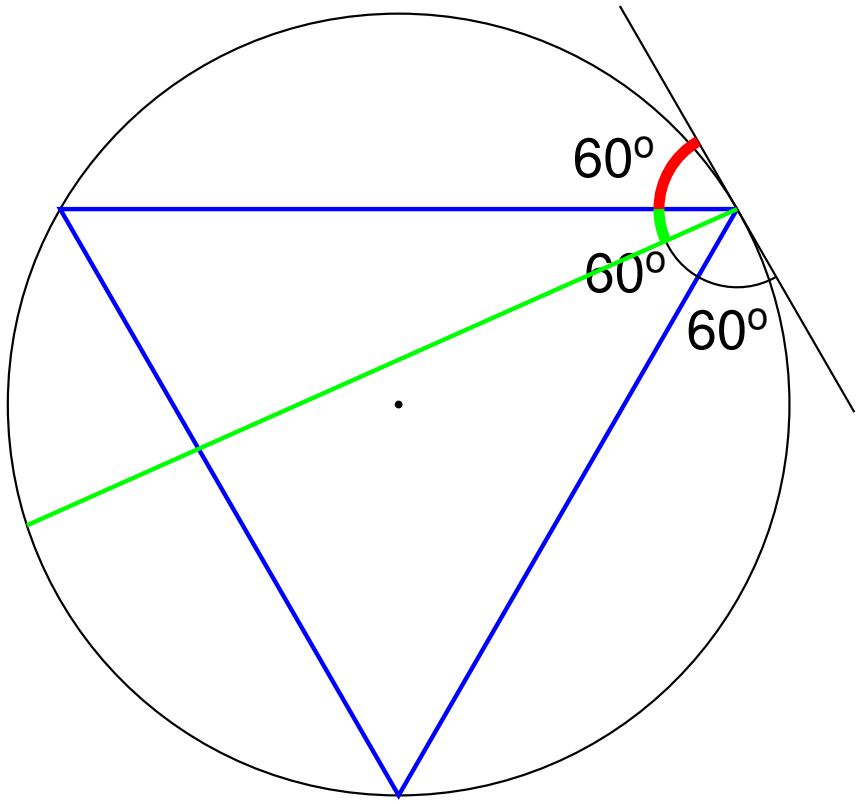
Angle:

Wait. . . it's $\frac{1}{3}$!



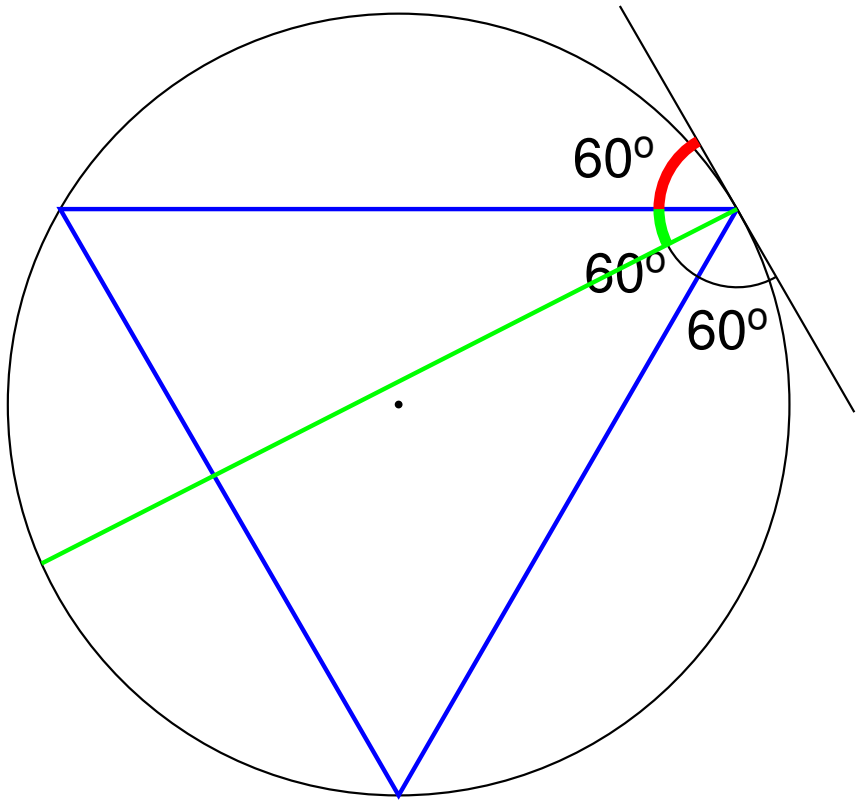
Angle:

Wait. . . it's $\frac{1}{3}$!



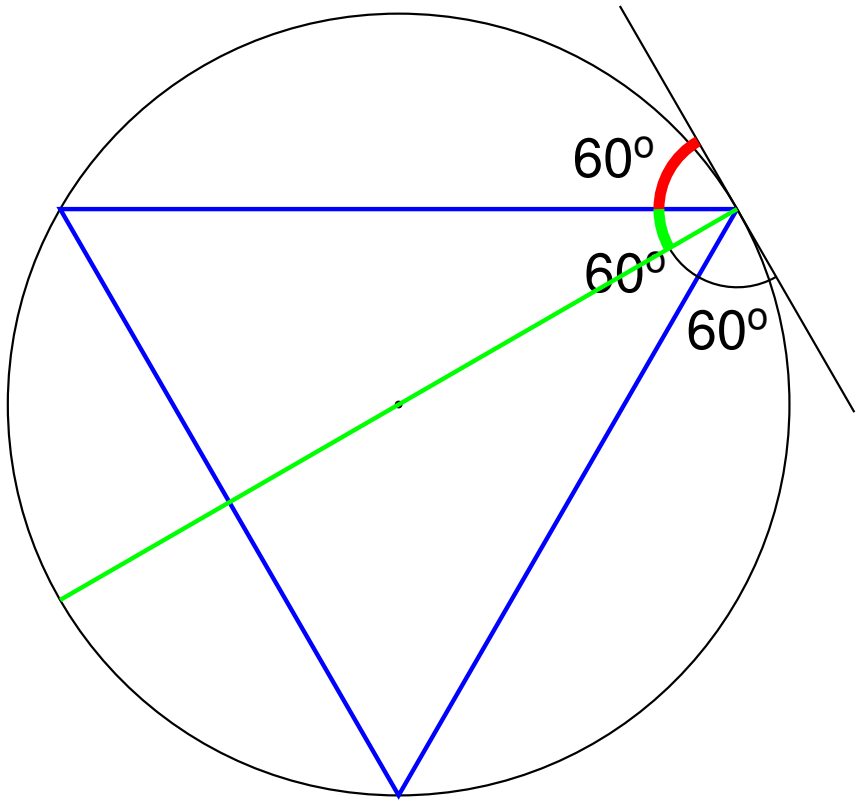
Angle:

Wait. . . it's $\frac{1}{3}$!



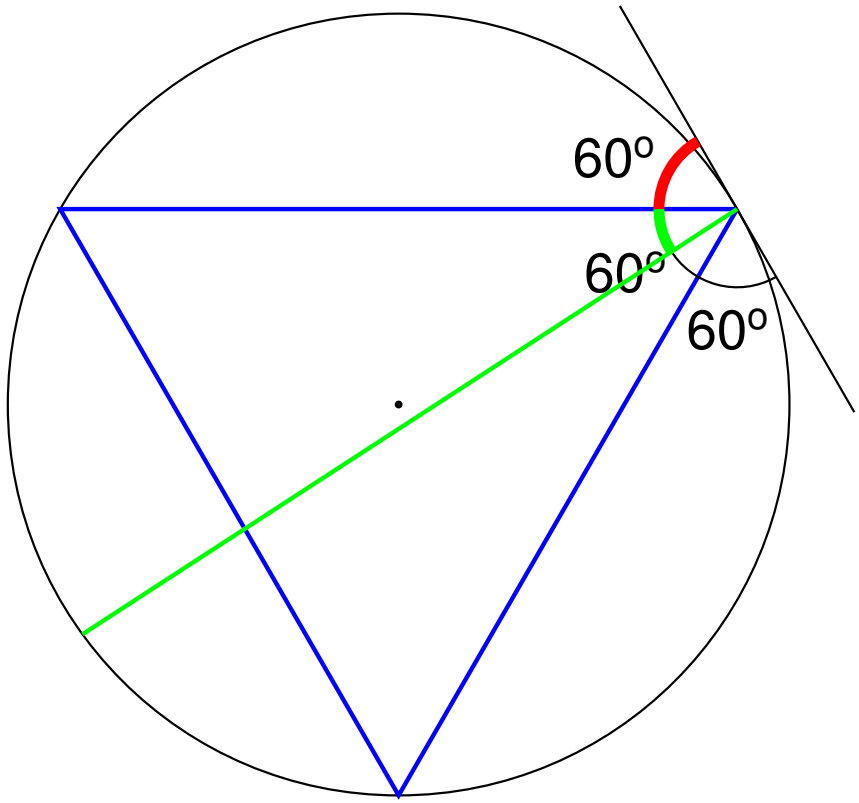
Angle:

Wait. . . it's $\frac{1}{3}$!



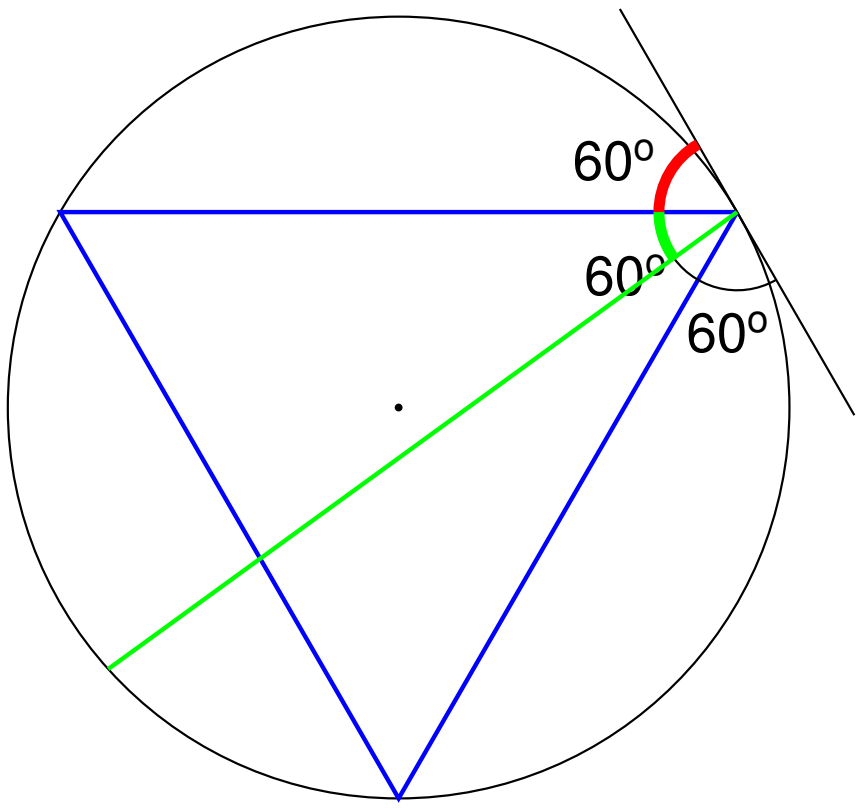
Angle:

Wait. . . it's $\frac{1}{3}$!



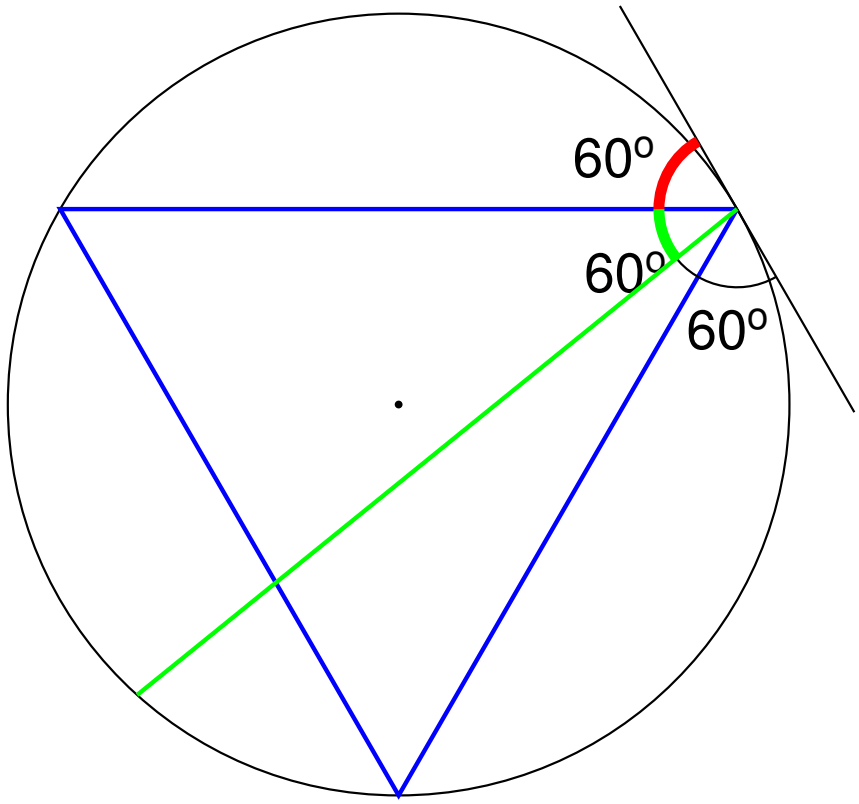
Angle:

Wait. . . it's $\frac{1}{3}$!



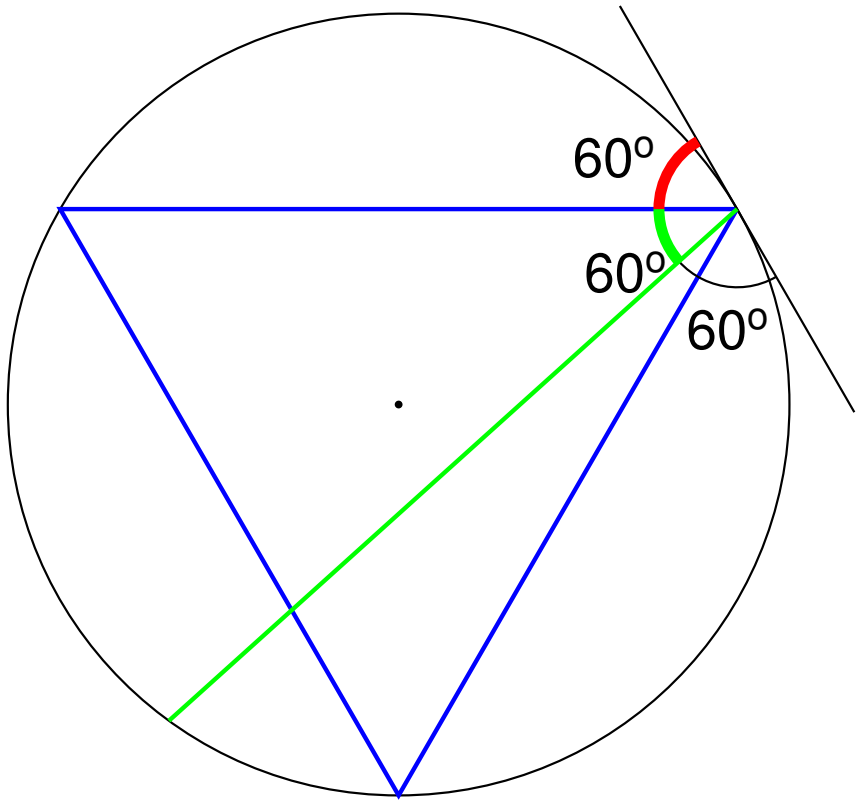
Angle:

Wait. . . it's $\frac{1}{3}$!



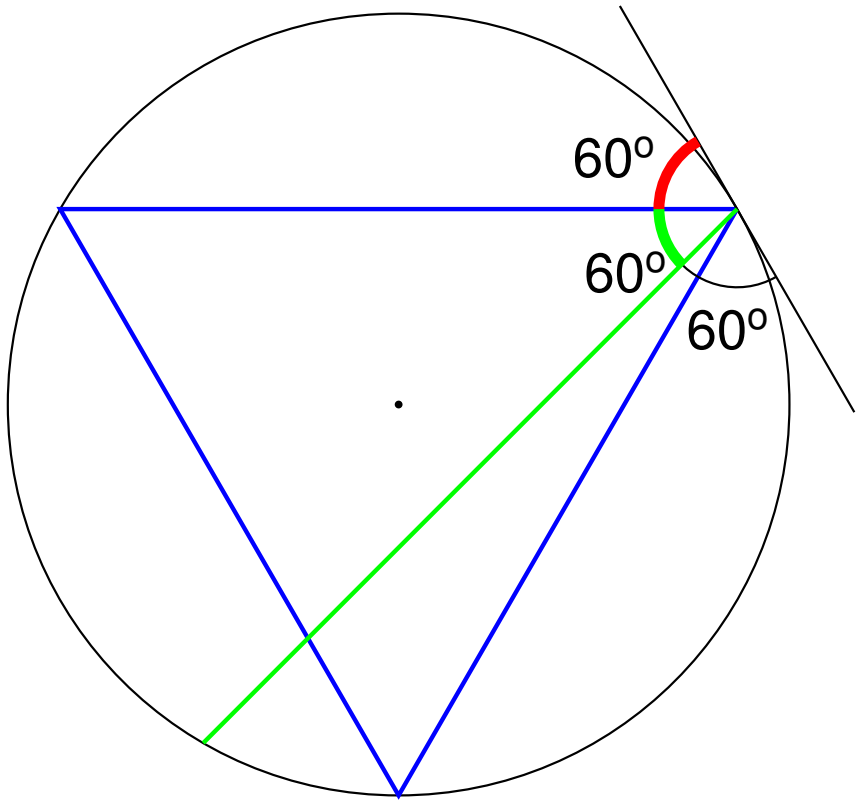
Angle:

Wait. . . it's $\frac{1}{3}$!



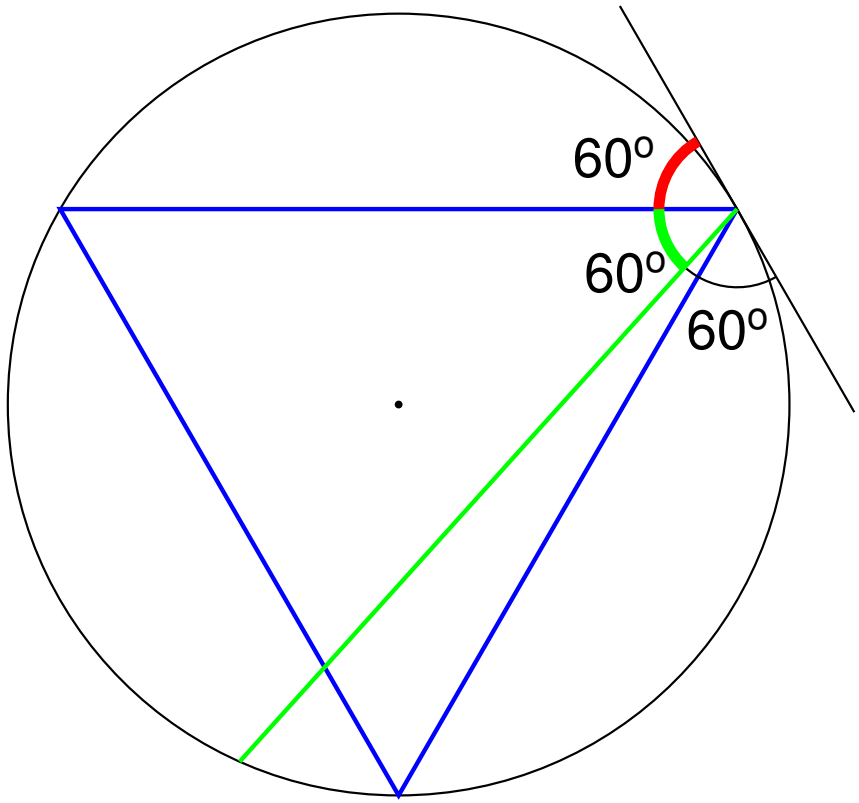
Angle:

Wait. . . it's $\frac{1}{3}$!



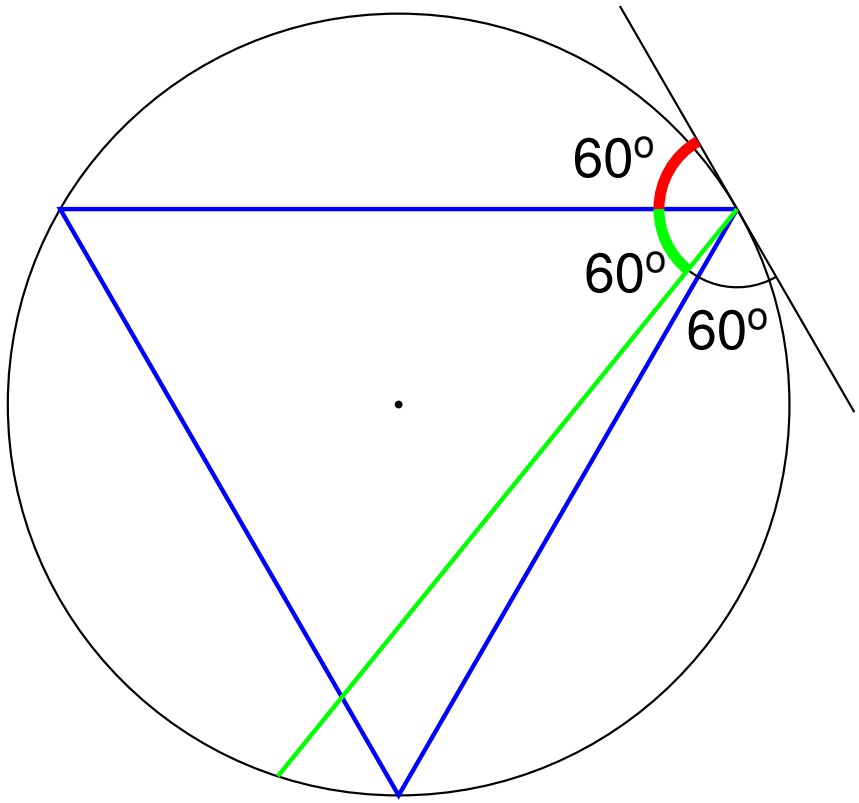
Angle:

Wait. . . it's $\frac{1}{3}$!



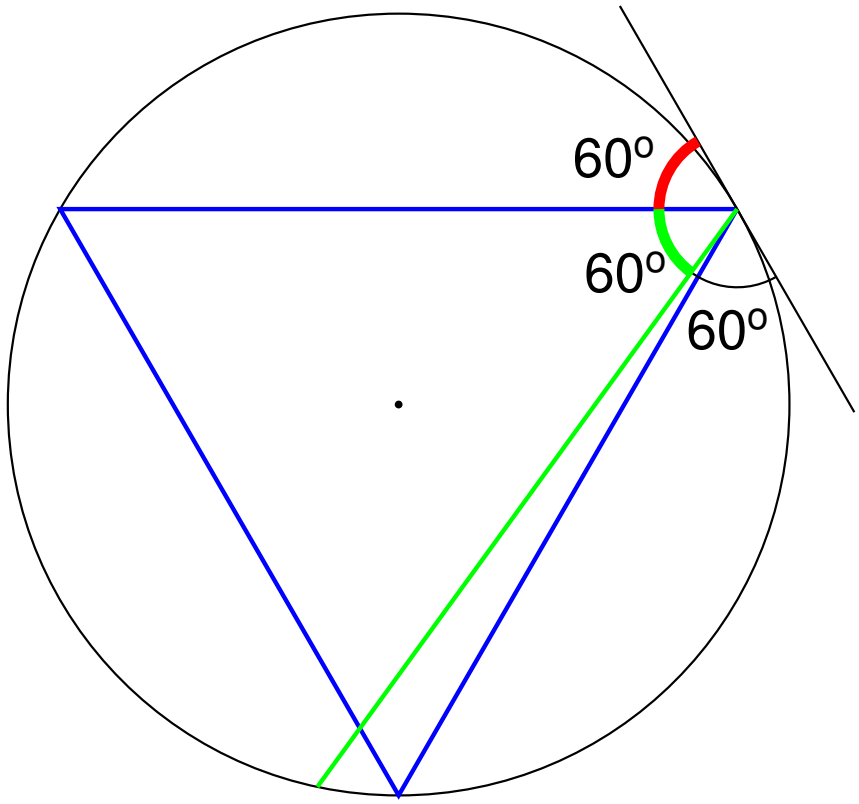
Angle:

Wait. . . it's $\frac{1}{3}$!



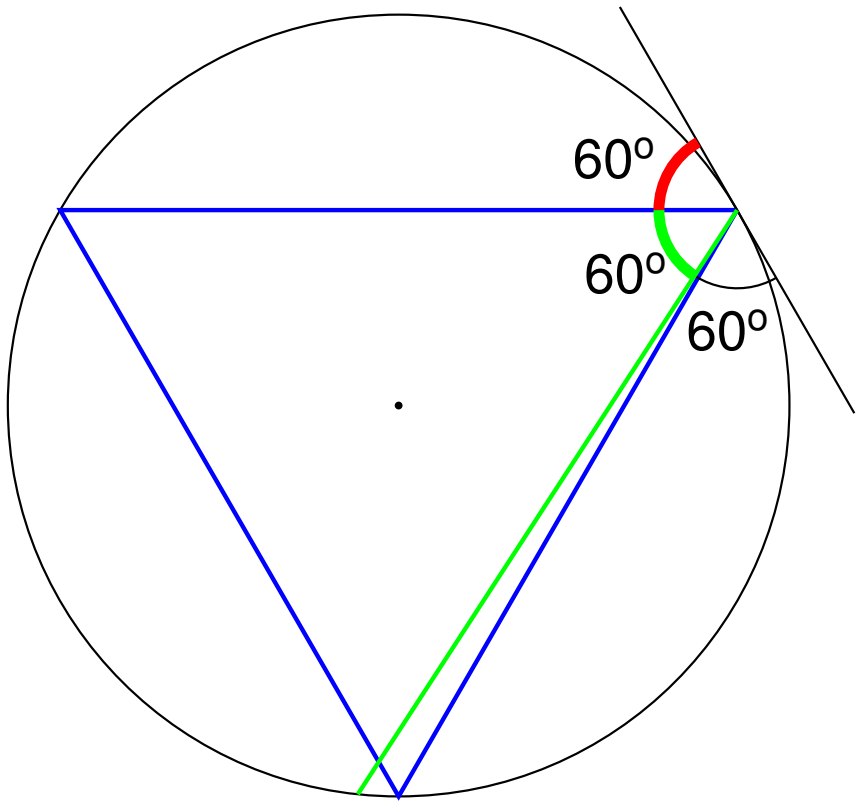
Angle:

Wait. . . it's $\frac{1}{3}$!



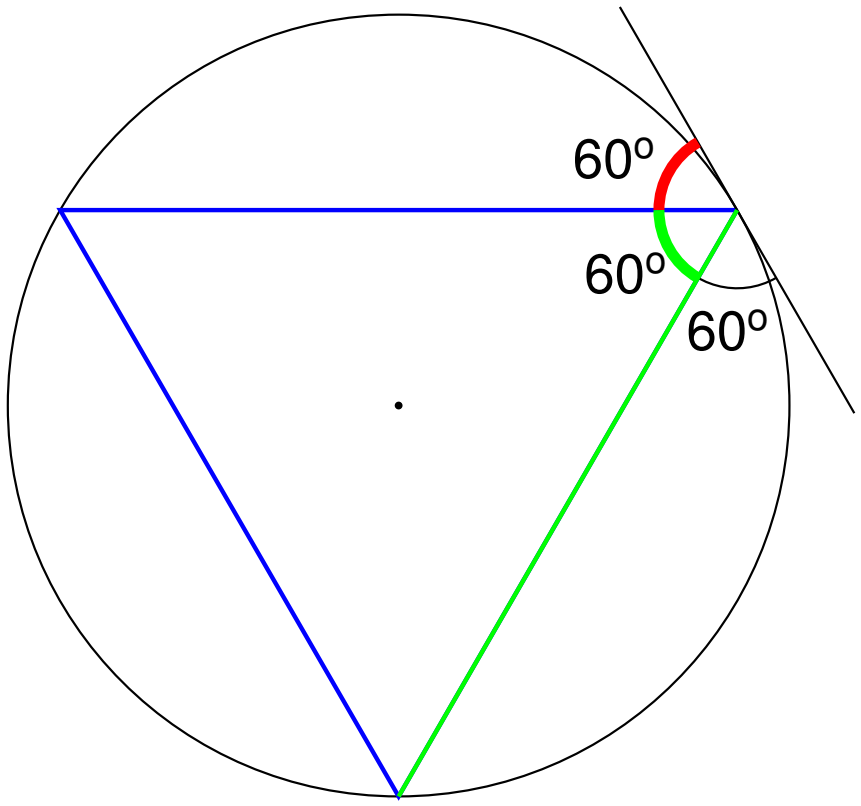
Angle:

Wait. . . it's $\frac{1}{3}$!



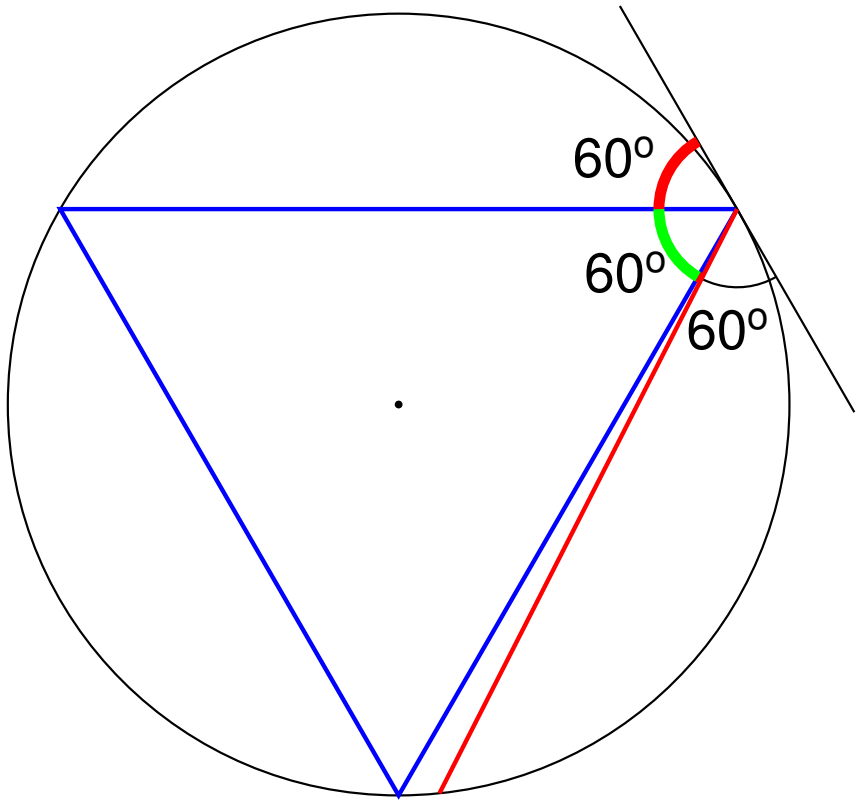
Angle:

Wait. . . it's $\frac{1}{3}$!



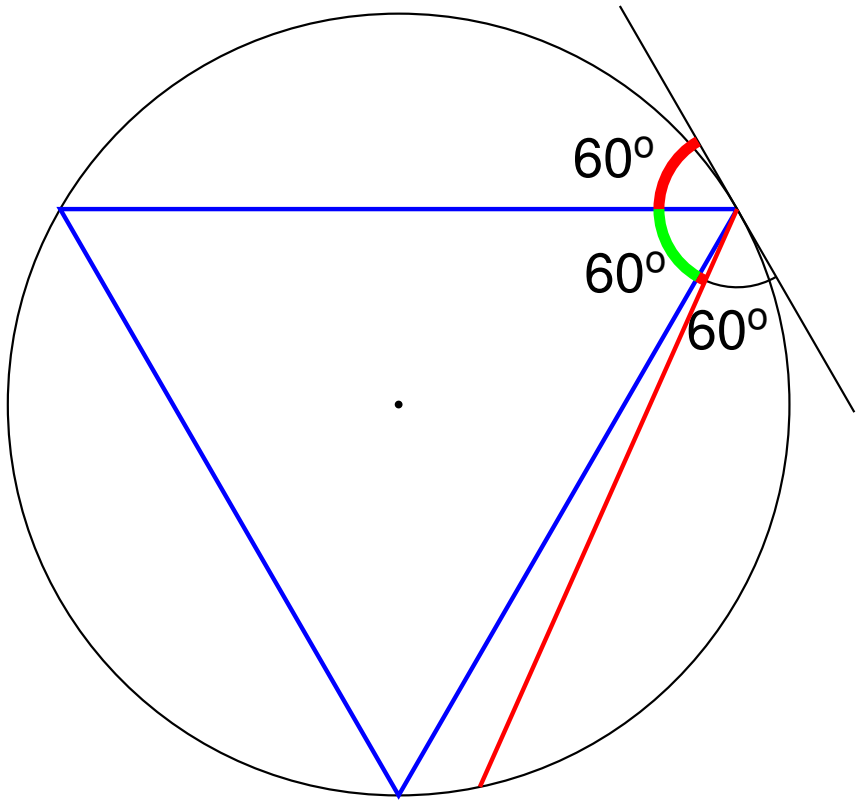
Angle:

Wait. . . it's $\frac{1}{3}$!



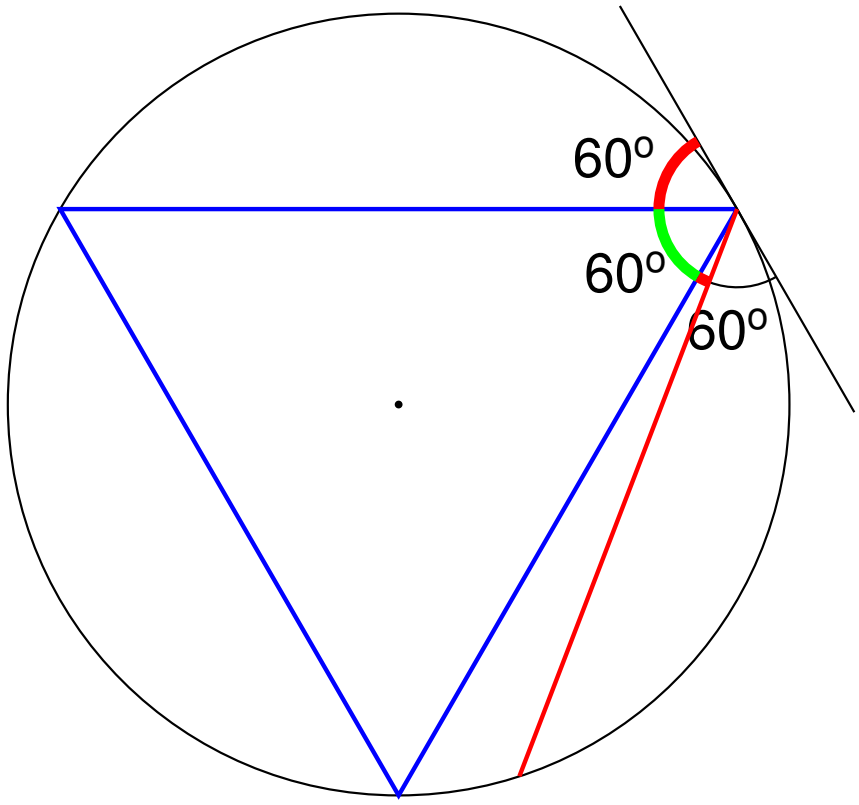
Angle:

Wait. . . it's $\frac{1}{3}$!



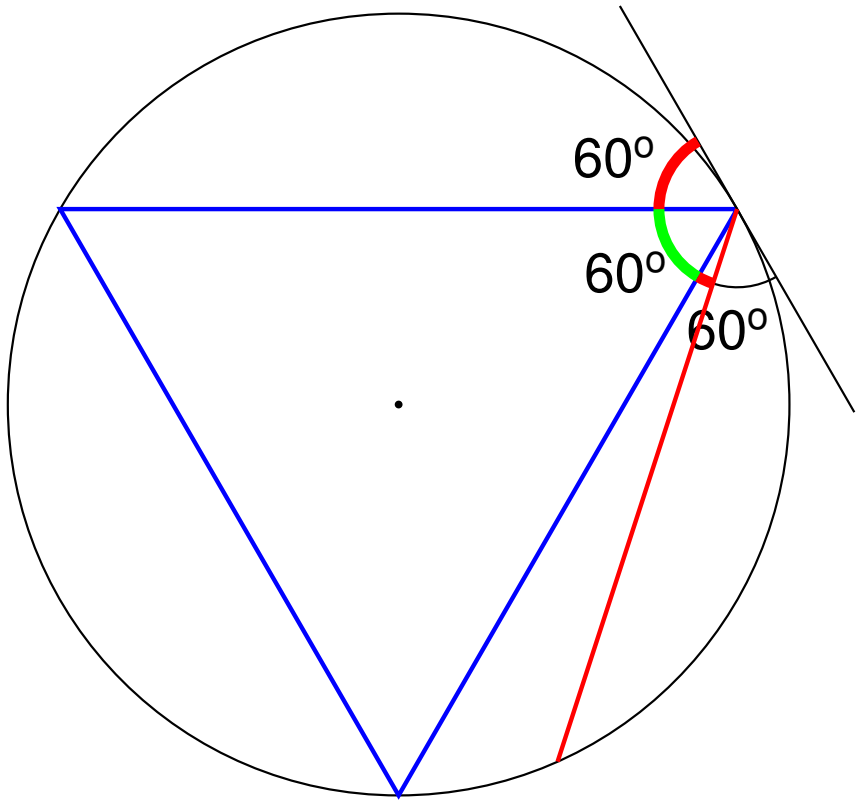
Angle:

Wait. . . it's $\frac{1}{3}$!



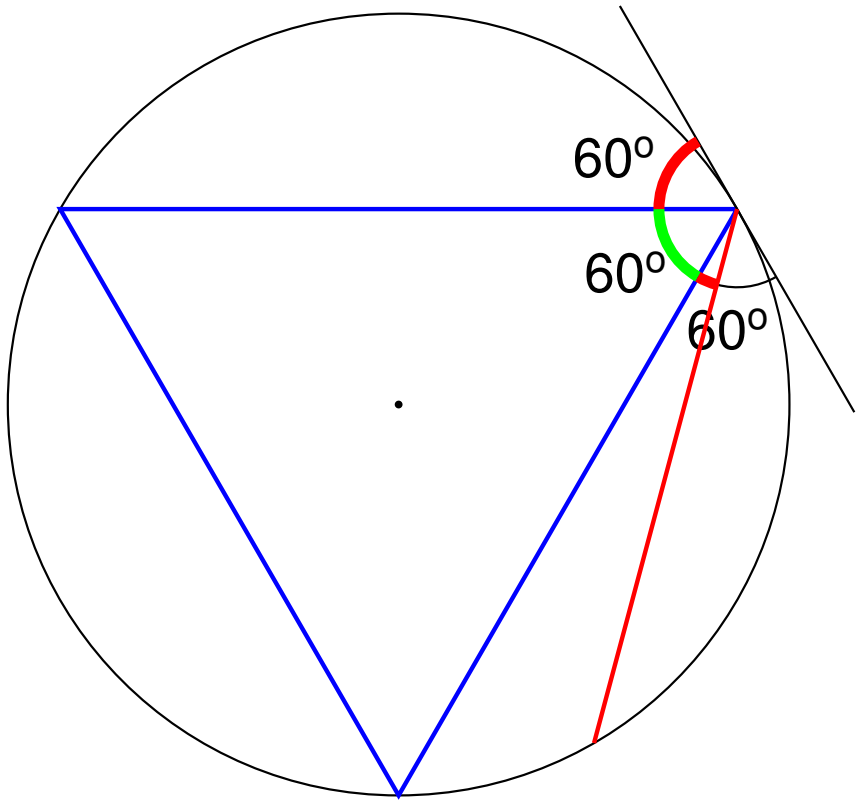
Angle:

Wait. . . it's $\frac{1}{3}$!



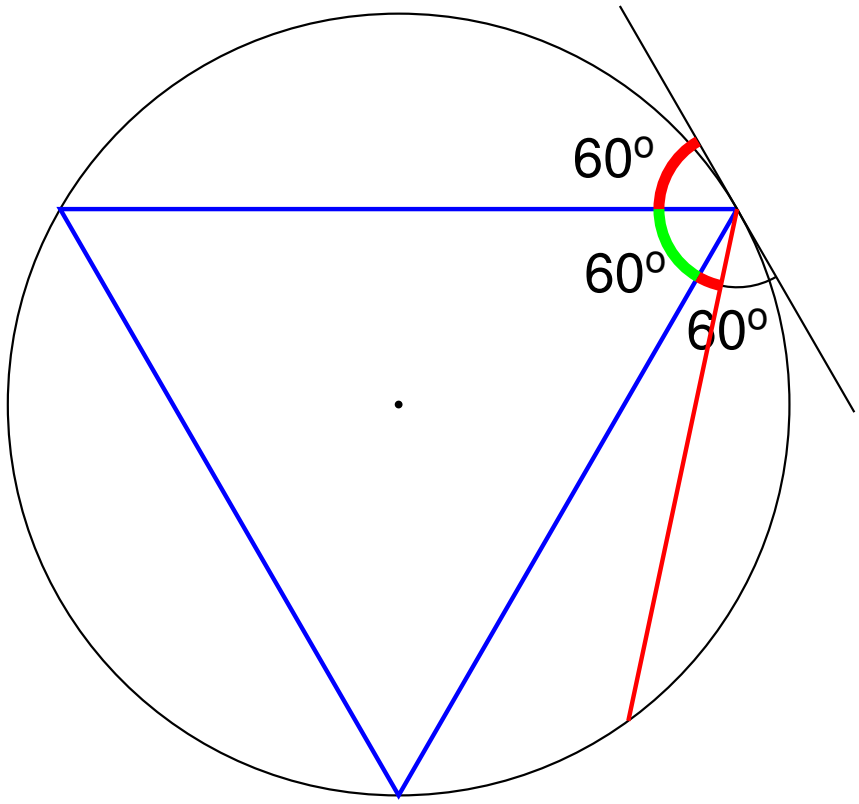
Angle:

Wait. . . it's $\frac{1}{3}$!



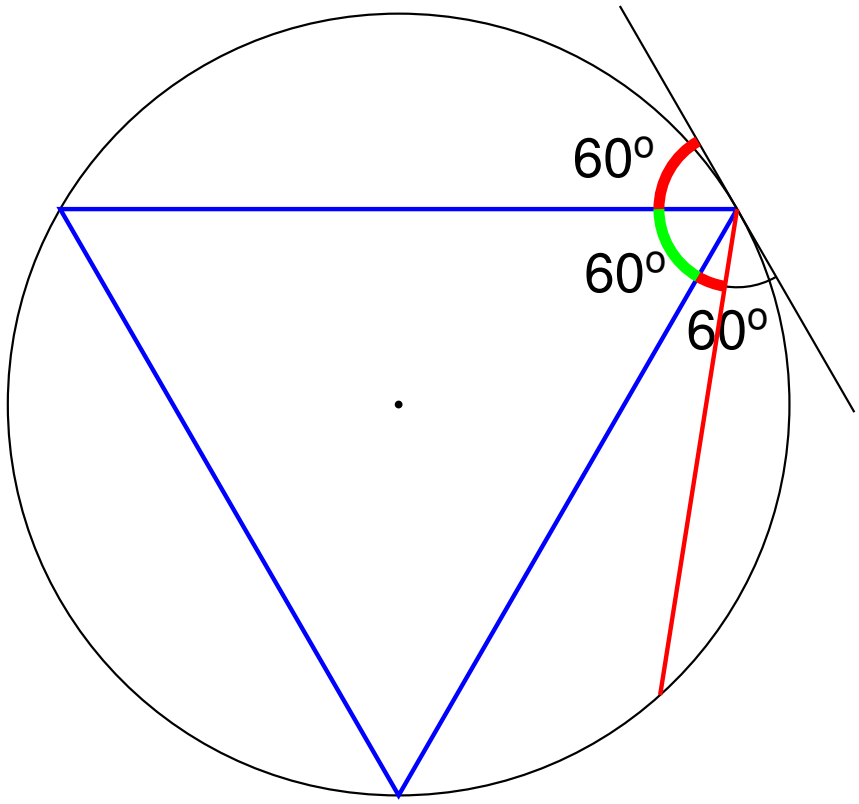
Angle:

Wait. . . it's $\frac{1}{3}$!



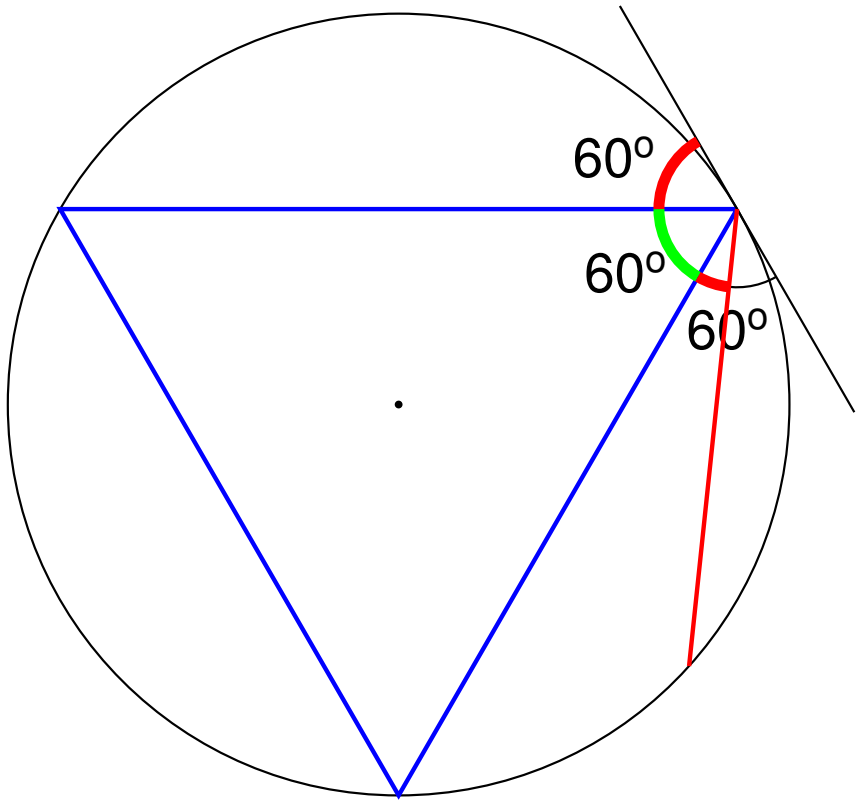
Angle:

Wait. . . it's $\frac{1}{3}$!



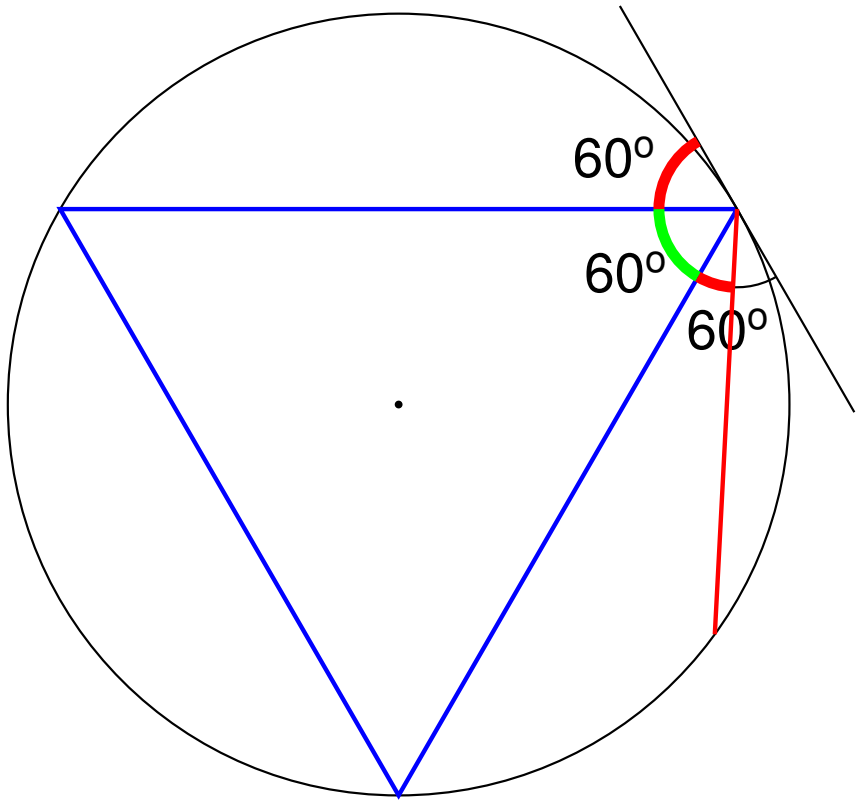
Angle:

Wait. . . it's $\frac{1}{3}$!



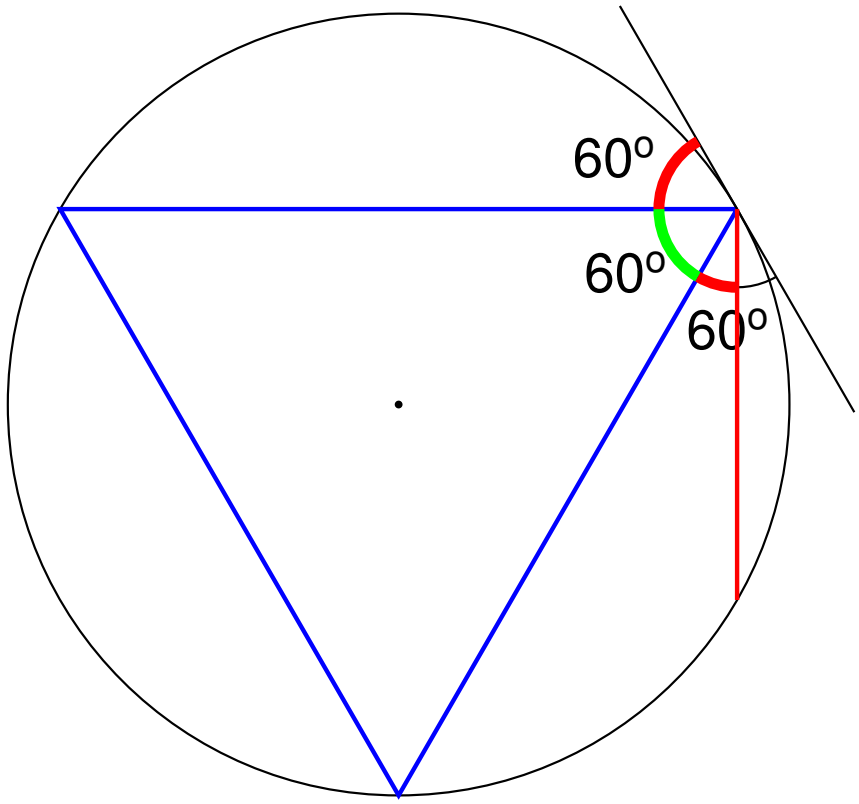
Angle:

Wait. . . it's $\frac{1}{3}$!



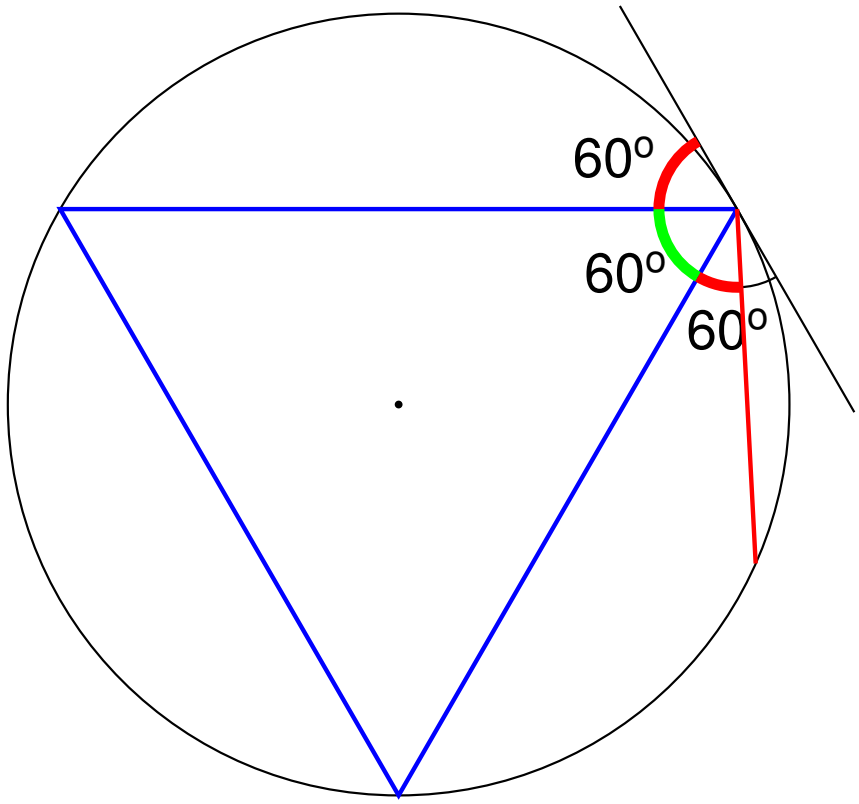
Angle:

Wait. . . it's $\frac{1}{3}$!



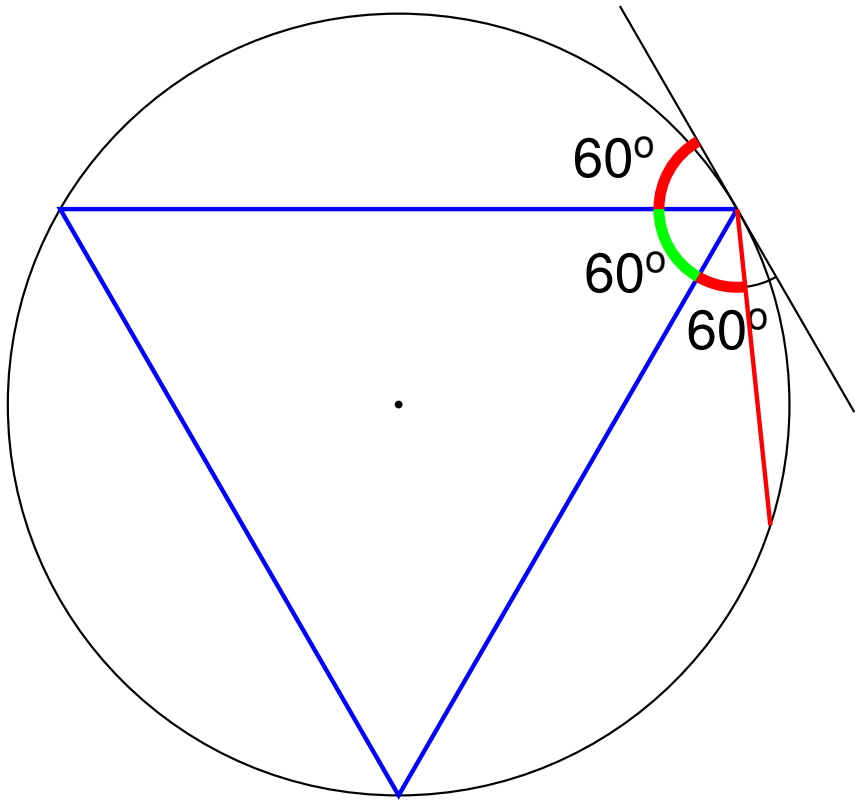
Angle:

Wait. . . it's $\frac{1}{3}$!



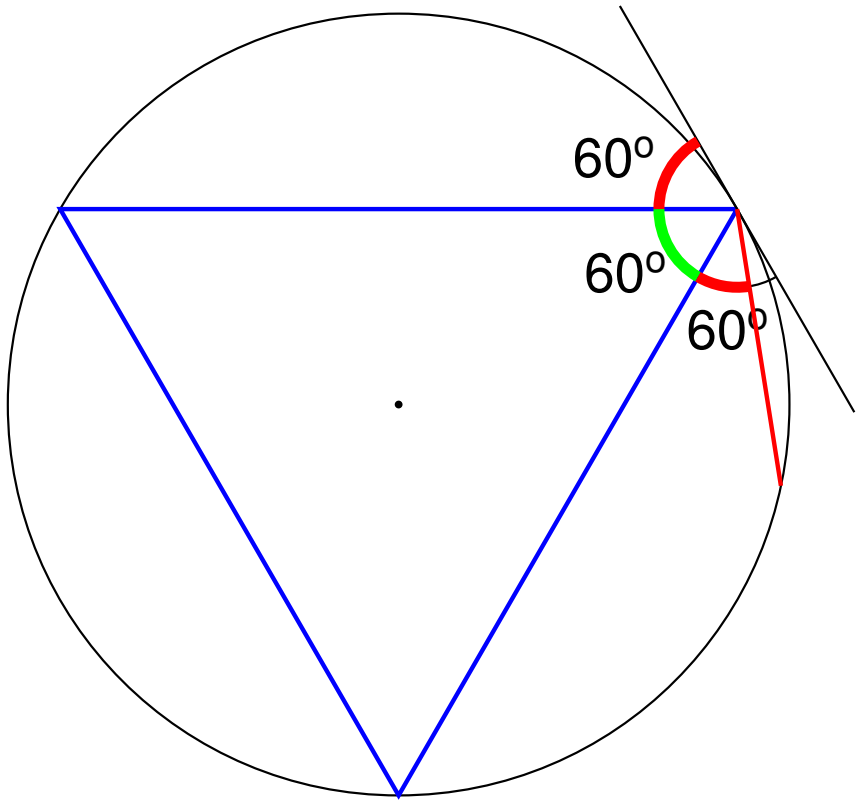
Angle:

Wait. . . it's $\frac{1}{3}$!



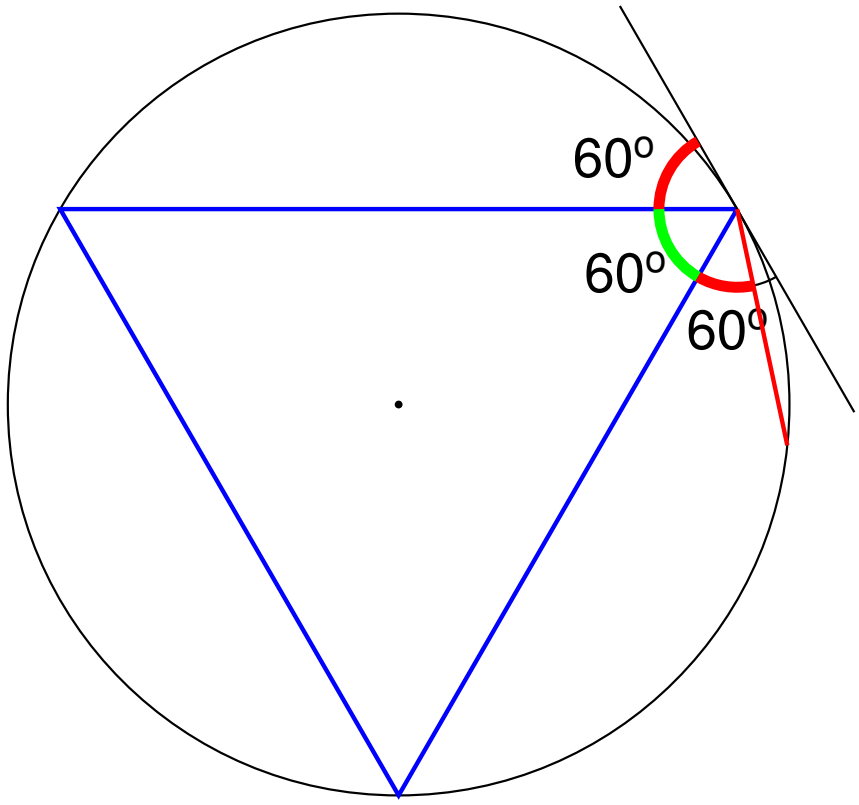
Angle:

Wait. . . it's $\frac{1}{3}$!



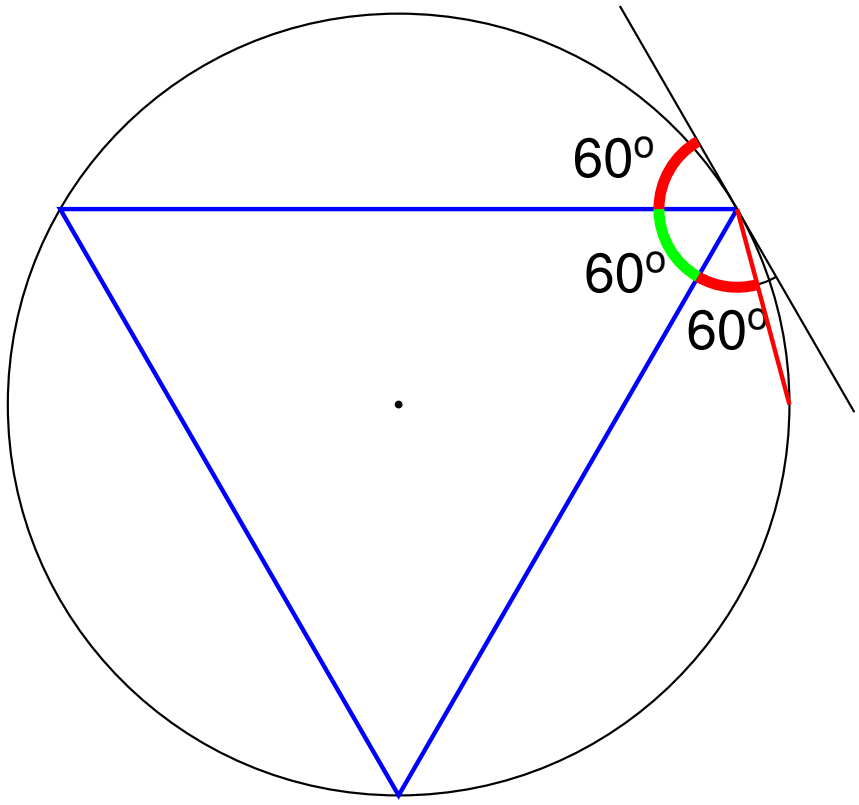
Angle:

Wait. . . it's $\frac{1}{3}$!



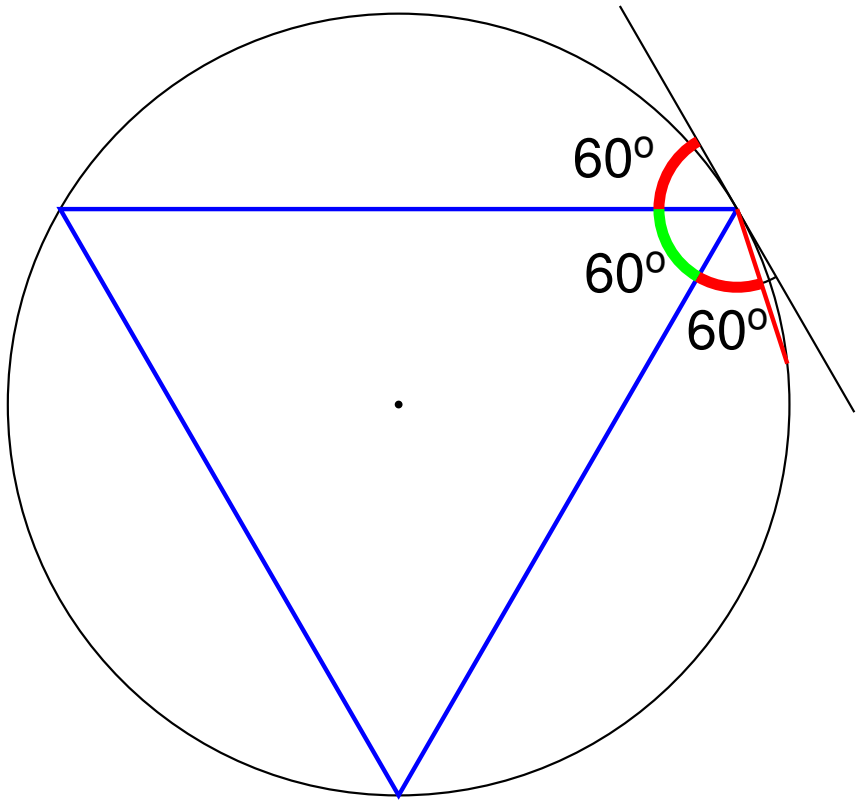
Angle:

Wait. . . it's $\frac{1}{3}$!



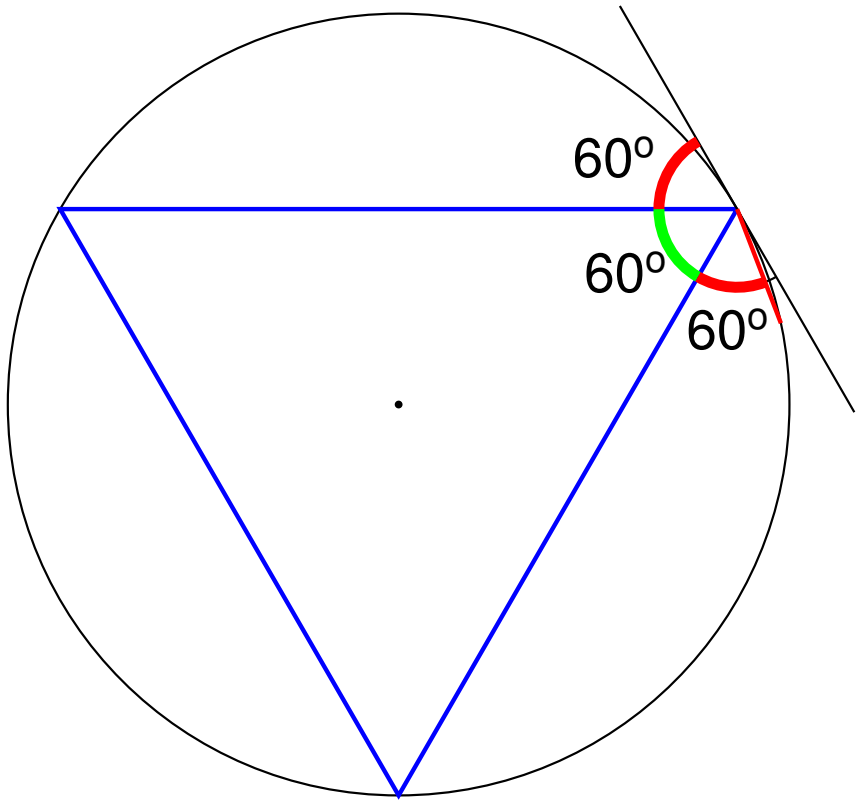
Angle:

Wait. . . it's $\frac{1}{3}$!



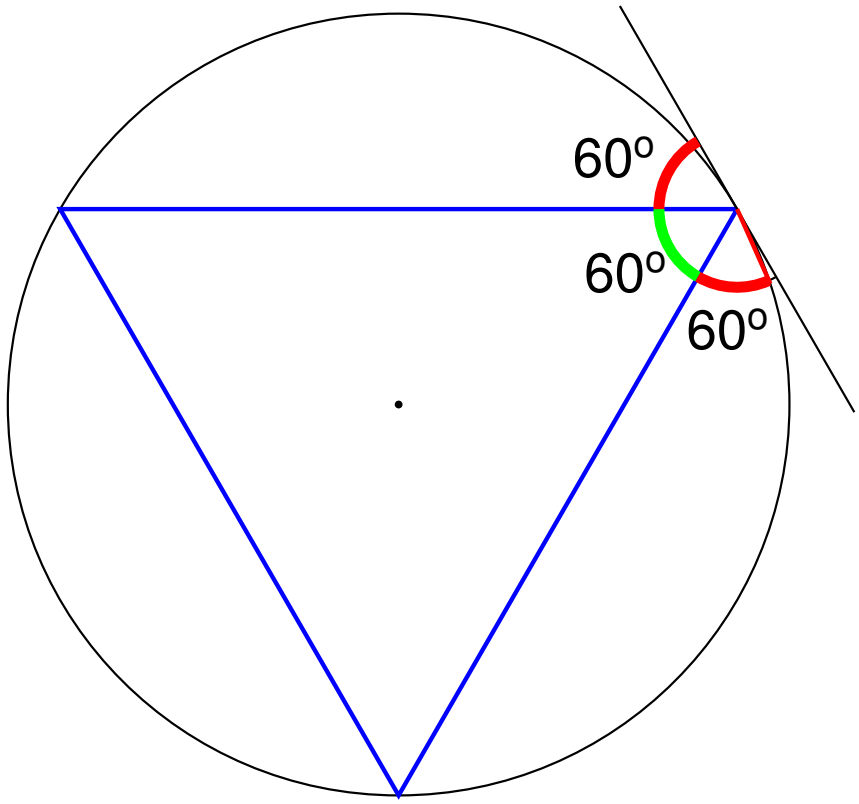
Angle:

Wait. . . it's $\frac{1}{3}$!



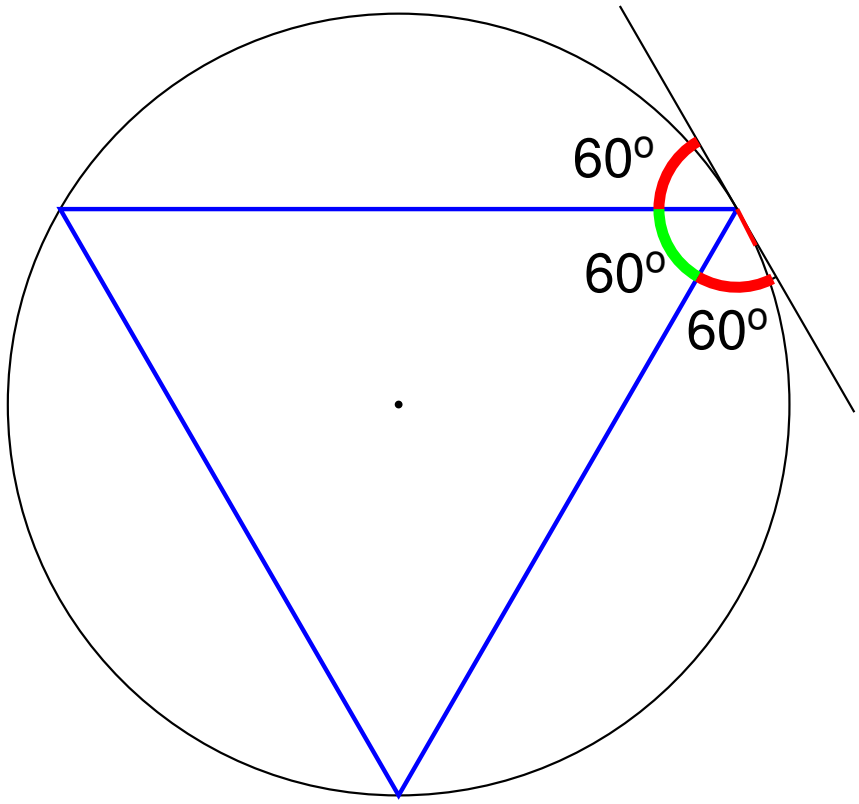
Angle:

Wait. . . it's $\frac{1}{3}$!



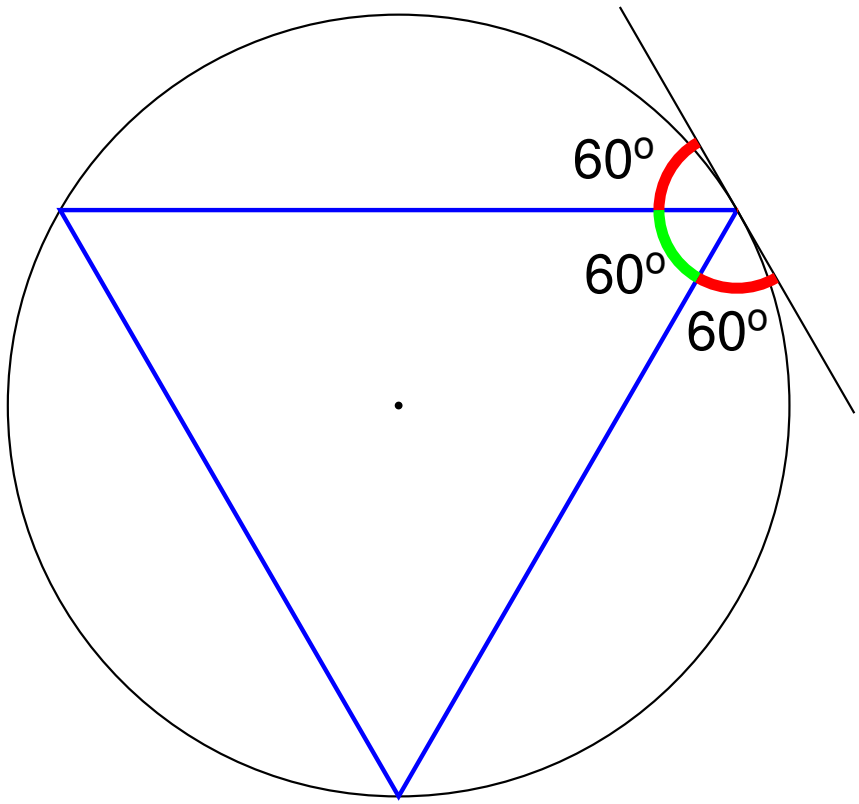
Angle:

Wait. . . it's $\frac{1}{3}$!



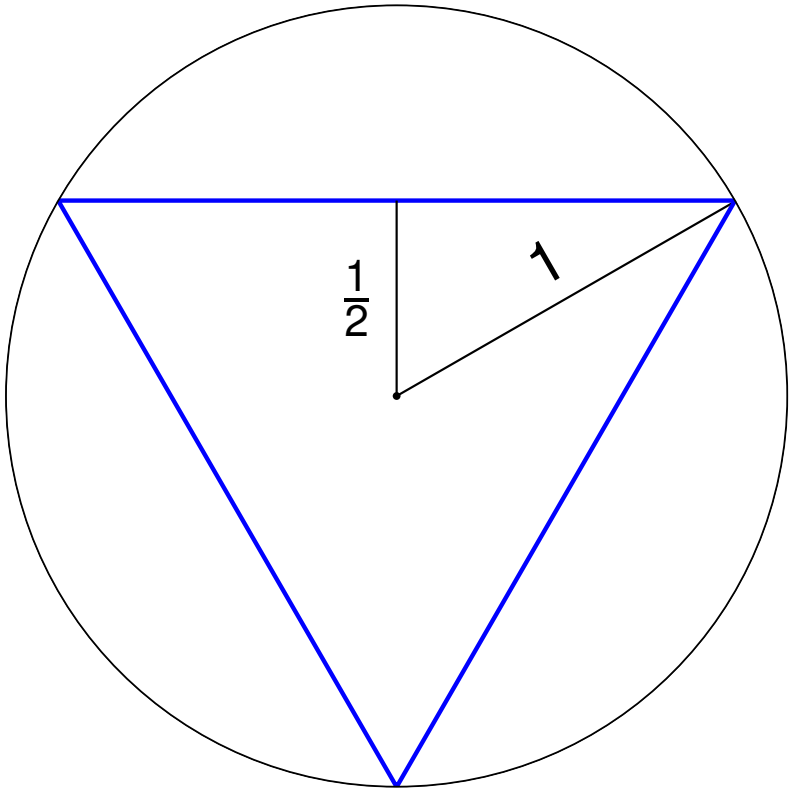
Angle:

Wait. . . it's $\frac{1}{3}$!

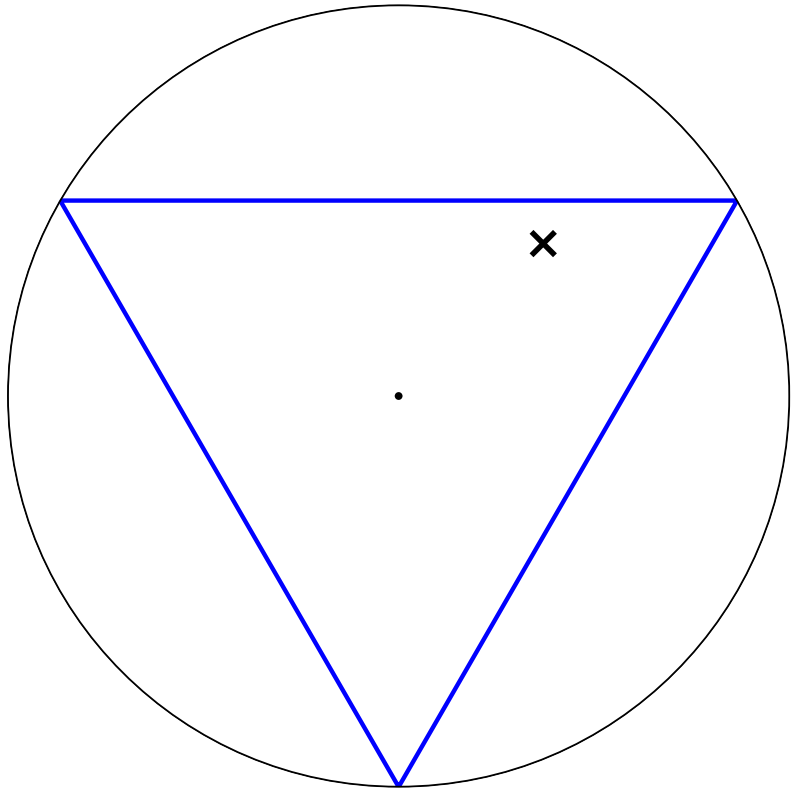


Angle: $\frac{60^\circ}{60^\circ+60^\circ+60^\circ} = \frac{1}{3}$.

Oh wait. . . it's $\frac{1}{4}$!

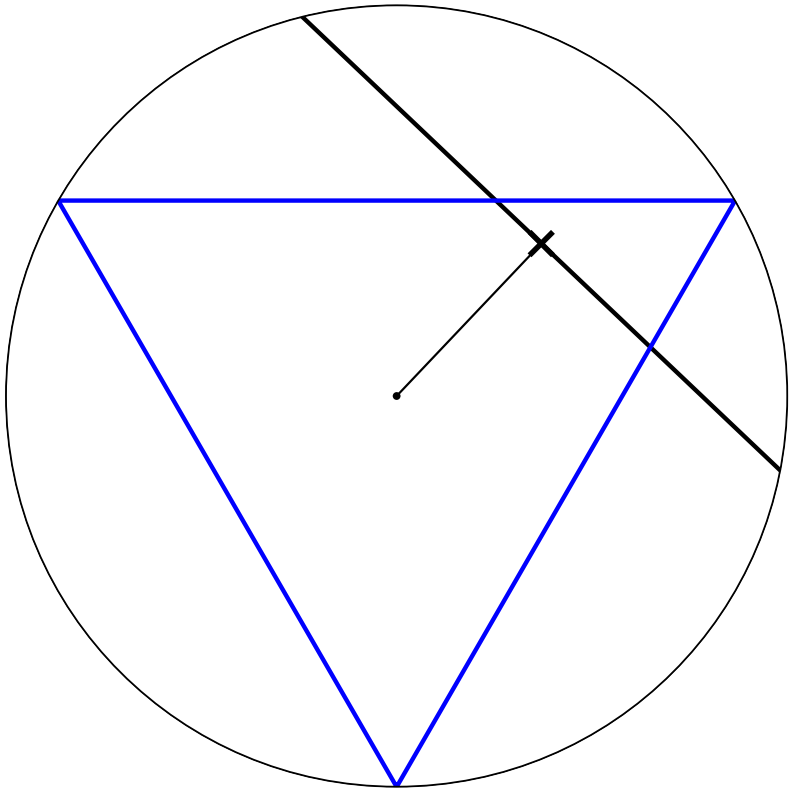


Oh wait. . . it's $\frac{1}{4}$!



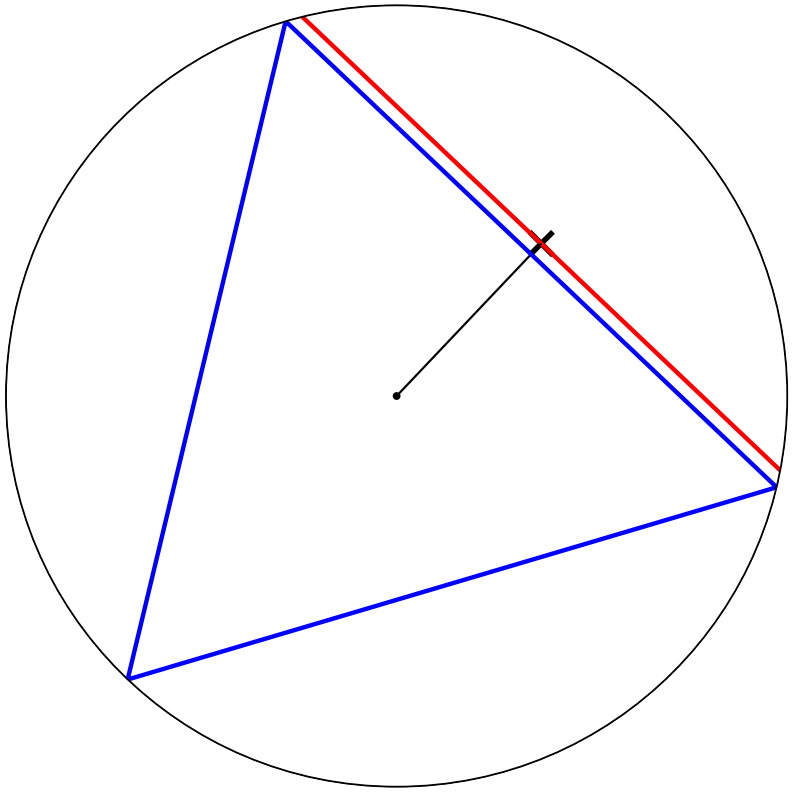
Midpoint:

Oh wait. . . it's $\frac{1}{4}$!



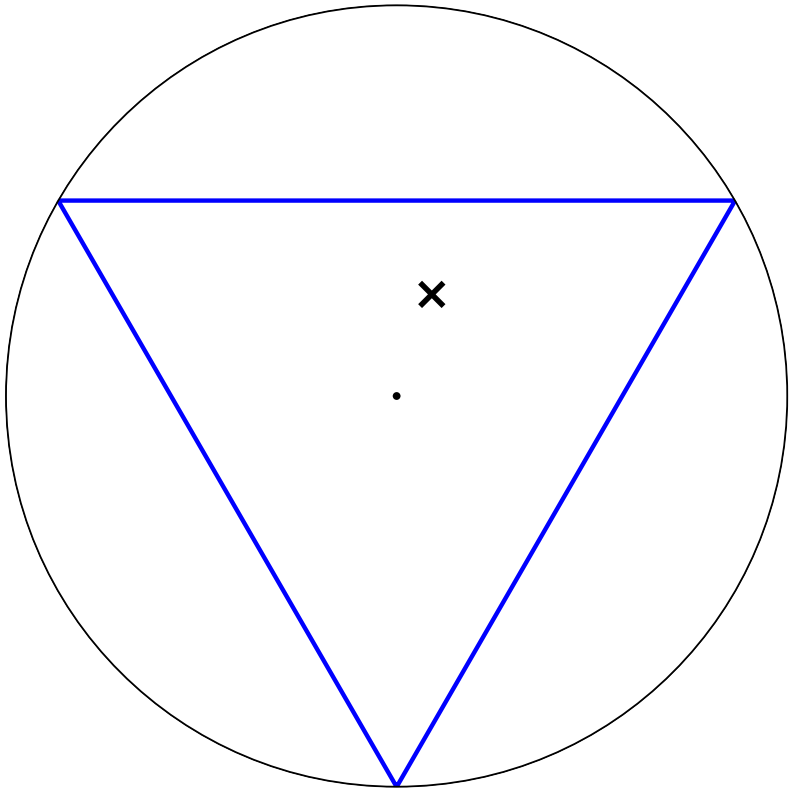
Midpoint:

Oh wait. . . it's $\frac{1}{4}$!



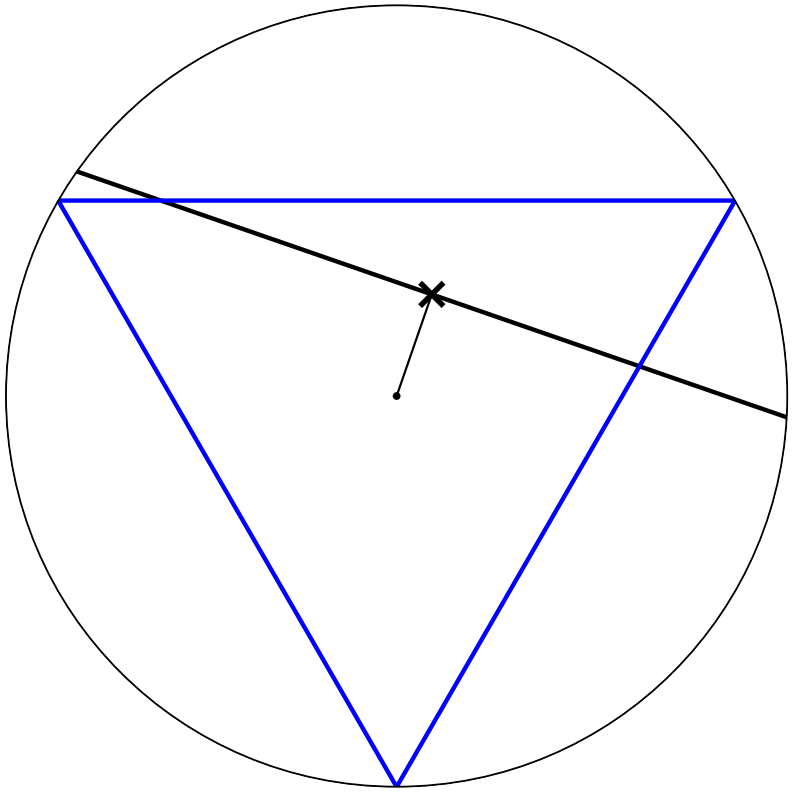
Midpoint:

Oh wait. . . it's $\frac{1}{4}$!



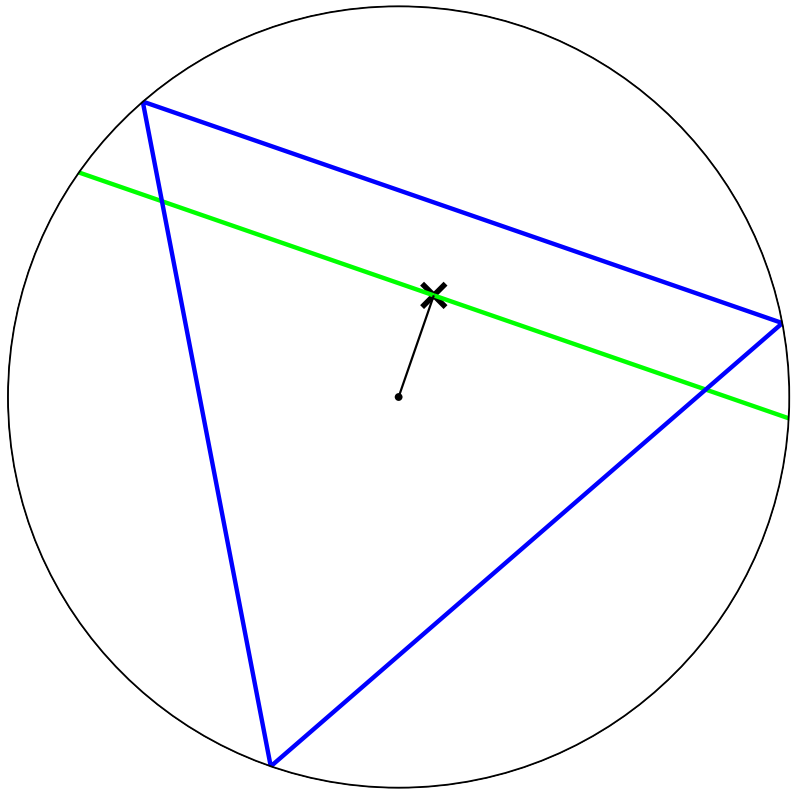
Midpoint:

Oh wait. . . it's $\frac{1}{4}$!



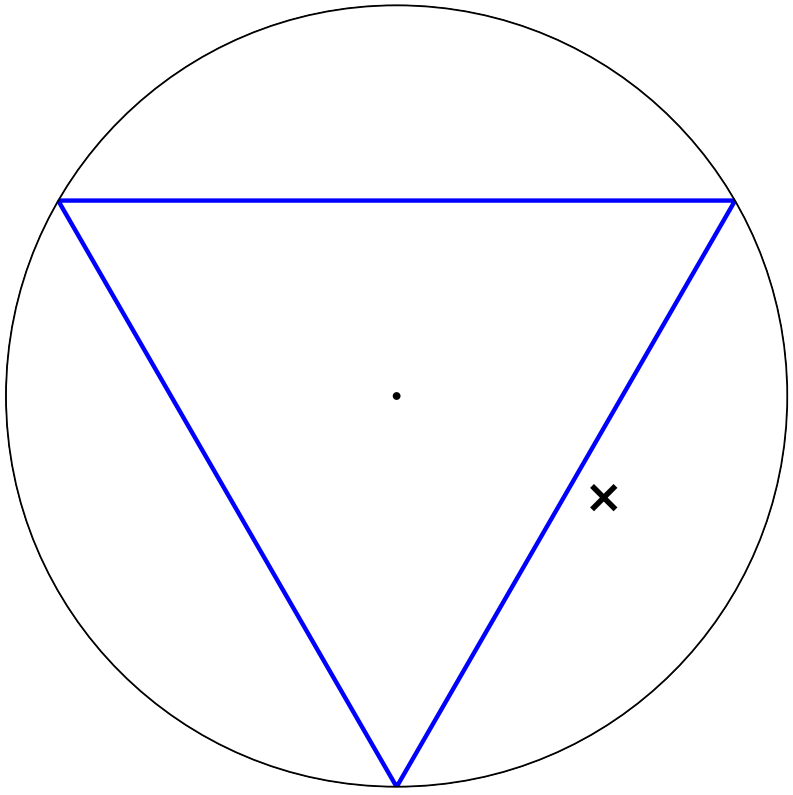
Midpoint:

Oh wait. . . it's $\frac{1}{4}$!



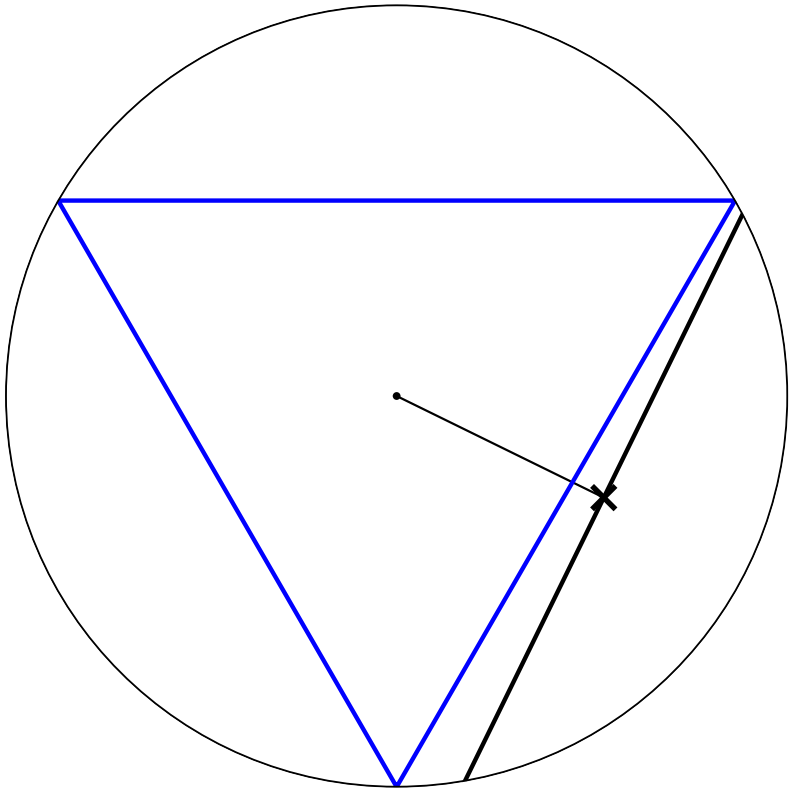
Midpoint:

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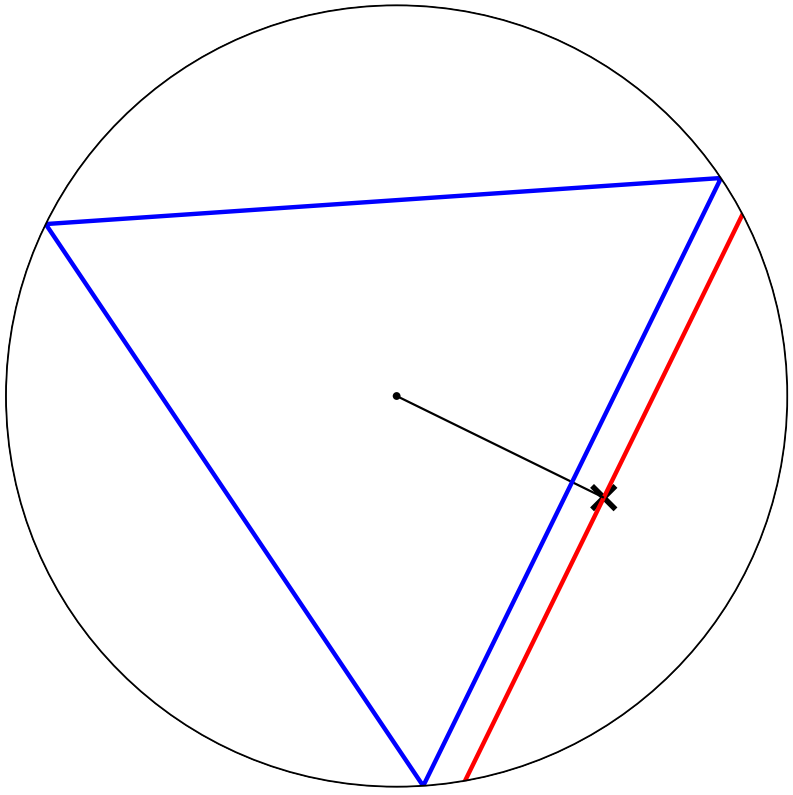
Midpoint:

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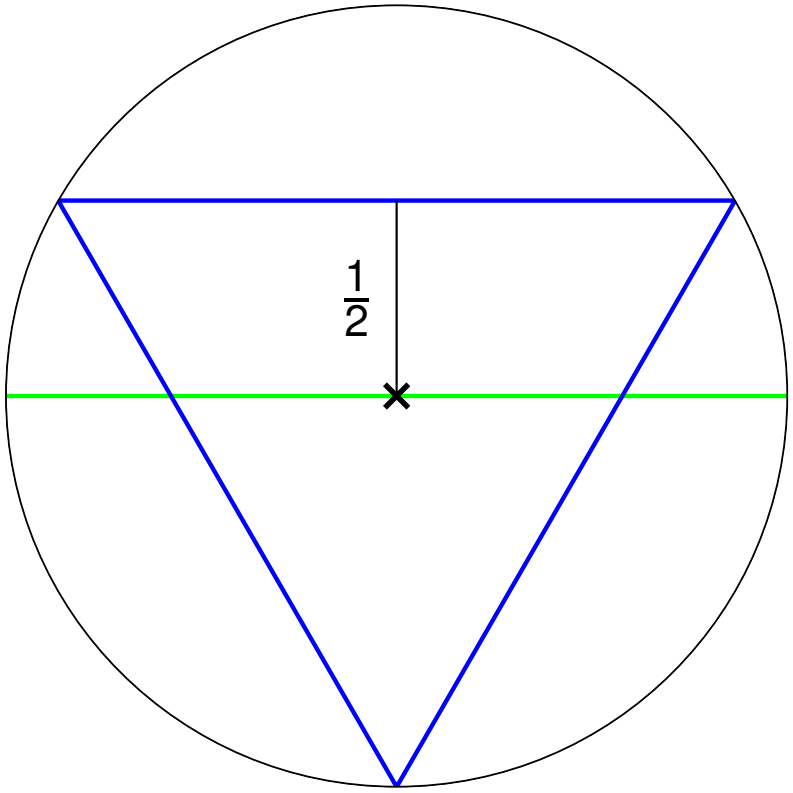
Midpoint:

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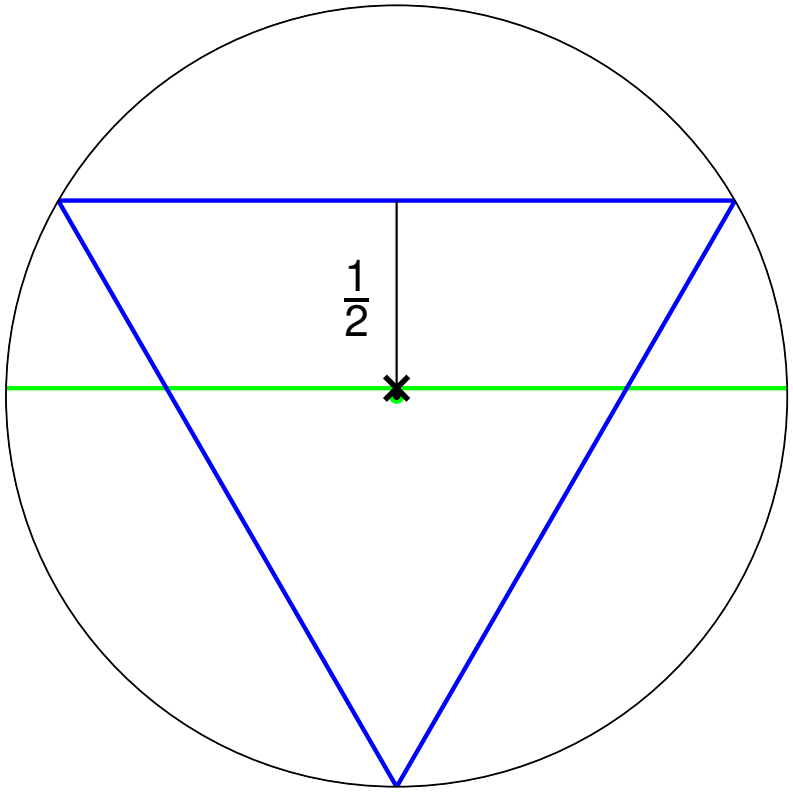
Midpoint:

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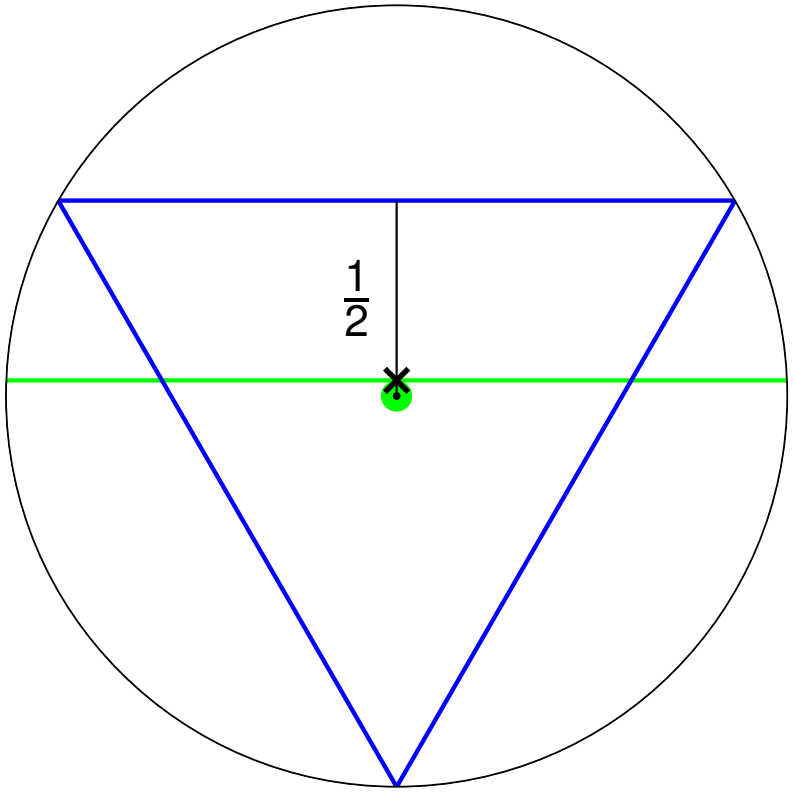
Midpoint:

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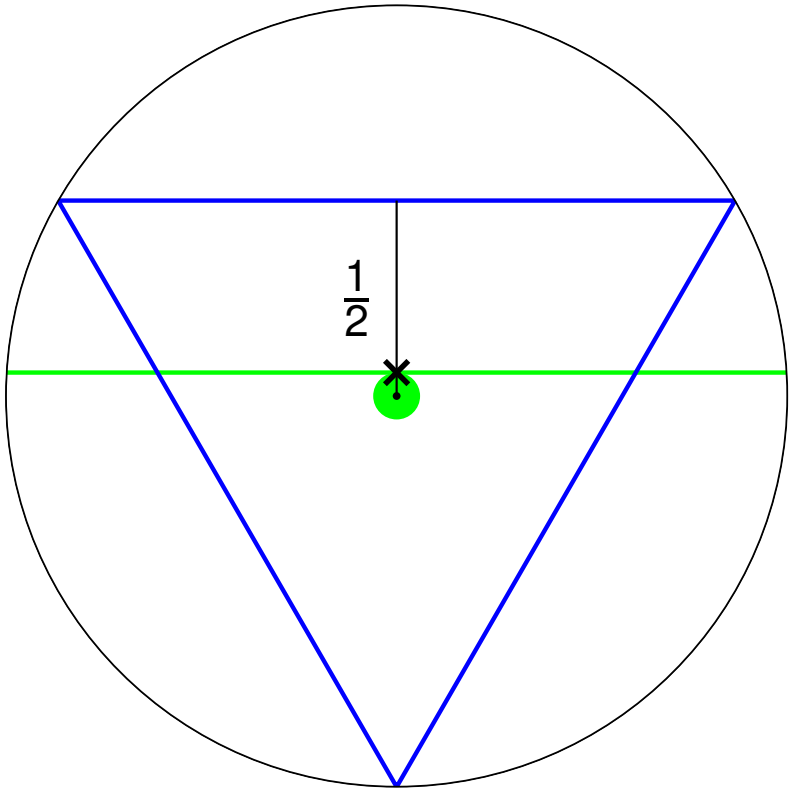
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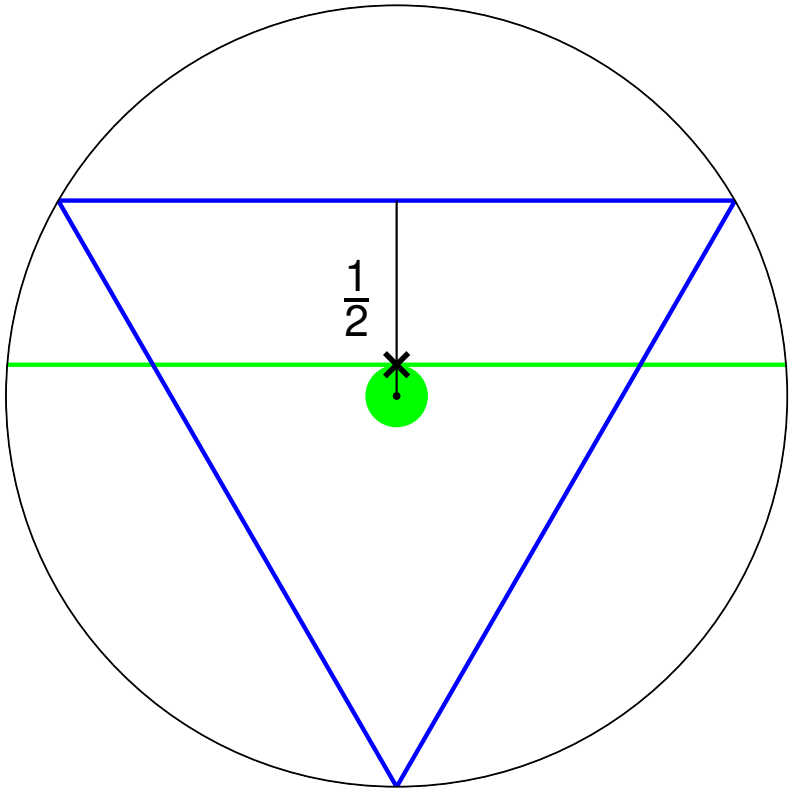
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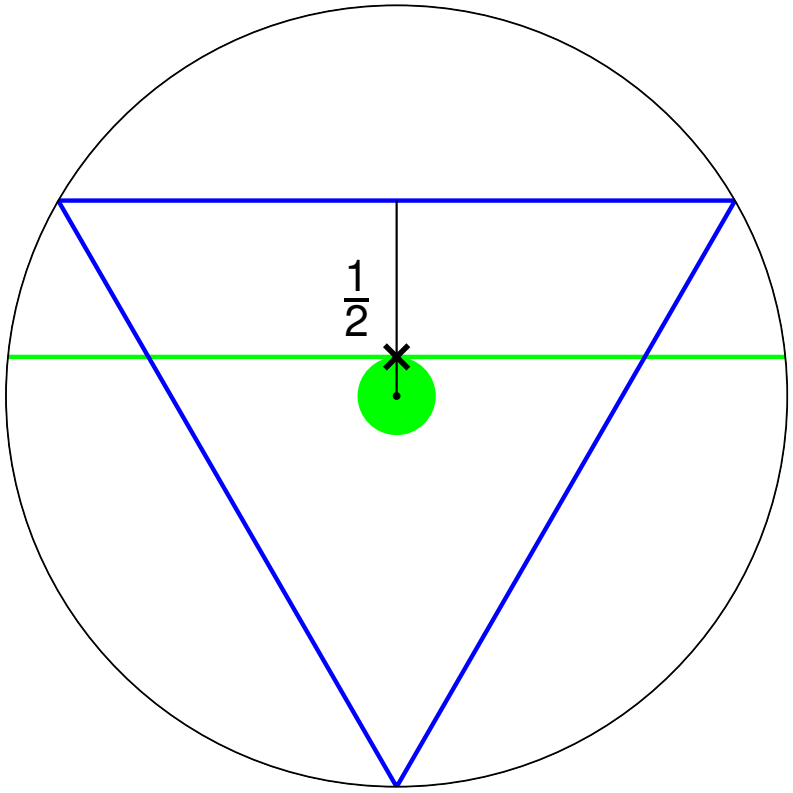
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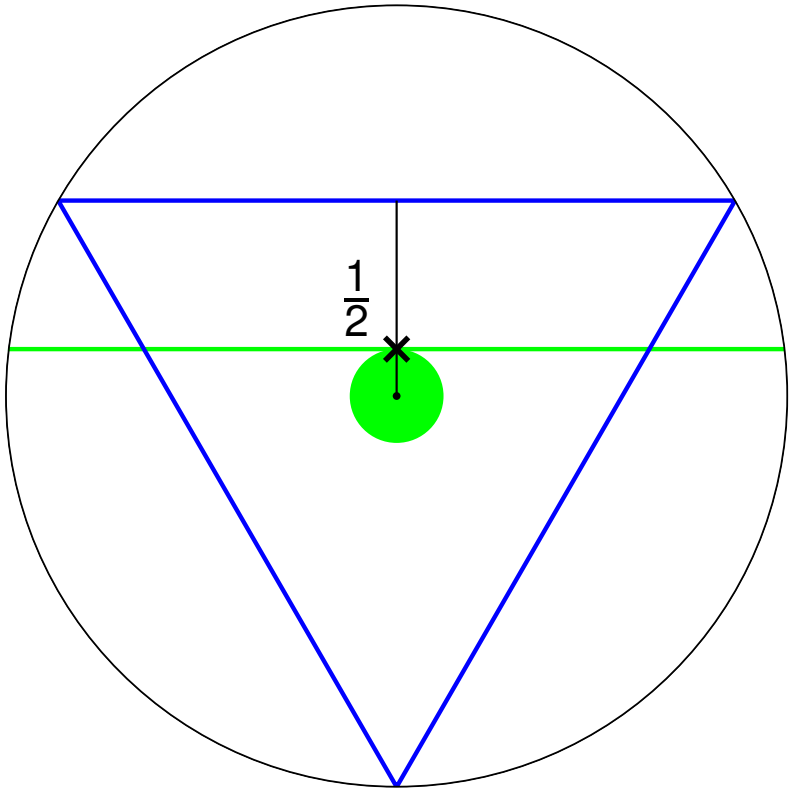
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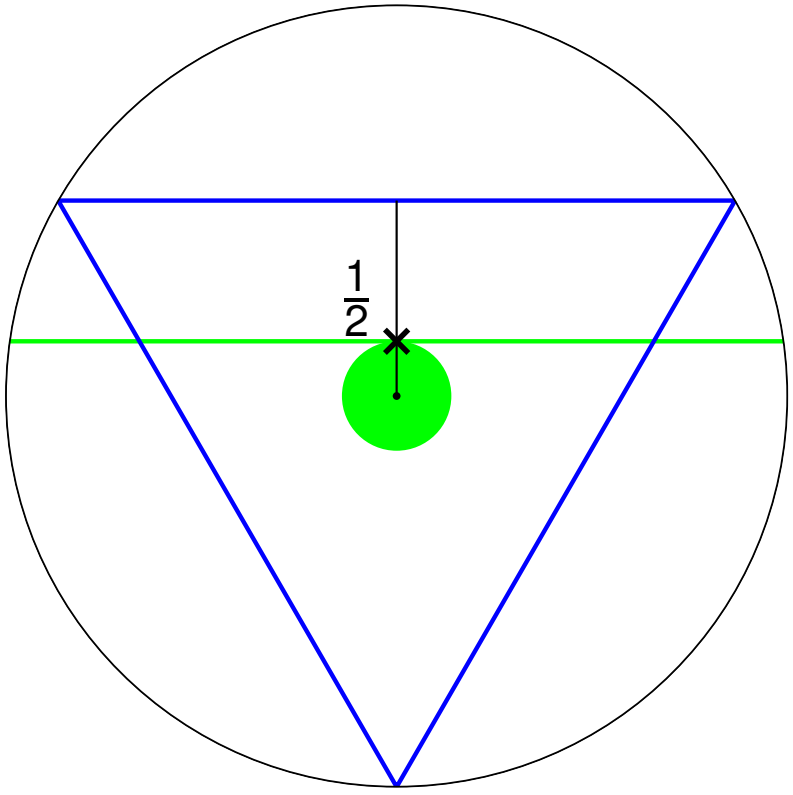
Midpoint:

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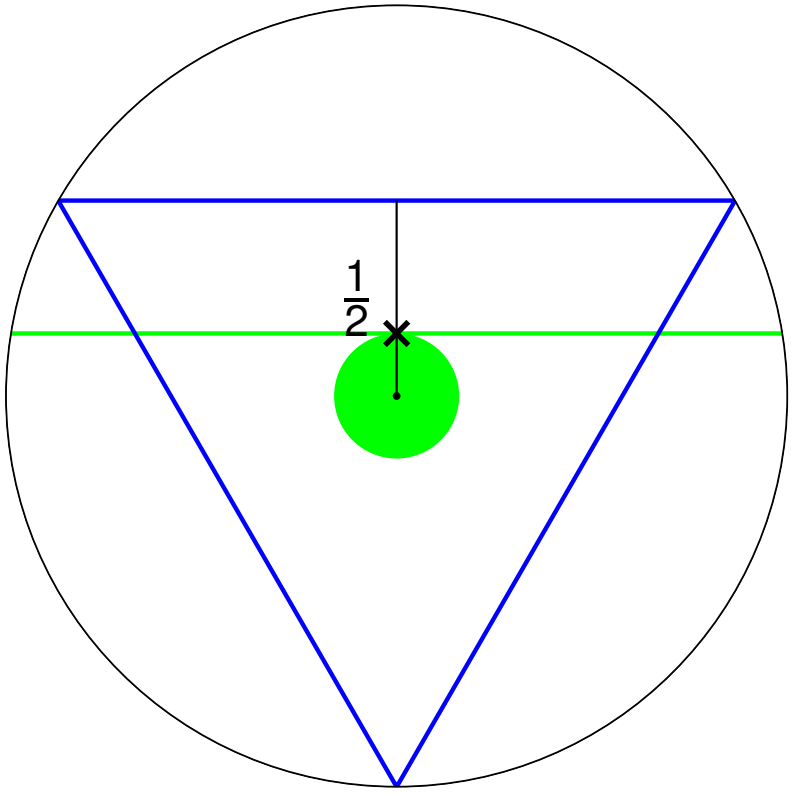
Midpoint:

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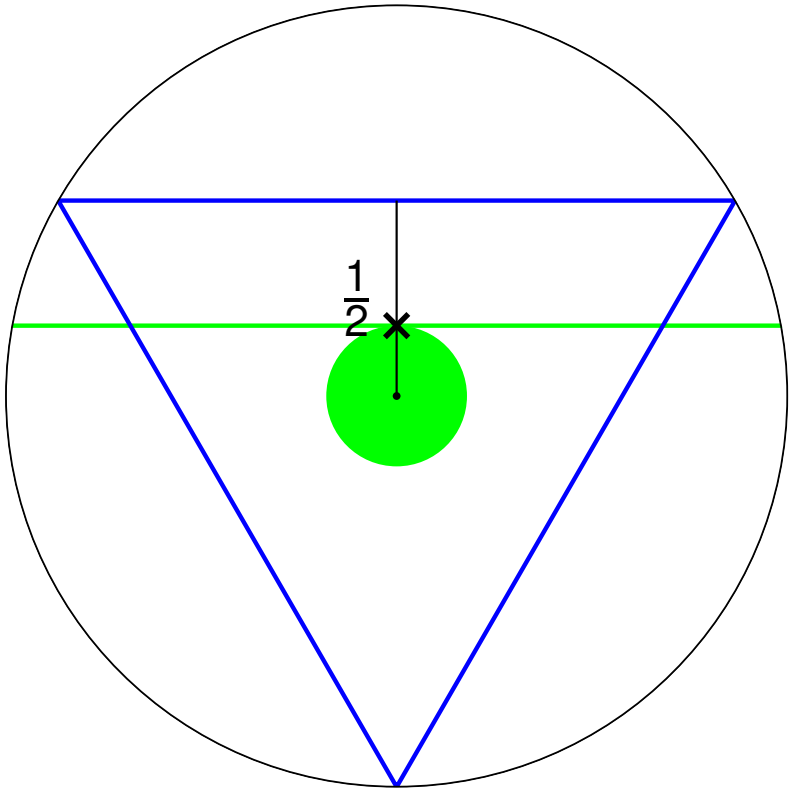
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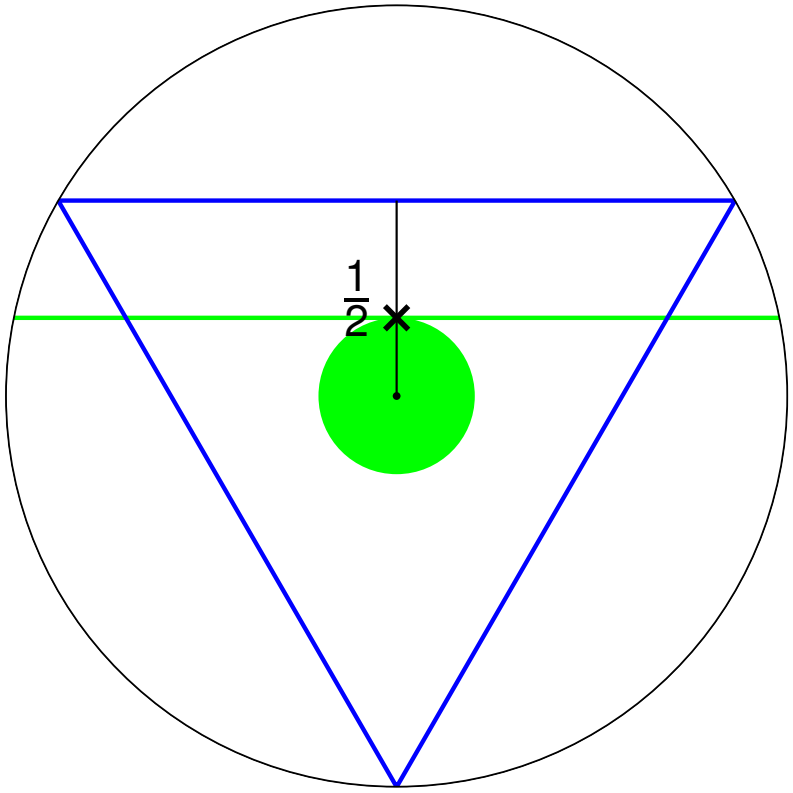
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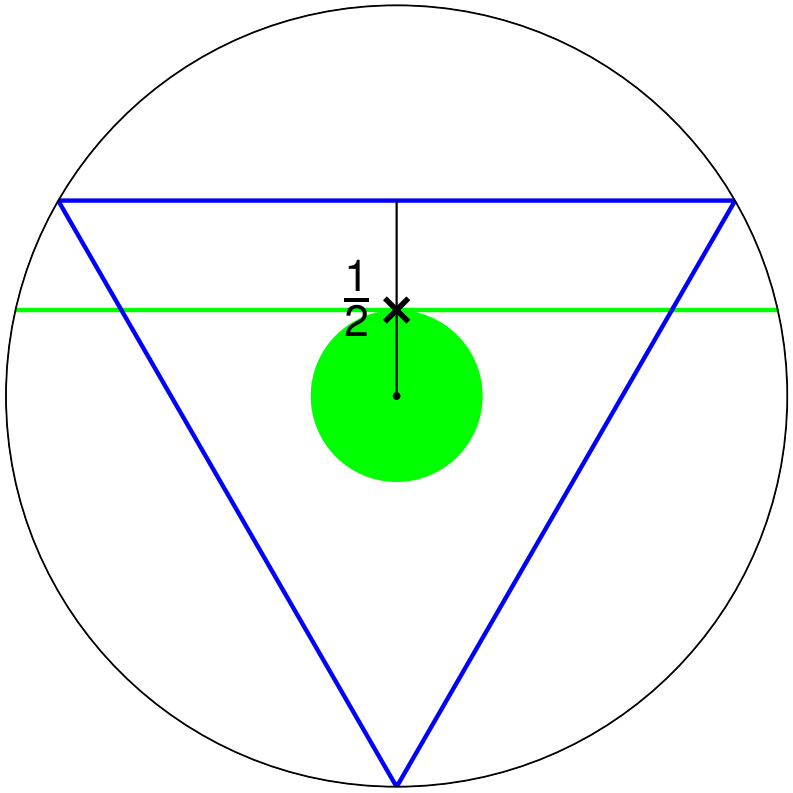
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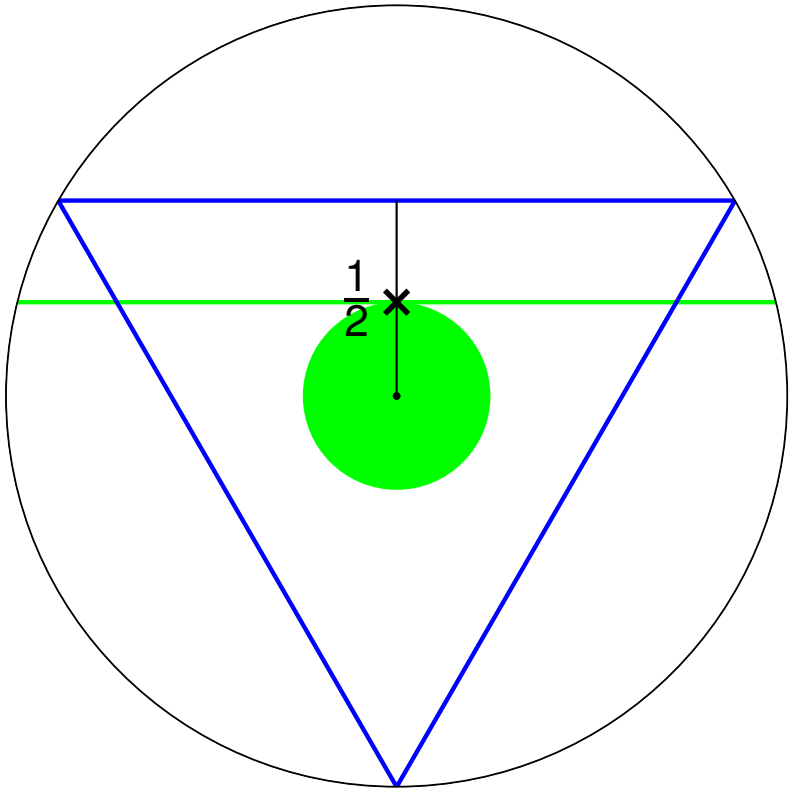
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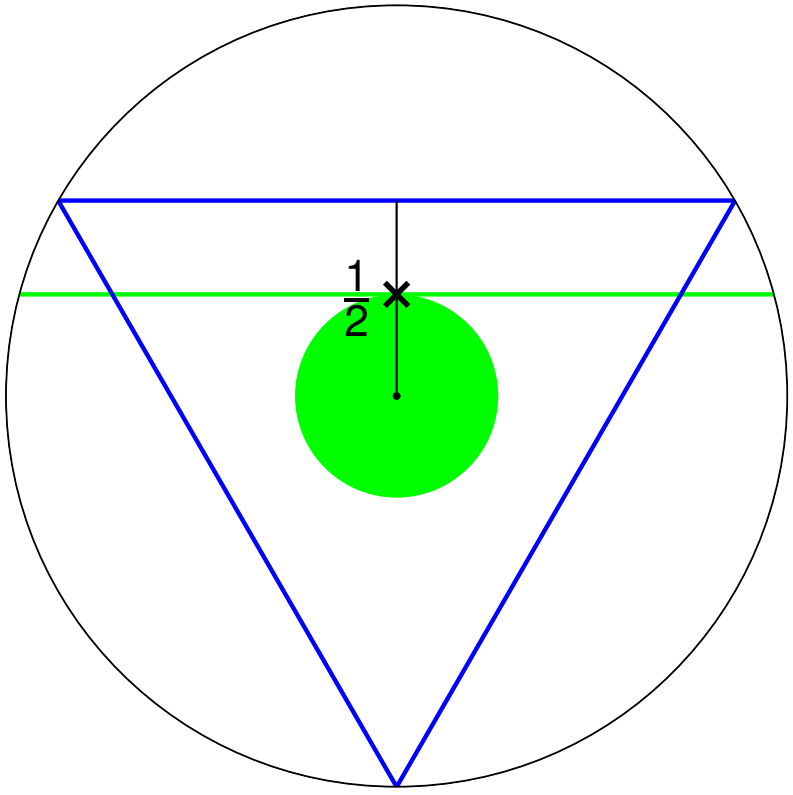
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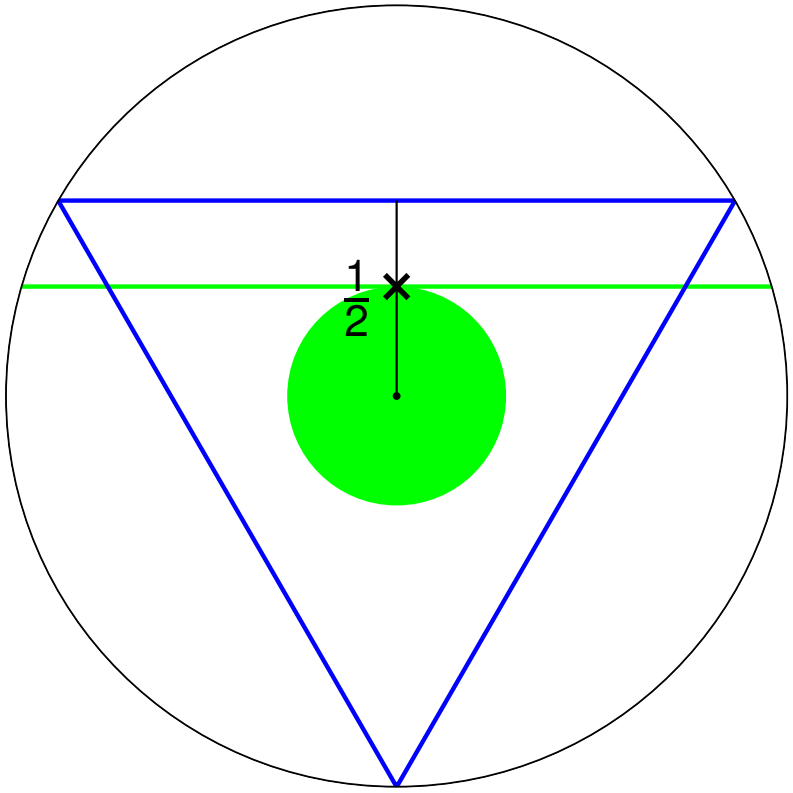
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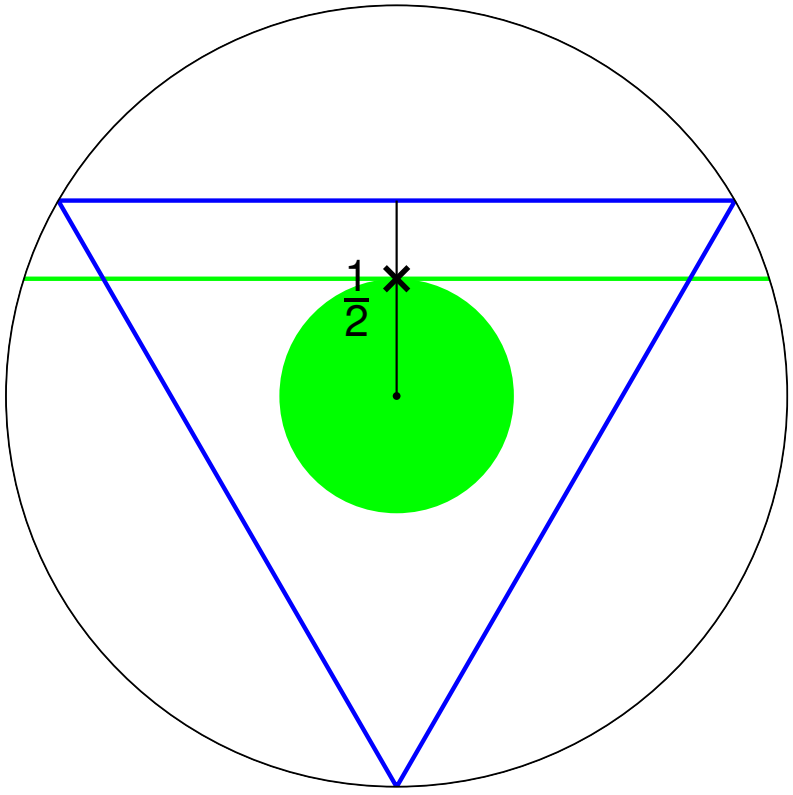
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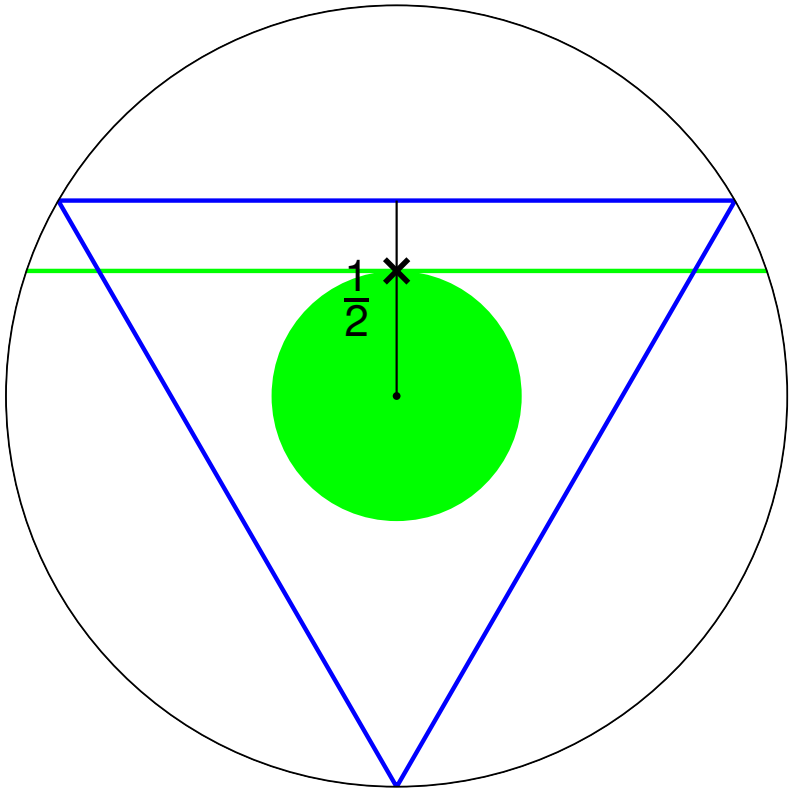
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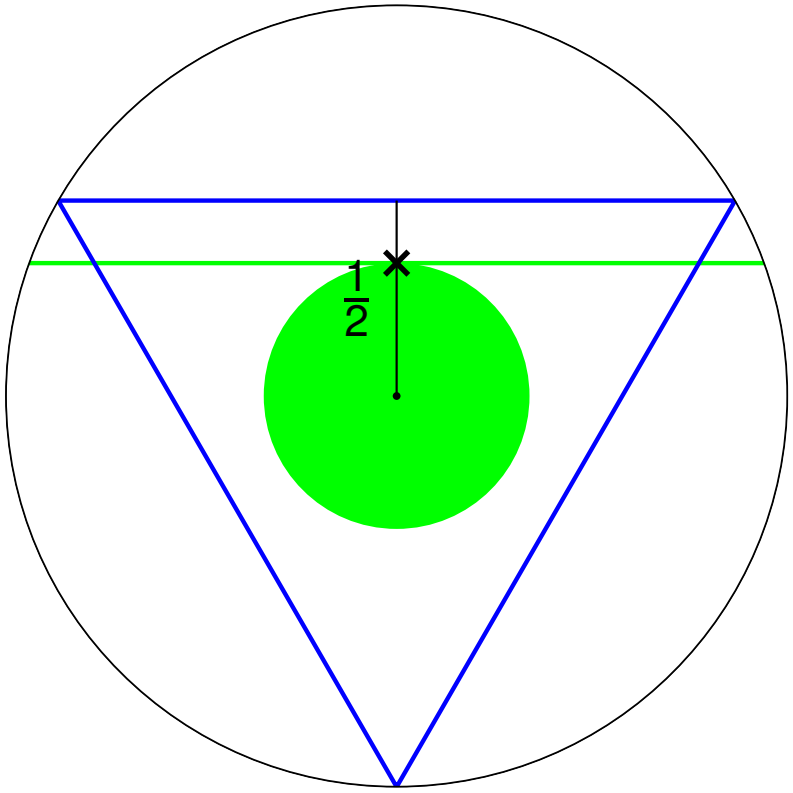
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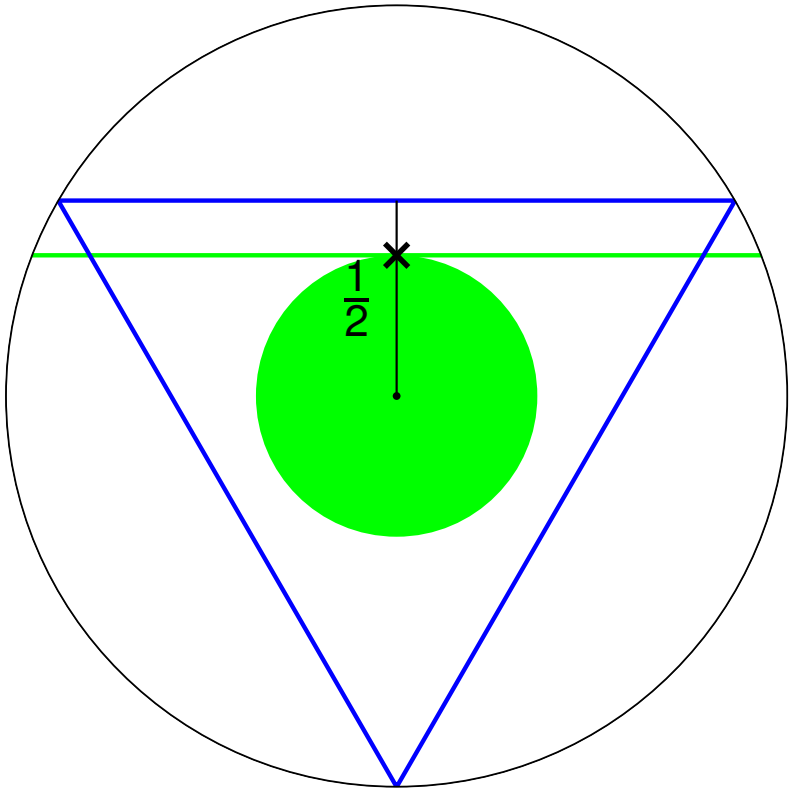
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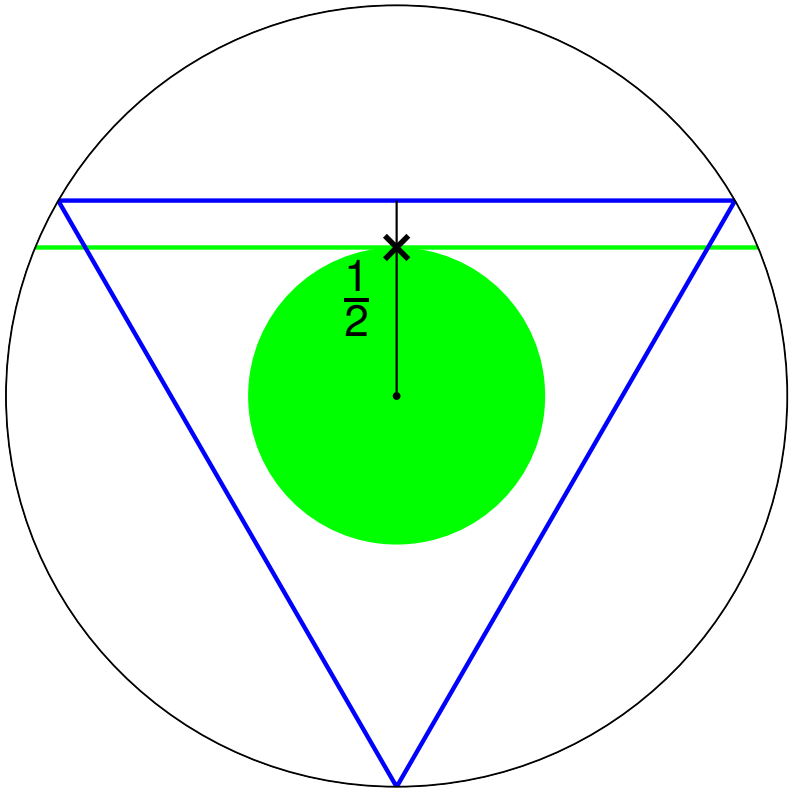
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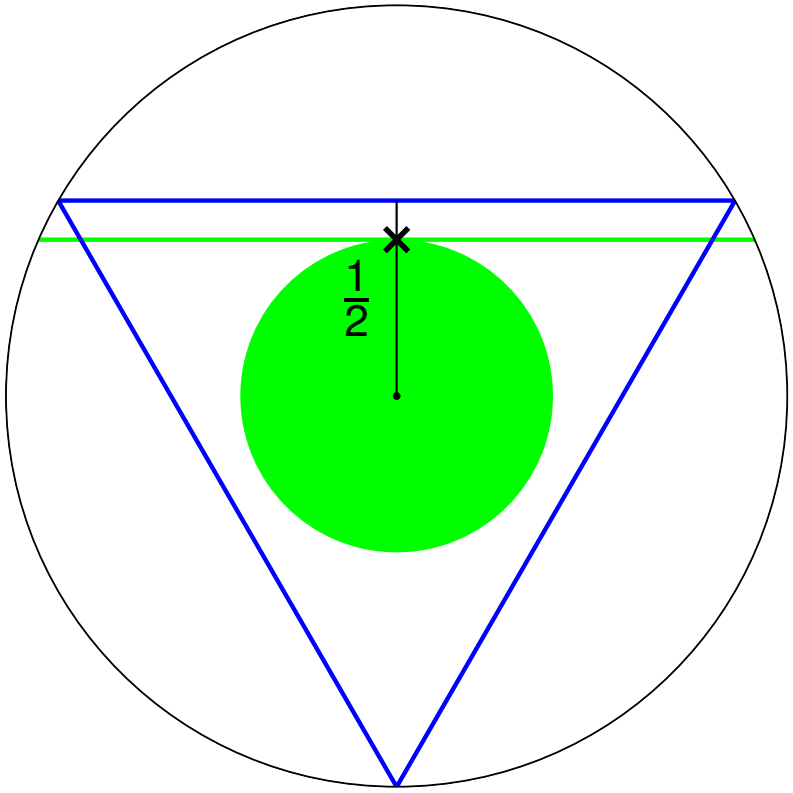
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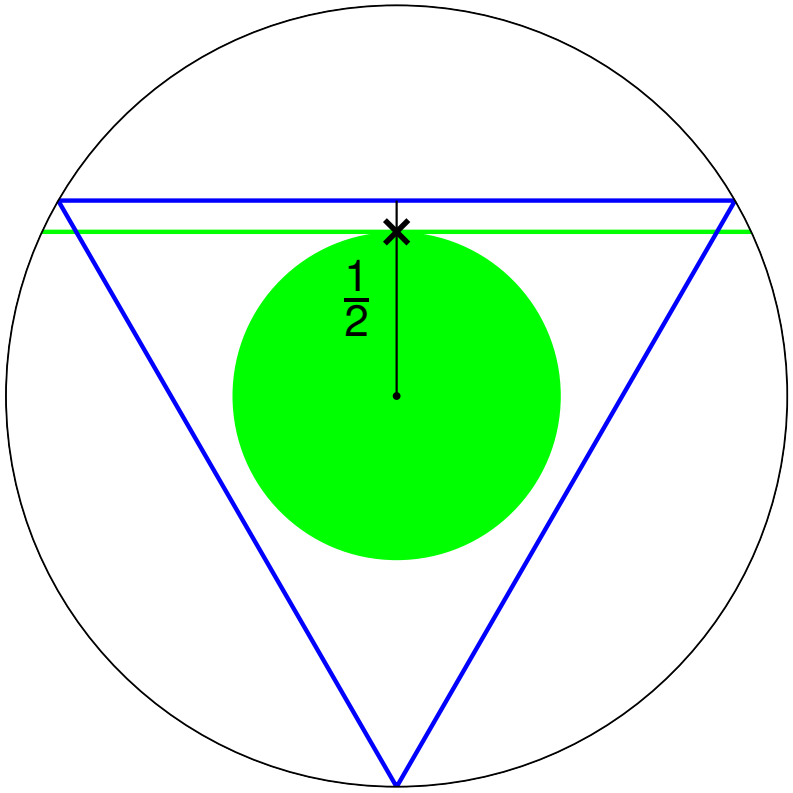
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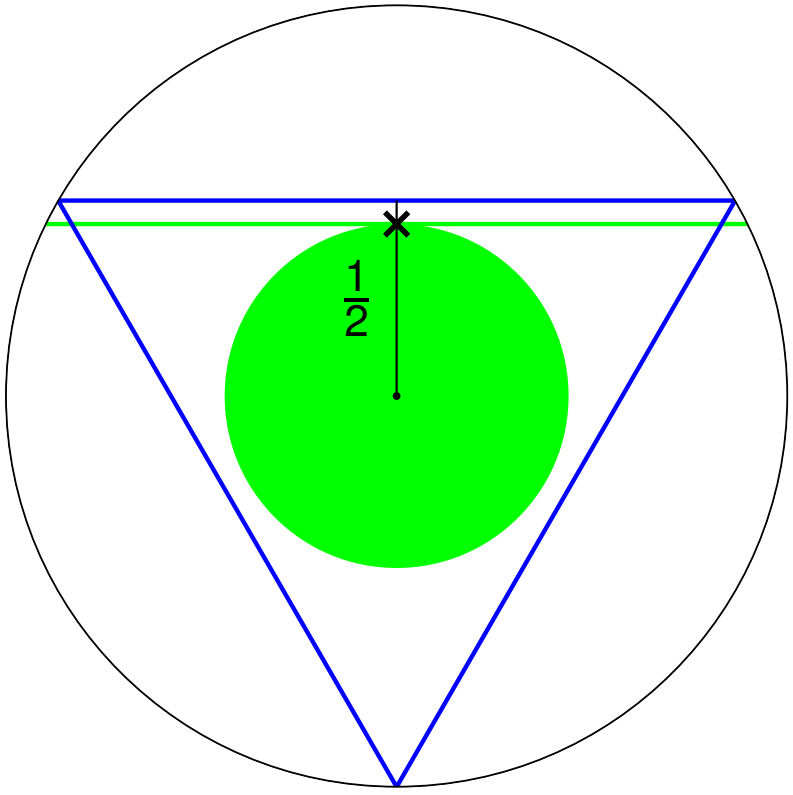
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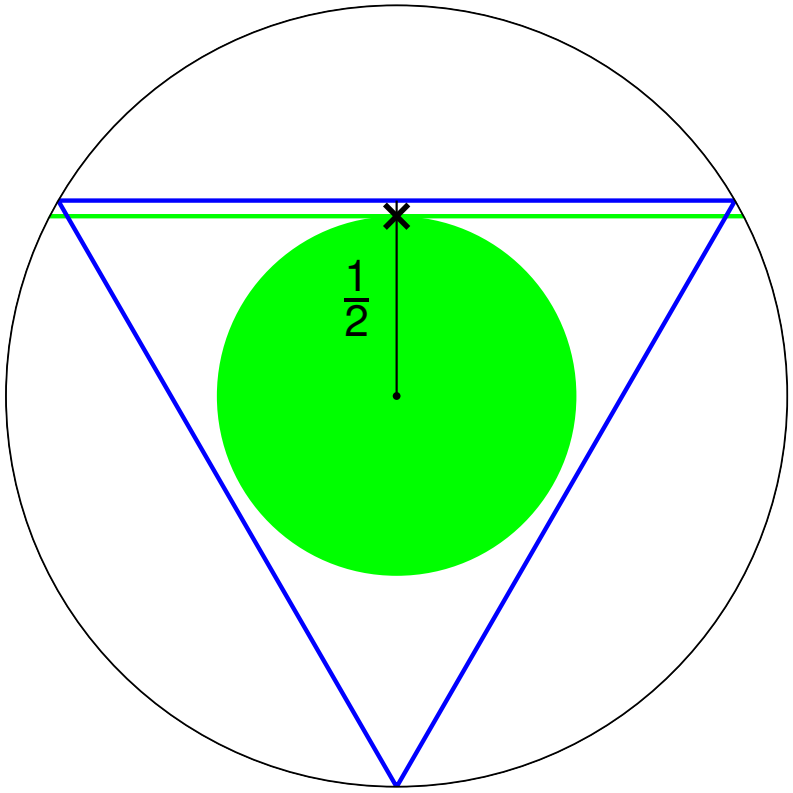
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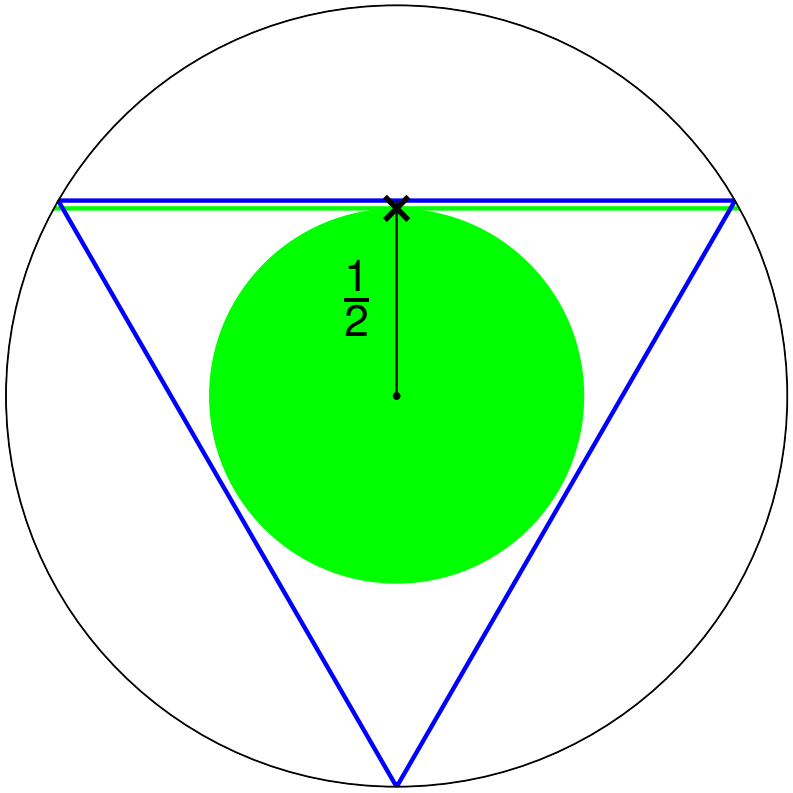
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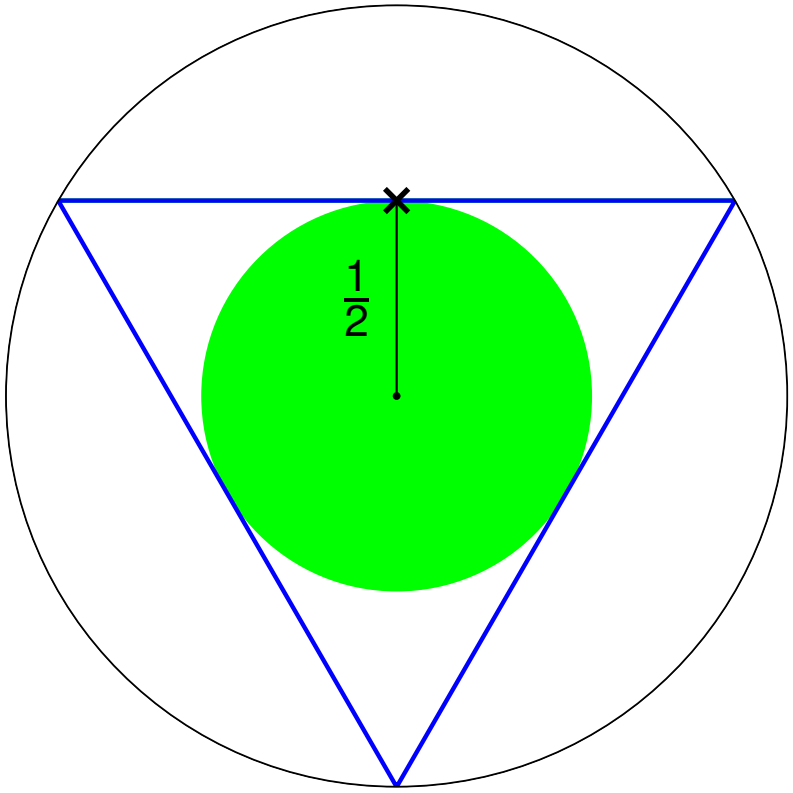
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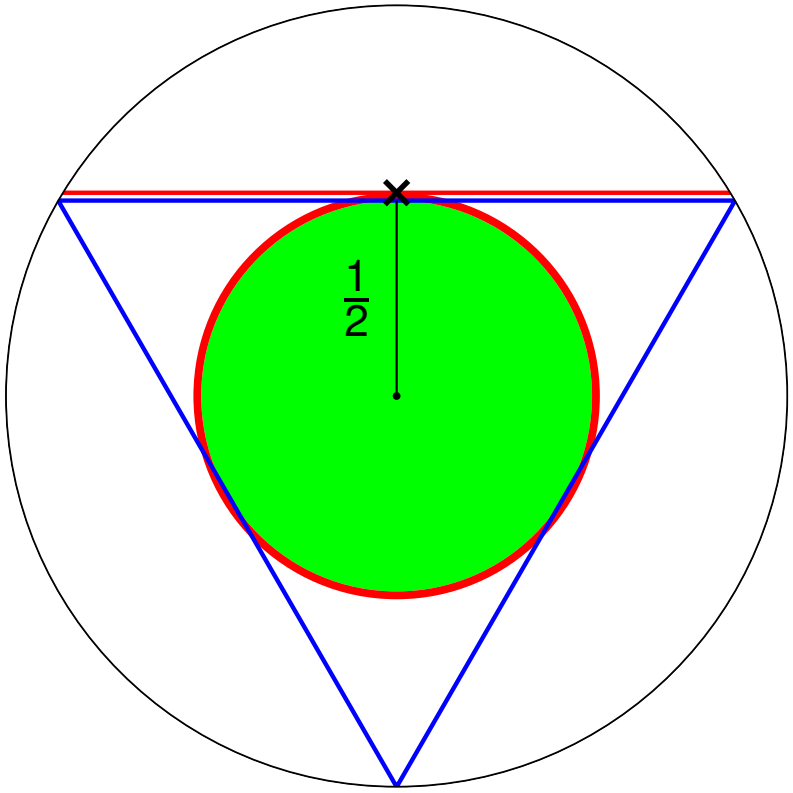
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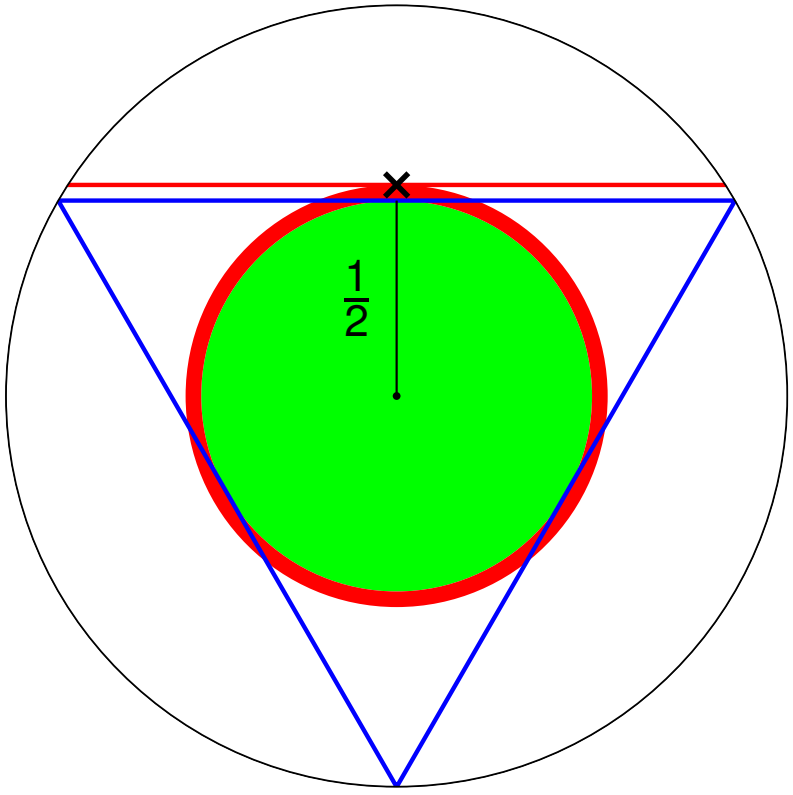
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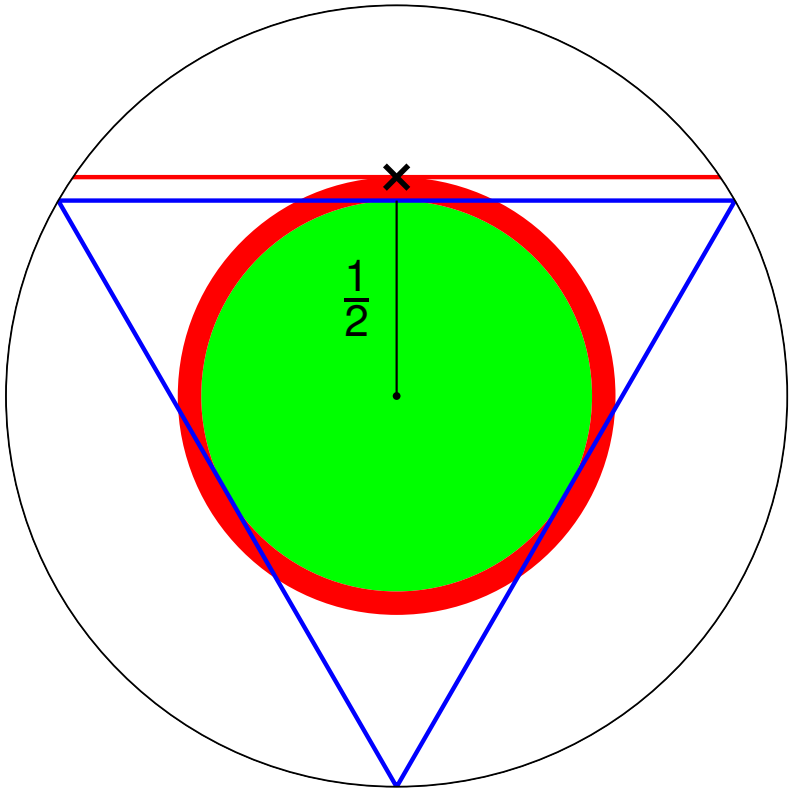
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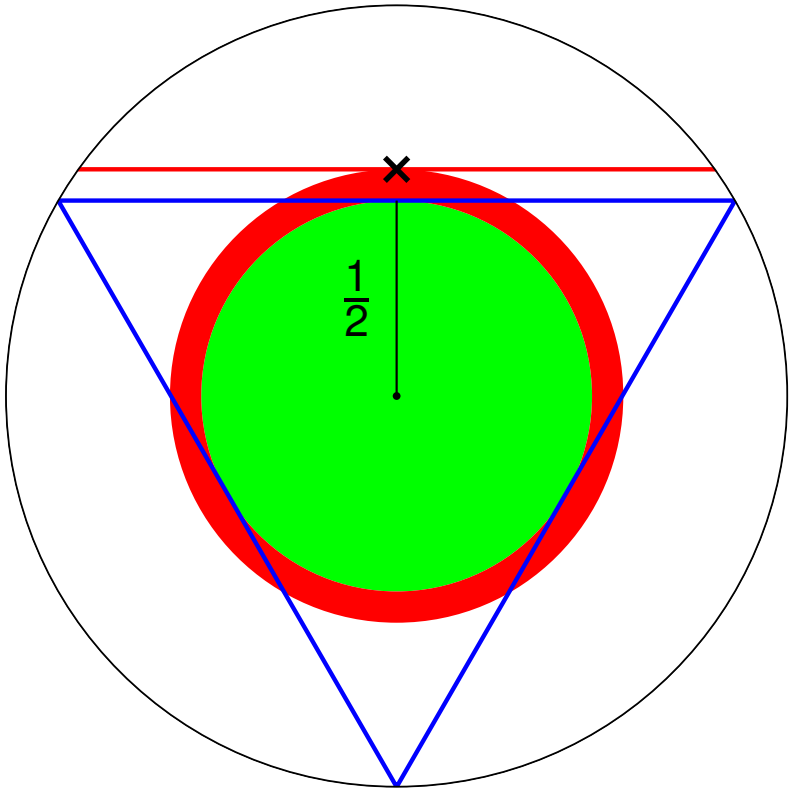
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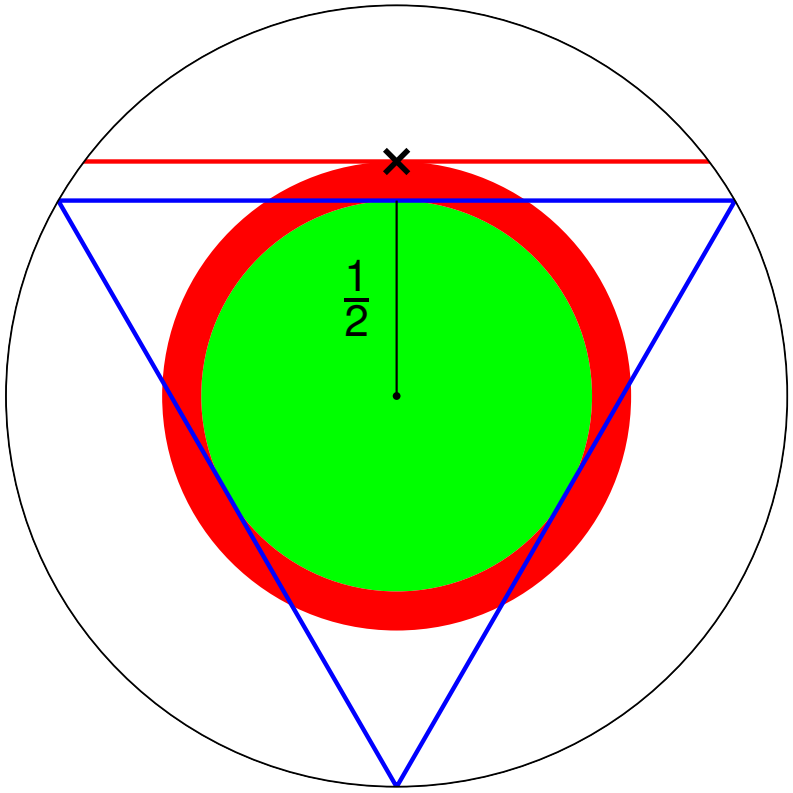
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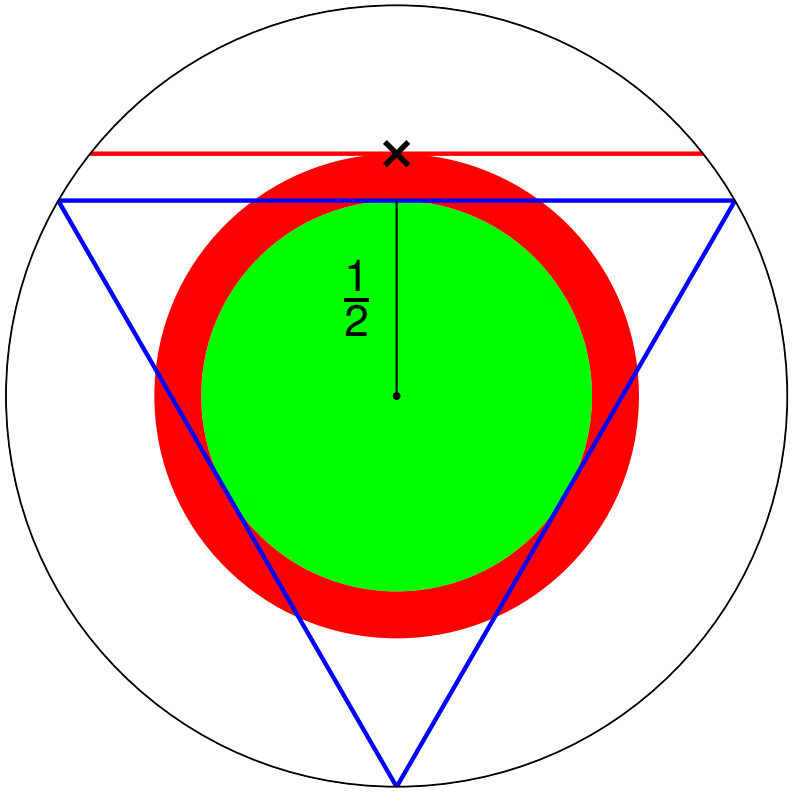
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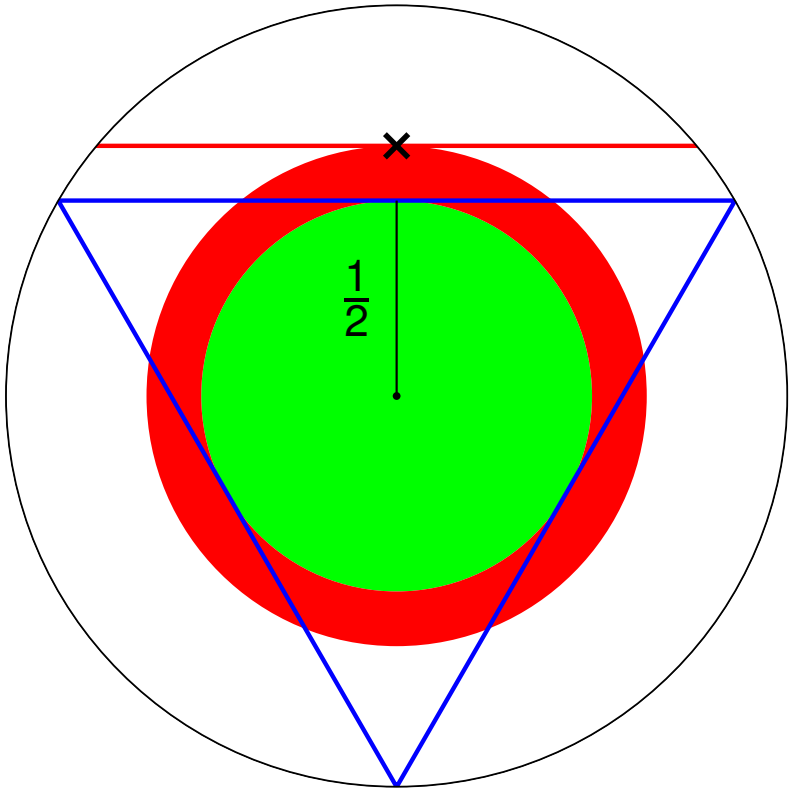
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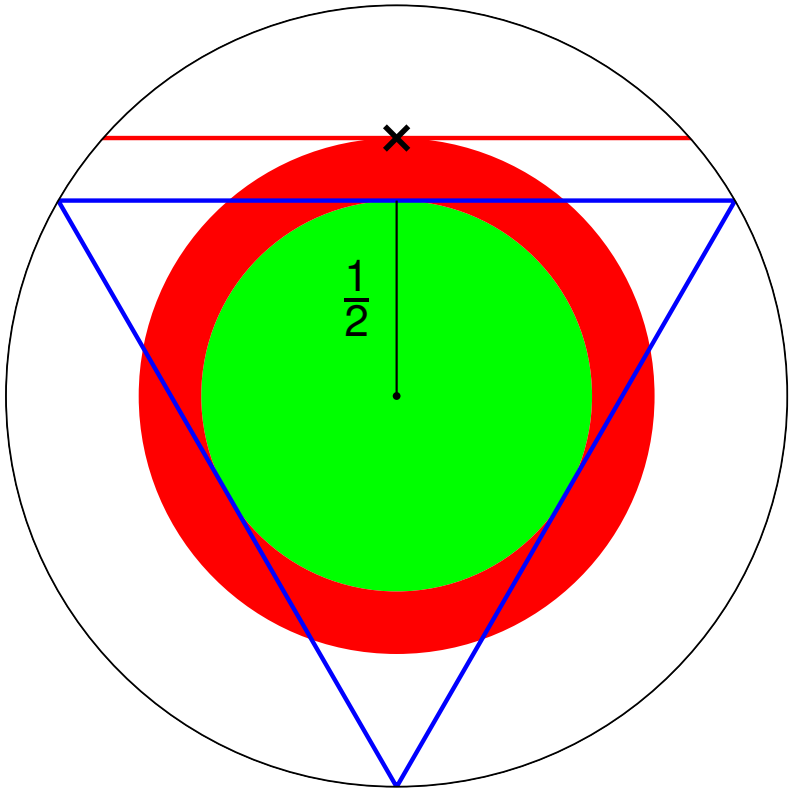
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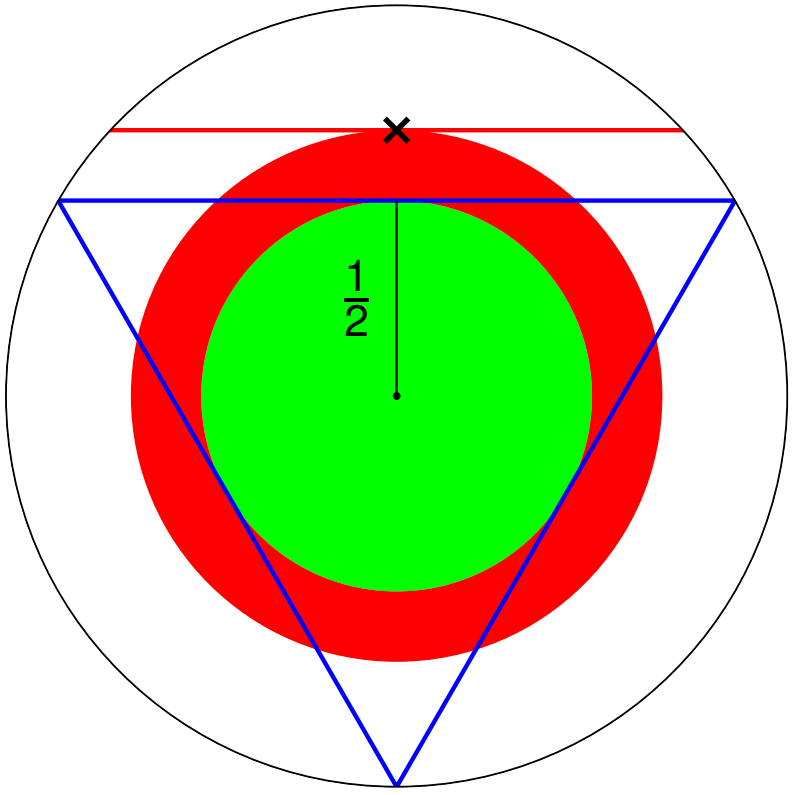
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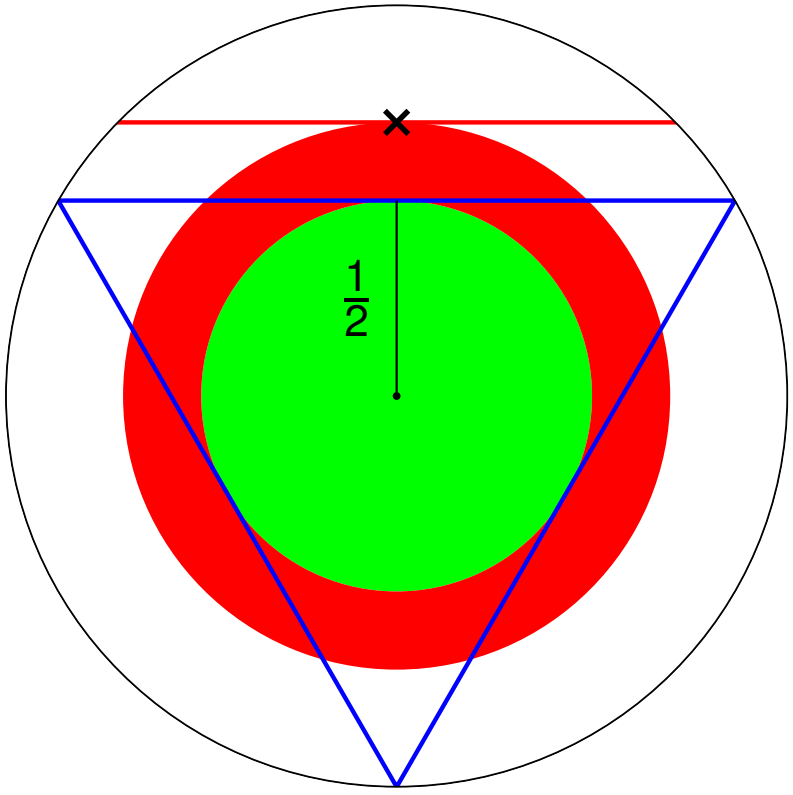
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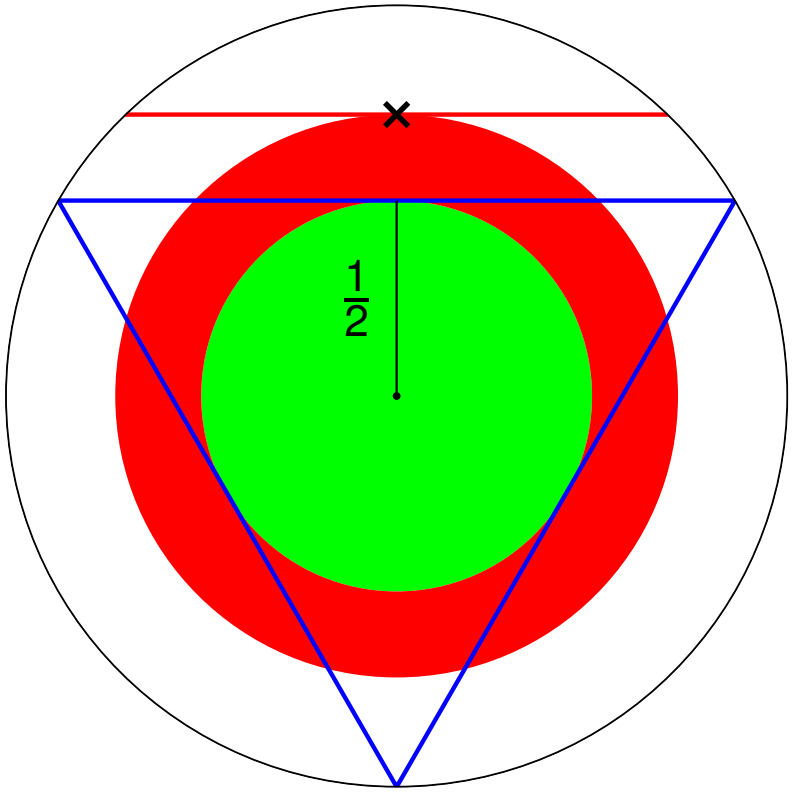
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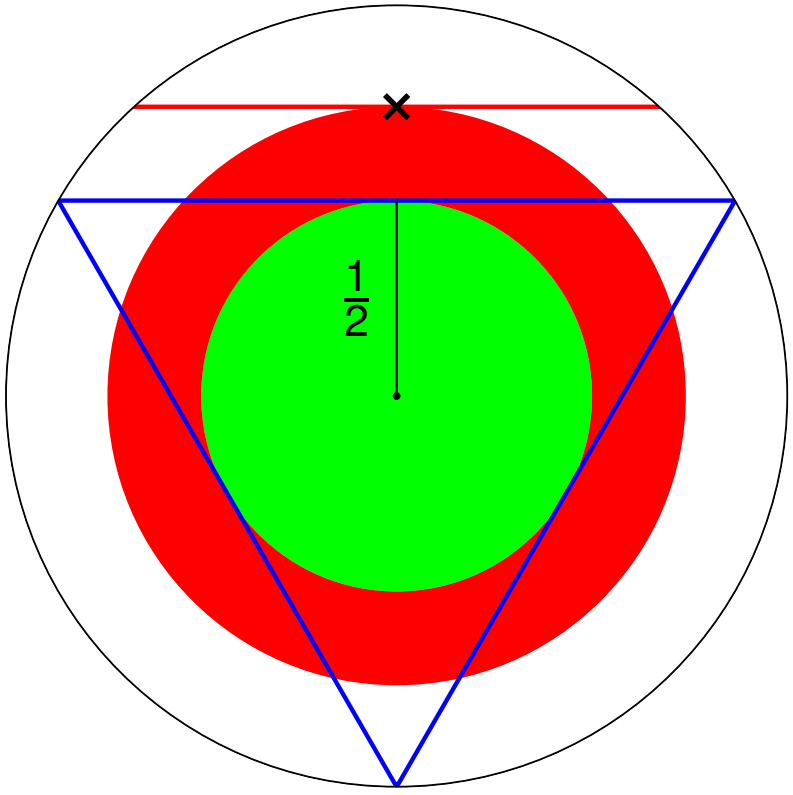
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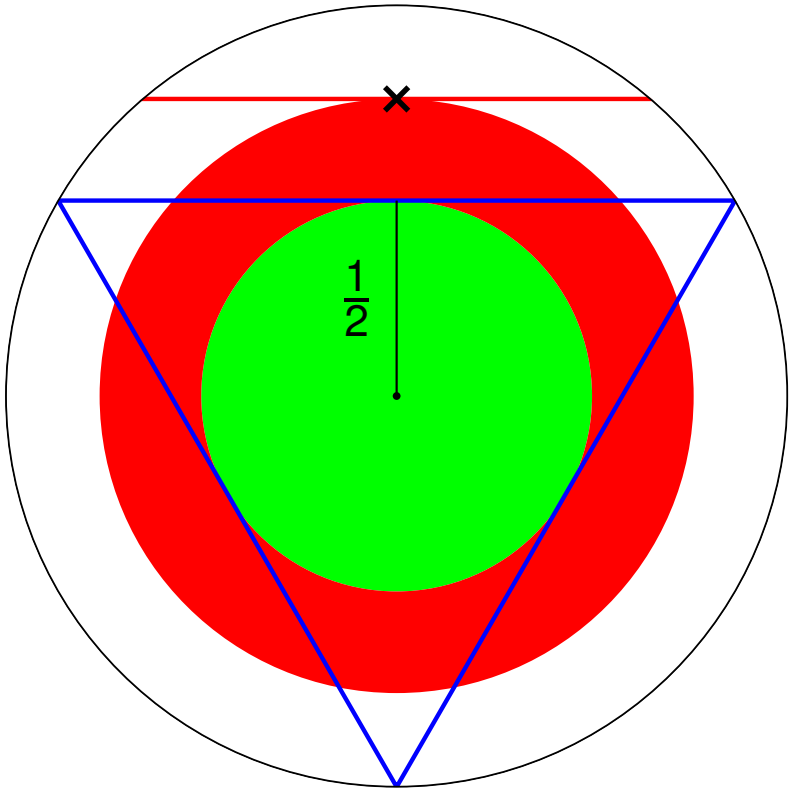
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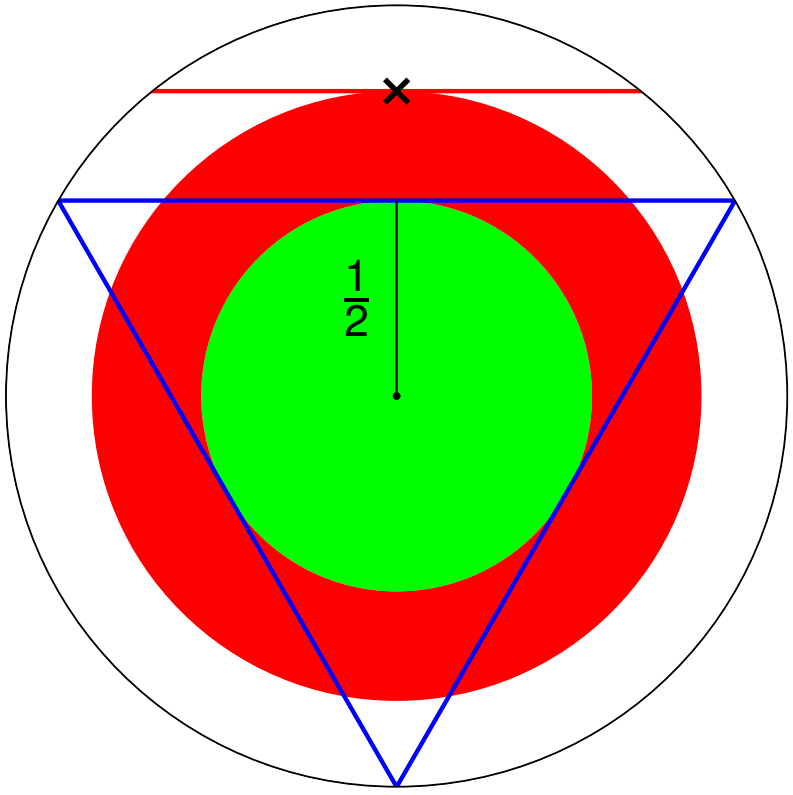
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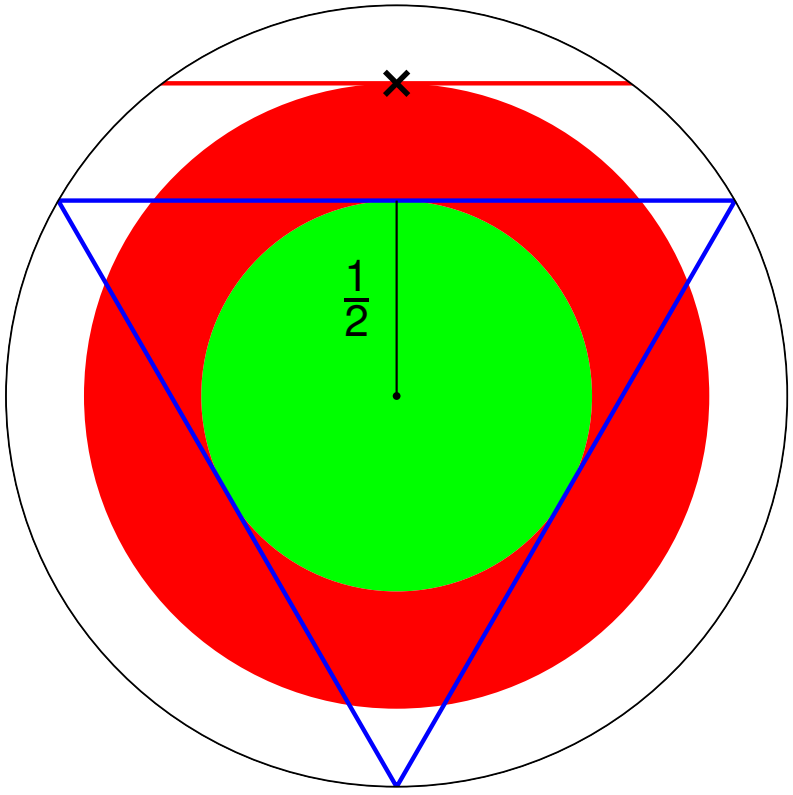
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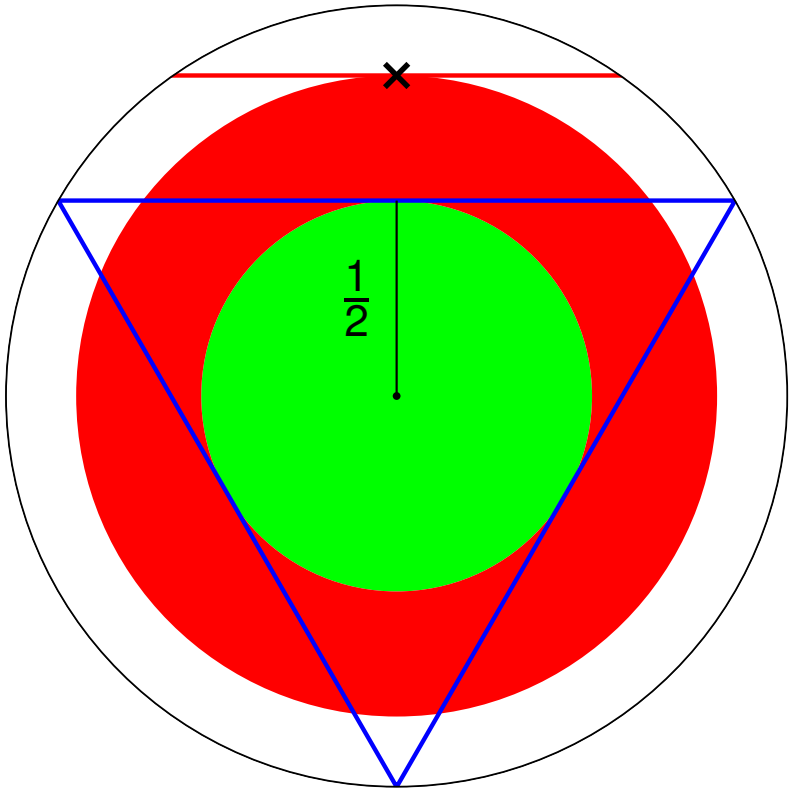
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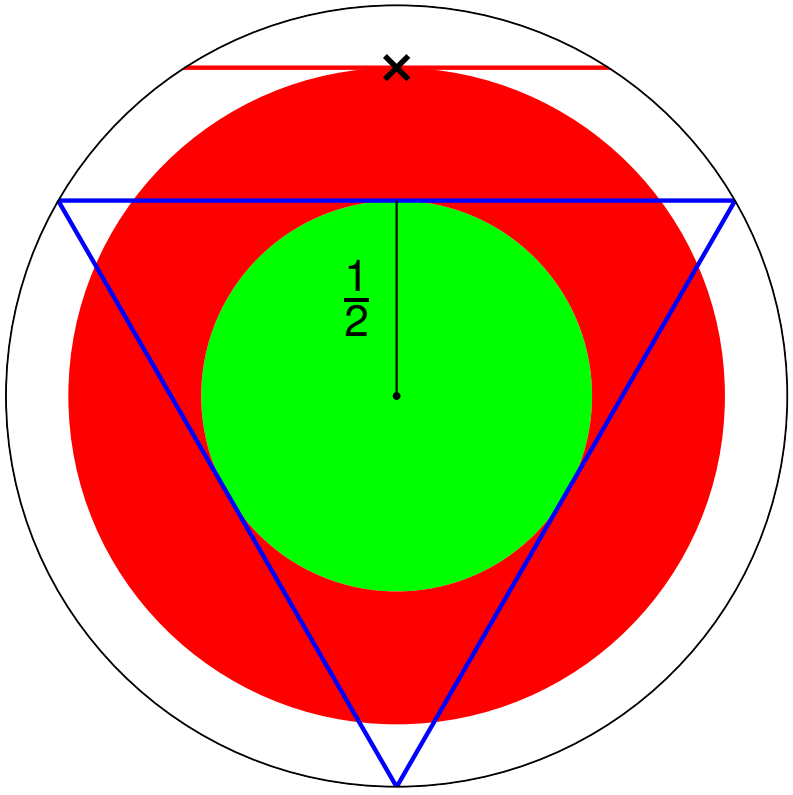
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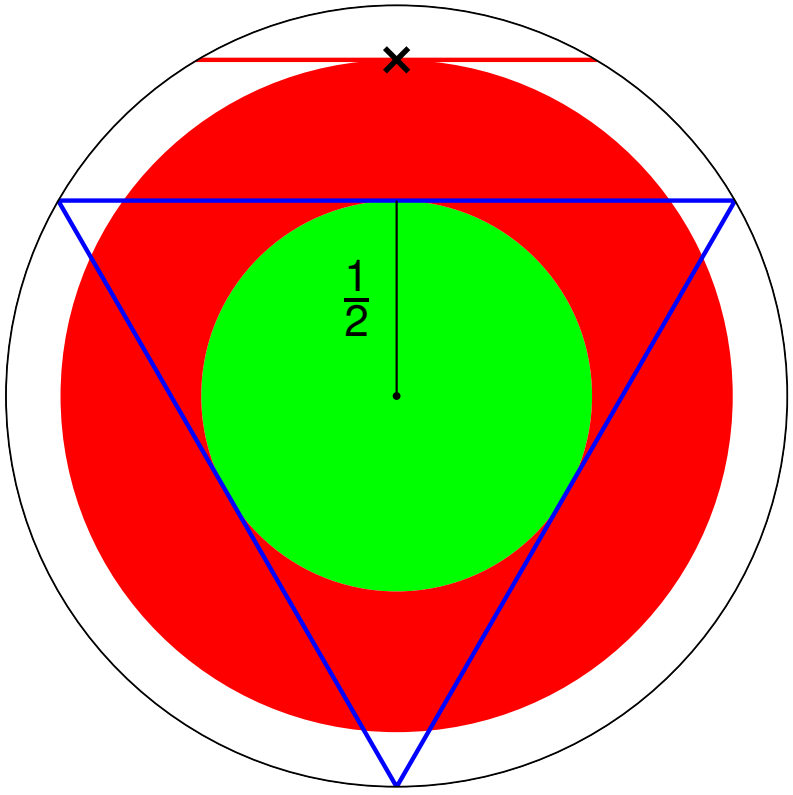
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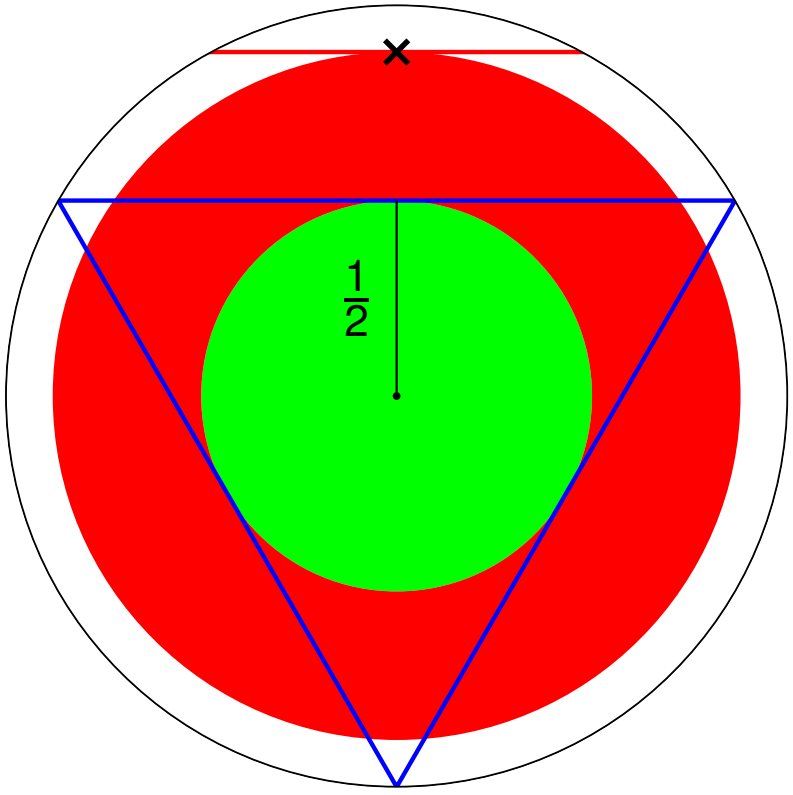
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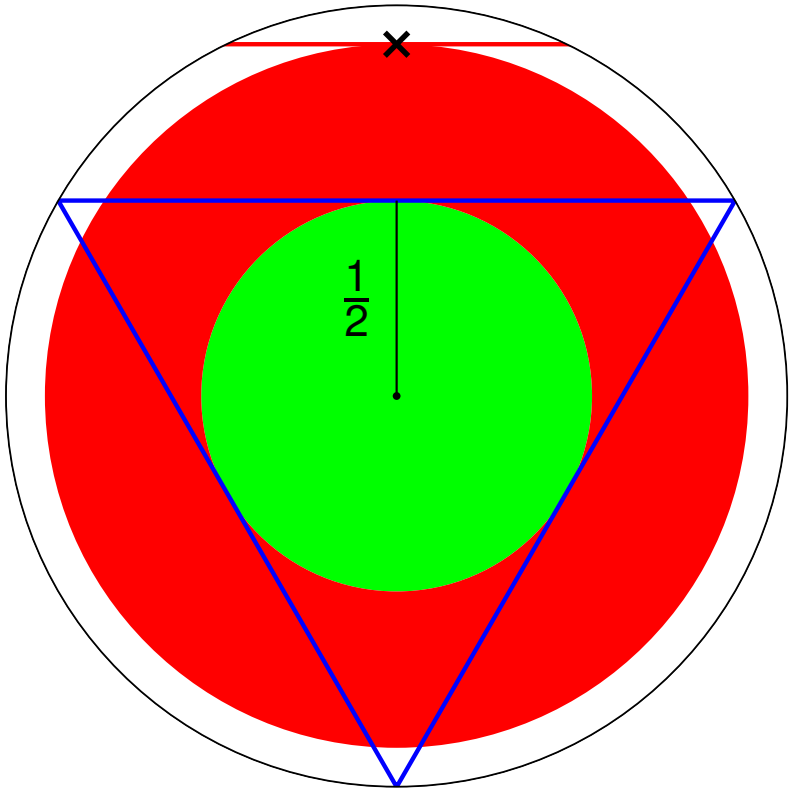
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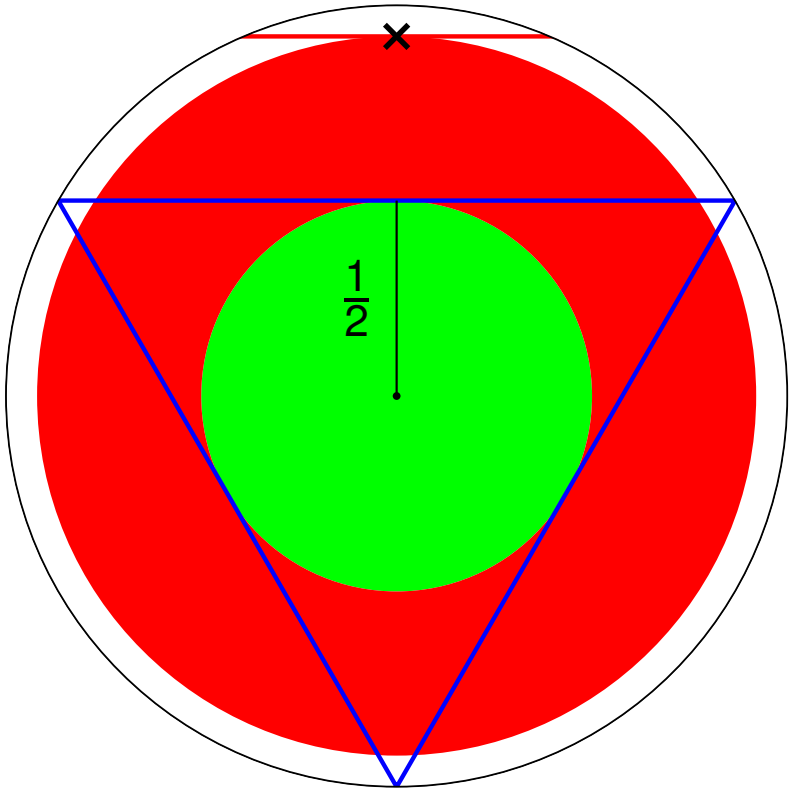
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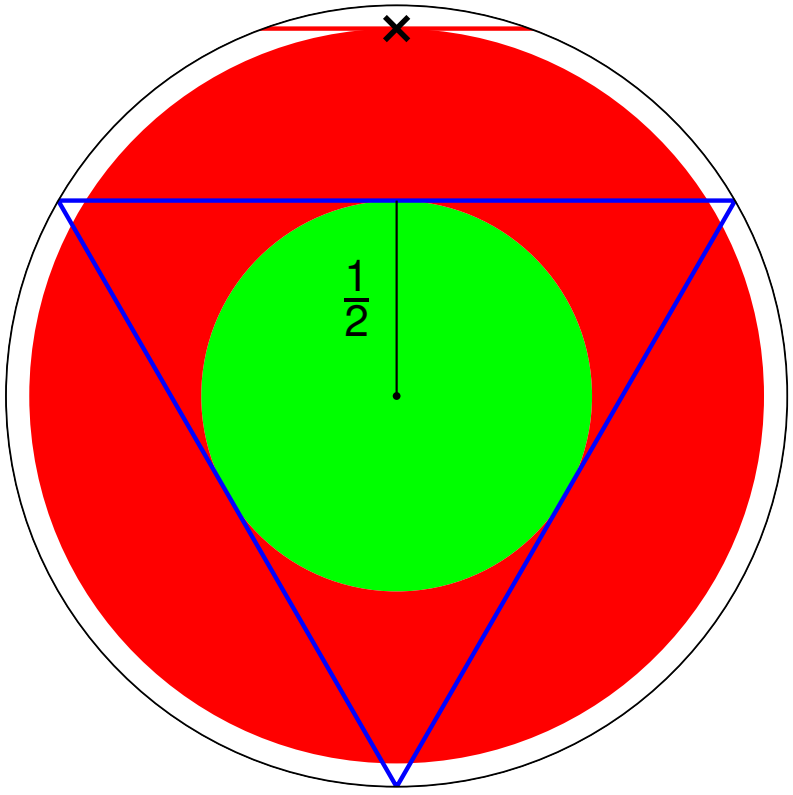
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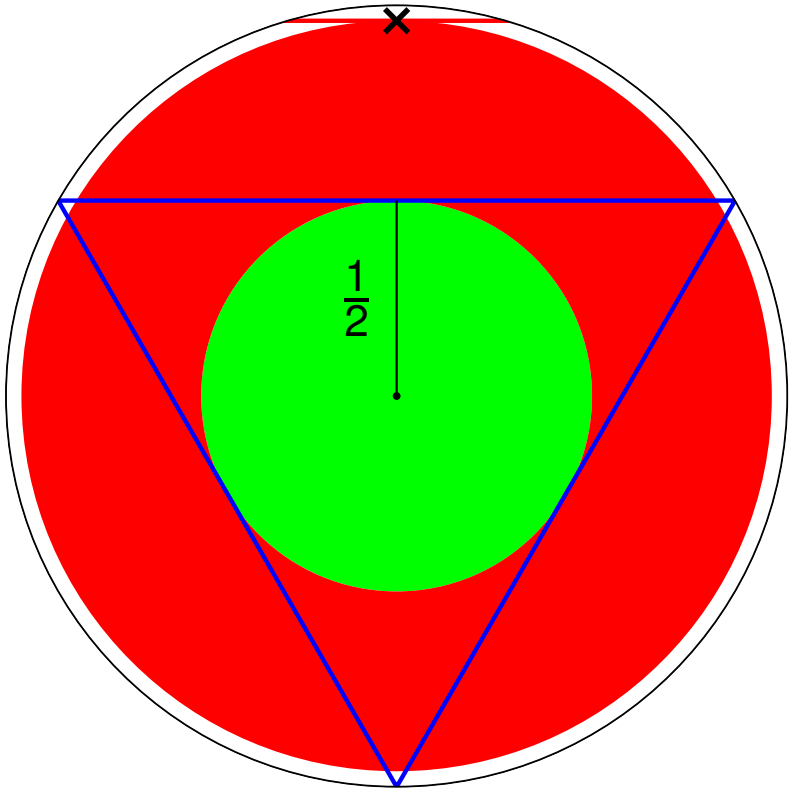
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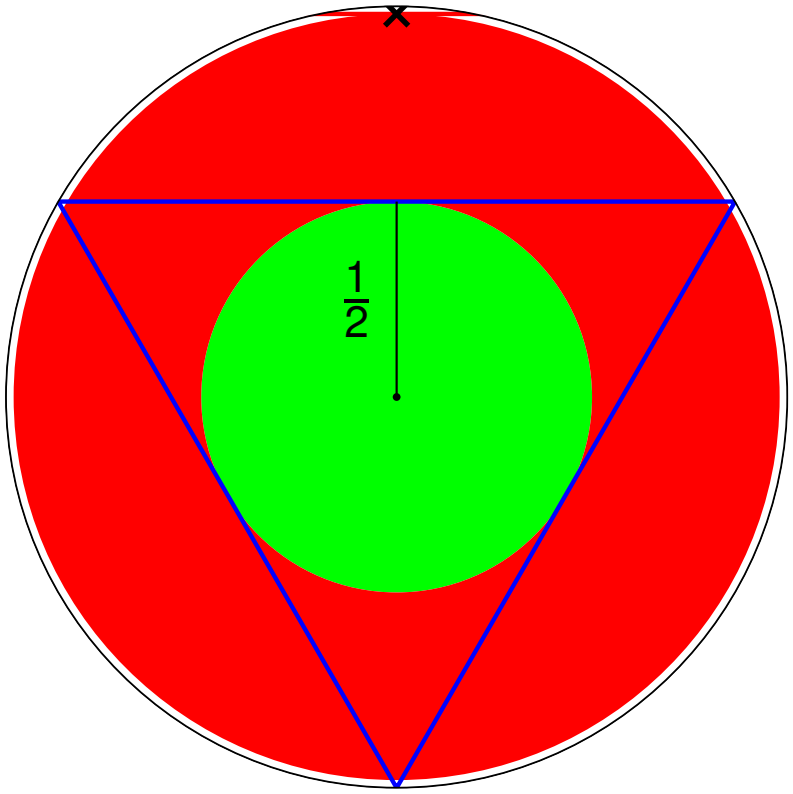
Midpoint:

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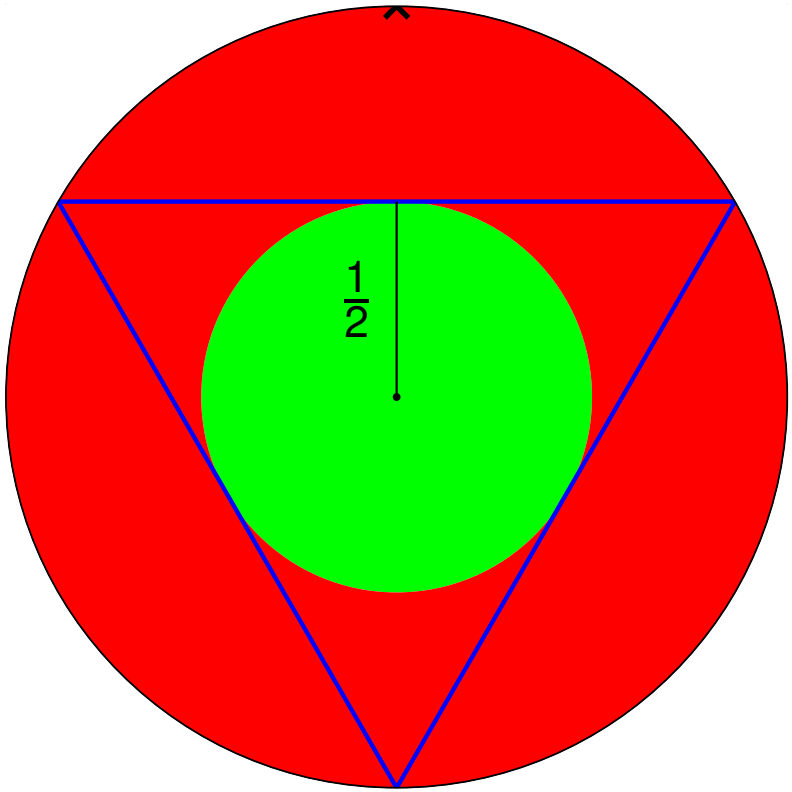
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Midpoint: $\frac{|\ \bullet\ |}{|\ \bullet\ |} = \frac{\left(\frac{1}{2}\right)^2 \cdot \pi}{1^2 \cdot \pi} = \frac{1}{4}.$

Random chords and a question

Answer: $\frac{1}{2}$? $\frac{1}{3}$? $\frac{1}{4}$?